



CHAPTER

ELECTRICAL SYSTEMS

LEARNING OBJECTIVES

By the end of this chapter you should be able to:

- recognize the electrical service drop and service entrance and how they should be arranged
- determine the size of the service and how to advise your client about it
- recognize the problems commonly found on the service drop and their implications
- identify common problems found on service entrance conductors and their implications
- understand the function of the service box or service panel
- be familiar with the arrangement and location of the service box
- recognize the common conditions found in service boxes and their implications
- know the functions of grounding and bonding
- understand the common problems found in system grounds and their implications
- identify 15 common problems with branch circuit wiring, and their implications
- recognize conditions found in all panels, as well as those unique to subpanels, fuses, breakers, and panel wires
- understand how wires should be connected and supported
- understand how to identify knob-and-tube wire and the issues associated with it
- know how to identify aluminum wiring and the issues associated with it
- be familiar with common problems at lights and receptacles and their implications
- be familiar with the strategies for inspecting the various components of household electrical systems

INTRODUCTION

The goal of this chapter is to ensure an understanding of the basics of residential electrical systems. You should be able to identify and describe common electrical components, understand the function and importance of each, and recognize adverse conditions that may occur. We will examine the causes and implications of these conditions and appropriate courses of action to be recommended to clients. The systems addressed include—

1. service drop and entrance
2. the grounding system
3. the electrical service box, including the main disconnect and fuses or breakers
4. distribution panels
5. branch circuit wiring
6. lights, outlets, switches, and junction boxes

Electricity has a language all its own. You should know the technical terms but also be able to communicate with clients clearly. For example, clients will understand “wires coming from the street to the house” better than “service drop,” and “breakers or fuses” better than “over current protection devices.” By the way, most people use the words wire and cable interchangeably, although wire is a single conductor and cable is a group of conductors.

Safety is such a critical issue with respect to inspecting electrical systems that we will mention up front several precautions you should follow:

1. Wear rubber-soled shoes.
 2. Do not stand in water or wet areas when touching electrical equipment (sometimes this can't be avoided on the outside of houses).
 3. Use insulated tools to remove covers.
 4. Do not let your clients stand beside a panel you are working on. They may do something dangerous before you can stop them.
 5. Use a voltage detector to ensure that any electrical equipment you are about to touch is not live. Some advocate using lineman's gloves when working on electrical systems.
-

7.1 SERVICE DROP

Function

The purpose of the service drop and the service entrance is to get electricity safely from the utility into your house.

Overhead Wires to House

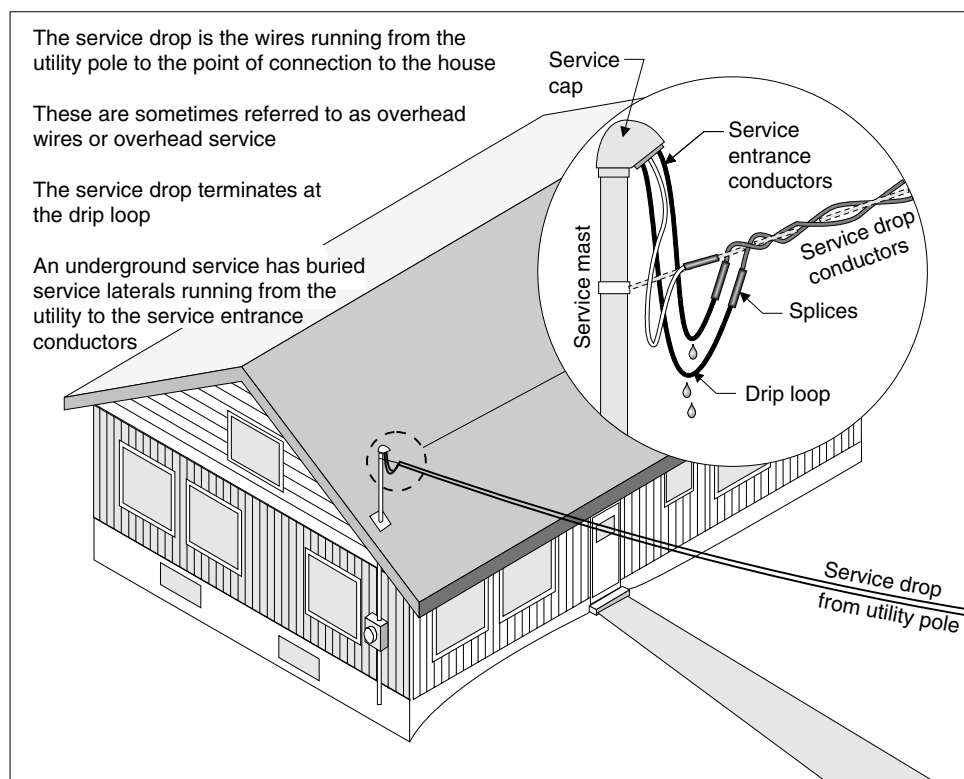
The **service drop** is the collection of overhead wires coming from the utility pole (often at the street) to the point of connection to the house (Figure 7.1). People call these the overhead wires or overhead service. Electricians and utilities use the words **service drop**.

Underground Wires

Some utilities use the word **service laterals** to refer to underground services. Here the wires come from the street underground to the house. They may come up out of the ground outside the house to go to an outside meter, or they may go straight into the house.

Drip Loop

Where the service drop or service lateral ends, there is a **splice**, sometimes called the **service point**. This means two wires are joined together. In a service drop this will be at the **drip loop**. The drip loop is a U-shaped bend in the wires that allows water to drip off, so it won't go into the **service entrance**. The drip loop is typically

FIGURE 7.1 Service Drop

part of the service entrance. The drip loop also shows that the wires are properly secured to the building and are relaxed.

The drip loop is at the top of the **masthead** (also called **service cap**, **entrance cap**, **pothead**, **weatherhead** or **servicehead**). The wires running from this connection down into the service box are called the **service entrance**. The wires may be in a conduit or may just be a cable. They may be above the roof or below it, attached to the house wall.

The utility usually provides the service drop and the homeowner is responsible for the service entrance, including the drip loop.

You have to decide whether you are going to get on the roof. This gives you a better look, although please resist the temptation to touch these wires. Electrical systems can look perfectly safe but be very dangerous.

7.1.1 Service Drop Description

Some home inspectors use the service drop wires to gauge the size of the electrical service. This may be a mistake. These wires are very often smaller than the service entrance wires and in-house wires and may mislead you. These wires are in open air and can carry more electricity than in-house wires because it's so easy for them to dissipate their heat. The wire sizes and ampacities that we normally use are for wires in conduit or cable, not for service drop wires.

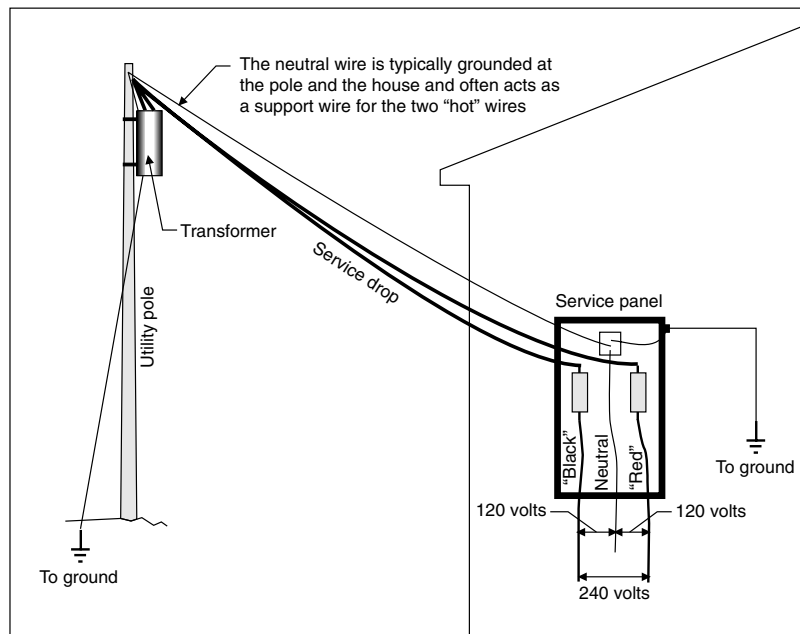
The number of wires coming in through the service drop will tell you some things about the house service. There will usually be three wires coming in through the service drop: a black, a red and a white (neutral). Sometimes the neutral is also

Responsibilities

Inspect from Roof? Careful

Often Smaller Wires Than Service Entrance Wires

Usually Three Wires in Service Drop

FIGURE 7.2 120/140 Volts

the support cable for the red and black. Sometimes there is a separate cable supporting the three conductors. The neutral wire may be bare (no insulation), and in some cases is smaller than the hot wires.

240 Volts

Two Wires for 120-Volt Services

Four Wires Mean 3-Phase Service (Commercial)

No Service Entrance Wires

Are there three wires spliced into the service entrance? If so, this will be a typical 240-volt, single-phase residential service (Figure 7.2).

Two wires spliced into the service entrance cable indicate a 120-volt service. This is rare and not adequate for most modern lifestyles.

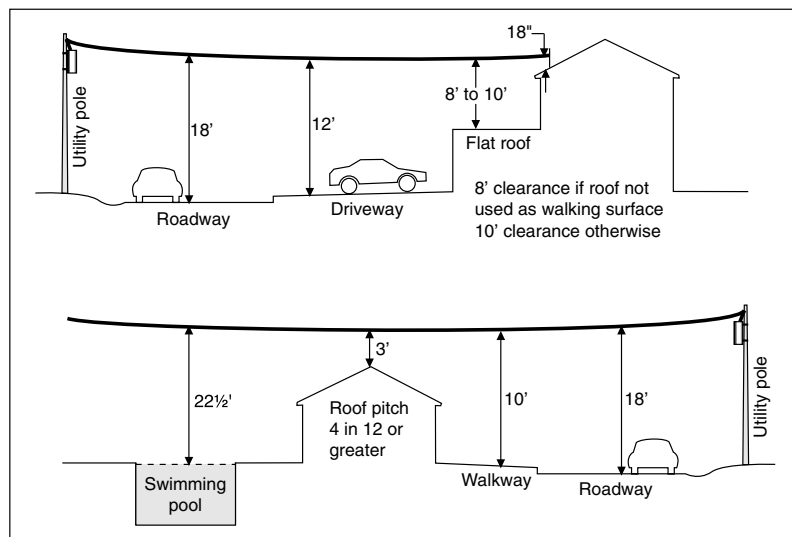
Four wires coming into the house indicate a three-phase system. This is a commercial electrical system and beyond the scope of a standard home inspection.

Underground service laterals may head straight into the house to the main disconnect. In this case, there are no service entrance conductors as such.

7.1.2 Conditions

Let's look at some of the service drop problems you'll find:

1. Overhead wires too low
2. Damaged or frayed wires
3. Trees or vines interfering with wires
4. Wires too close to doors or windows
5. Wires not well secured to the house
6. Poor connection between the service drop and service entrance
7. Inadequate clearance from roofs

FIGURE 7.3 Service Drop Clearances

Overhead Wires Too Low

Wires Should Be out of Reach

The wires should be out of reach of people, and safely above places where they may be hit by vehicles, ladders and other equipment (Figure 7.3). The wires need only be 10 feet above a walking surface, 12 feet above a driveway and 18 feet above a roadway. Wires near swimming pools must be 22½ feet away in any direction from the water.

Wires over Roofs

Wires running over a flat roof that is used as a walking surface should be 10 feet above the roof. Otherwise, wires should be 8 feet above the roof. If the roof has a slope greater than 4 in 12, the wires have to be only 3 feet above the roof.

Decks and Balconies

Wires should be out of reach of people on decks, porches and balconies. The wires should be 8 to 10 feet above the floors of these structures.

IMPLICATION

There's an electrical safety issue for people. You don't want to touch service drop wires because of the danger of electrical shock. Wires running just above roofs are dangerous for roofers, and may be damaged by snow or ice in cold climates.

STRATEGY

Check the height of overhead wires coming in, and use common sense as to whether they could be touched by people or vehicles.

Damaged or Frayed Wires

These are usually (but not always) the responsibility of the utility.

IMPLICATIONS

There is the possibility of a short circuit and loss of electrical service to the house. There is also a risk of electrical shock, especially if the wires fall.

STRATEGY

Visually examine these wires from the ground (ideally with binoculars) and look for evidence of mechanical damage or loose insulation. Pieces of insulation hanging off the wires are the most common visible defect.

Trees or Vines Interfering with Wires

Service drop wires should be clear of anything that will cause abrasion.

IMPLICATIONS

The implications are mechanical damage to the wires and reduced ability to dissipate heat.

Check that the wires are clear of any obstructions. Where tree branches or vines interfere with the wires, inform your client. It is possible that the utility will correct this problem.

Wires Too Close to Doors or Windows

Modern electrical codes require that wires be 3 feet away from the sides or bottoms of windows, doors or fire escapes. This is true in some jurisdictions whether the windows open or not, as fixed windows may be replaced with operable windows in the future.

IMPLICATIONS

Again, there is a hazard wherever anyone may be able to touch wires.

STRATEGY

Make a note of situations where wires are too close to windows, doors, or fire escapes. Be careful, since utilities may have accepted this installation. Simply point out the common sense issues and advise your client that the utility is the authority.

Wires Not Well Secured to the House

In most modern installations, the neutral wire supports the red and the black wires. The neutral wire should be attached to the house or to an electrical mast. Rather than looking at all the ways this may be done, you should simply look for a successful connection.

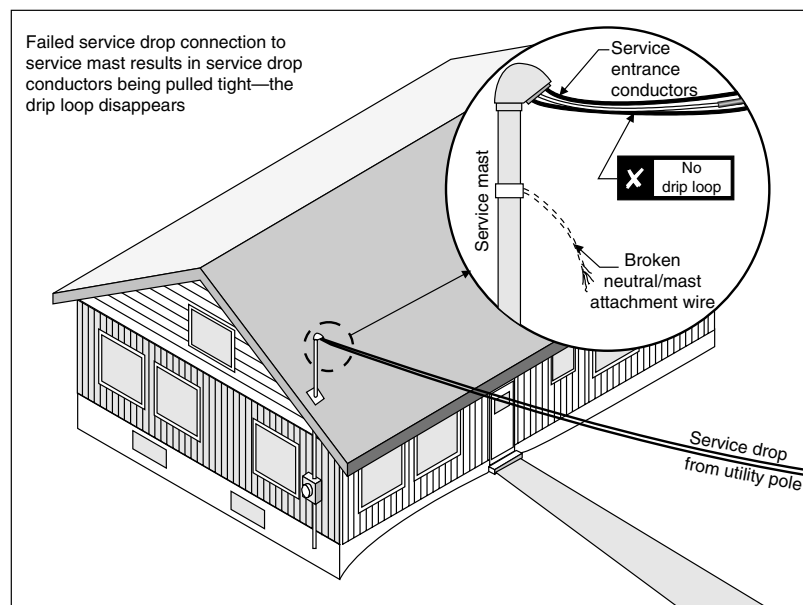
IMPLICATIONS

If the wires come loose, they may fall. This will cut off power to the house, and may result in live electrical wires lying on the ground.

STRATEGY

Look for evidence of movement at the point of connection to the house or mast. Connectors that are starting to pull away, or service drop wires with more sag than is typical, should be noted for further investigation. Watch for the absence of a drip loop (Figure 7.4). This suggests wires under tension.

FIGURE 7.4 Wires Not Well Secured to the House



Poor Connection Between the Service Drop and Service Entrance

At the end of the service drop, the wires are joined to the **service entrance wires**. These splices are most often made by the utility, but they can come loose. Connections should be made at a drip loop, which is a relaxed, sagging part of the service entrance cables. The connection should not be at the low point of the loop, since this is where water may collect.

A poor connection may result in a loss of electrical power to the house and, if the wires fall, there is an electrical hazard.

Look for evidence of the connection pulling apart. Electrical tape on connections may indicate amateurish work. Many utilities use a heat-shrink seal on this splice.

IMPLICATIONS

STRATEGY

Inadequate Clearance from Roof

As we discussed earlier, wires can be run over house roofs if they are at least 8 feet above the roof or 10 feet above a walk-out deck. Wires over a roof with a slope of more than 4 in 12 only have to be 3 feet above the roof. If the wires only project 4 feet onto the roof, they only have to be 18 inches above the roof.

Implications include mechanical damage and possible electrical shock during normal repair and maintenance activities.

Know what is accepted in your area, and advise your clients of the risks of wires that are too close to the roof.

In some jurisdictions, it was permitted to attach wires virtually to the roof surface. While this is not a desirable situation, the utility may tell your client that this is accepted. The utility is the authority, and you will lose any argument with them.

IMPLICATIONS

STRATEGY

7.1.3 General Strategy

With electrical systems, there are many numbers you can memorize. It's more important to understand the rationale behind things and to use common sense. Overhead service wires should be intact and out of the way. They should be well-supported and the connections should be secure. Rather than citing code, it's perfectly acceptable for a home inspector to say, "These wires are within reach of people. This should be checked by an electrician and the wires may need to be moved."

Use Common Sense

7.2 SERVICE ENTRANCE WIRES

From Service Drop to Service Equipment

Service wires pick up where the **service drop** wires end. They extend down to the **service equipment (main disconnect, service panel, service box)**.

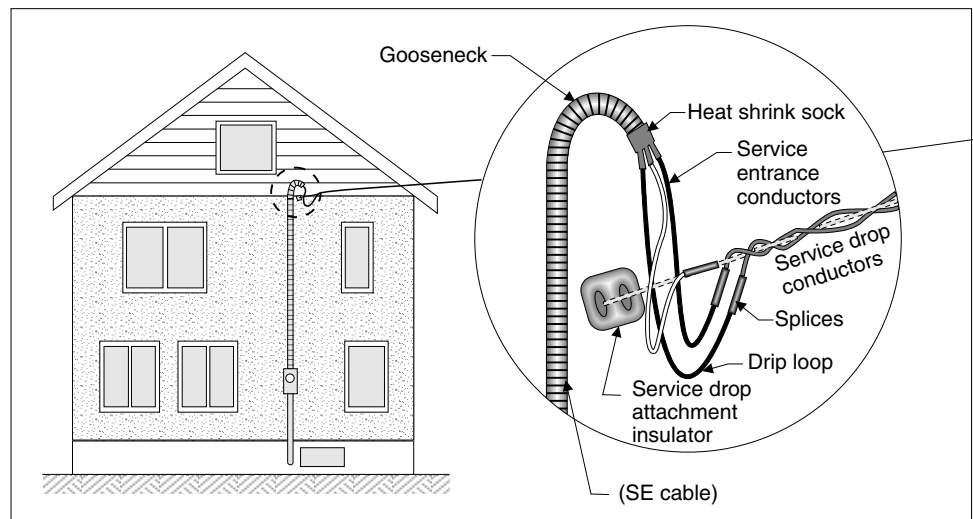
The electrical meter is often part way along the **service entrance wires**. The correct name for these wires is **service conductors**. Where these wires pass through a wall from outdoors to the service equipment, they are properly called (according to some authorities) the **service entrance conductors**.

The service conductors include a black and a red wire, which are hot (ungrounded), and a white wire, which is neutral (grounded).

Many jurisdictions require the service entrance wires to extend at least 30 inches out of the **service cap** at the top of the conduit or masthead. This gives the utility enough room to make their connections and leave a drip loop.

Three Wires

30 Inches at Top

FIGURE 7.5 Using Service Entrance Cable*Drip Loop*

A drip loop should be provided whether there is a mast above the roofline or a conduit or cable below the roofline.

Gooseneck on SE Cable

Where service entrance (SE) cable is used without conduit or masthead, the cable itself is bent downward (like a candy cane or **gooseneck**) before connecting to the service drop wires (Figure 7.5). This discourages water entry into the cable between the wires and the exterior sheathing.

Conduit or Service Entrance Cable

The service entrance wires may be in a conduit on the outside of the building, or there may be a special **Service Entrance (SE) cable** that does not require a conduit. Either situation is permitted.

Some service drops are attached to the side of a building. Here the conduit will extend up the building and terminate in a service cap.

Mast

Where the service drop wires are above the roofline, a **service mast** is required. This mast extends up through the roof and terminates in a **service cap**. This mast often forms a conduit that contains the service entrance wires. It is also the mechanical support for the service drop wires. Depending on the height of the mast, guy wires may be necessary to hold the mast straight. The weight of the service drop wires is significant, and poorly supported masts can be bent or even broken.

Flashing

Where the mast comes up through the roof, the hole must be properly flashed. Good practice dictates that the service drop wires are attached to the mast within about 12 inches of the top. We don't want the service wires too close to the roof. Many jurisdictions call for them to be 2 feet above the roof surface. This means that the mast usually extends at least 3 feet above the roof.

*Wires Well above Roof**Service Entrance Wire Type*

If you can see the service conductors at the service cap, you can sometimes check the gauge and find out whether they are copper or aluminum.

Service Size

The Standards ask you to report the service size (amperage and voltage rating). Based on the gauge of the wire, you can determine whether you have wires suitable for a 60-amp, 100-amp or 200-amp service. The number of service drop wires told you already whether it's a 120 volt or 240 volt service.

Meter Weather-Tight

You can look to see whether the connections at the meter base are secure and weather-tight.

Let's look at some of the problems that you'll find with service entrance wires:

1. No drip loop
2. No masthead (entrance cap, service cap, pothead, weatherhead)
3. Masthead not weather-tight
4. Service entrance wires too close to the roof
5. Mast or conduit bent
6. Mast or conduit rusted
7. Mast rotted (if wood)
8. Mast, conduit or cable not well secured
9. Mast, conduit or cable not weather-tight
10. Conduit or cable not well sealed at house wall penetration
11. Cable frayed, mechanically damaged or covered by siding

7.2.1 No Drip Loop

We've talked about how the drip loop is designed to prevent water getting into the conduit and to show that the service drop wires are not straining the splices.

The implications are water entry into the electrical system and possible mechanical stresses on the wires and splices.

Look for the service entrance wires to have a low spot where they connect to the service drop wires. The wires should be visibly slack.

IMPLICATIONS

STRATEGY

7.2.2 No Masthead or Service Cap

The implication is water getting into the electrical system.

Make sure there is a weather-tight cap on the conduit or mast. This should be arranged so that wires enter the underside of the cap.

Common sense should tell you whether the connection to the service drop is likely to allow water in.

IMPLICATION

STRATEGY

7.2.3 Masthead Not Weather-Tight

Sometimes there is a service cap (masthead), but it is not tightly installed or not installed at the right angle. It should be weather-tight.

The implication is water entering the electrical system.

Using common sense, make sure the top of the mast, conduit or cable is weather-tight.

IMPLICATION

STRATEGY

7.2.4 Mast or Conduit Bent

The masts or conduits should be straight and, in most cases, vertical.

The wires may sag reducing overhead clearance or fall, creating a dangerous situation. Weather-tightness may also be lost.

IMPLICATIONS

STRATEGY

Make sure the masts and conduits are not bent or pulled out of position. Look at clamps for evidence of movement. Masts taller than 5 feet should have guy wires.

Where there is a mast, look at the roof flashing connection to see if it has been displaced. The drip loop may or may not be intact, depending on how it fails.

7.2.5 Mast or Conduit Rusted

IMPLICATIONS

The implications of rusted mast or conduit are—

1. mechanical failure, allowing wires to fall
2. water entry into the electrical system

STRATEGY

Look for rust at electrical connections. Rust is most likely to occur at horizontal surfaces where water may collect. Pay particular attention to threaded connections where the pipe walls have been made thinner by cutting threads into them.

7.2.6 Mast Rotted

Rot is common on wooden masts, particularly near horizontal surfaces where water may accumulate, and near roof flashings. Wood masts are not allowed on new work in many areas.

IMPLICATIONS

The implications are a loss of strength of the mast and collapse of the wires.

STRATEGY

Look carefully at wooden masts for rot or insect damage. Suggest painting the wood if necessary.

7.2.7 Mast, Conduit, or Cable not Well Secured to the Building

Figure 7.6 summarizes the support requirements for cable, conduit and mast implementations.

IMPLICATIONS

The mast, conduit or cable may pull away from the building, resulting in fallen wires and a dangerous situation. Open joints will allow moisture into the electrical system.

STRATEGY

Rather than memorizing numbers for cable and conduit supports, look for evidence of movement or failure of connectors.

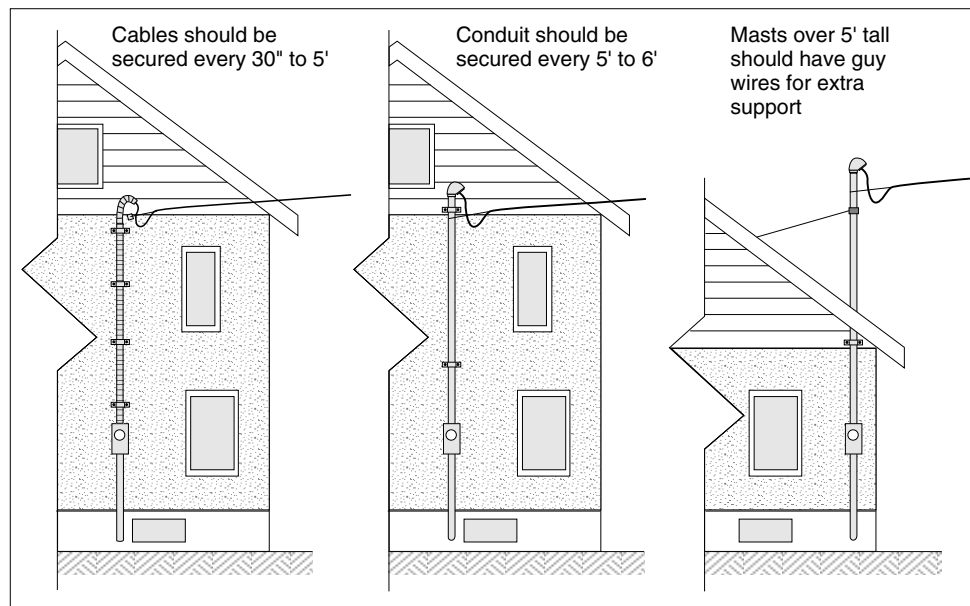
7.2.8 Mast, Conduit, or Cable not Weather-Tight

We've talked about some of the conditions that may cause this already. Common spots where weather-tightness may be compromised include (Figure 7.7)—

1. joints in conduits
2. the service cap (masthead)
3. the flashing at the roof level
4. the meter base

IMPLICATIONS

Water in the electrical system is hazardous. In a best case scenario, the electrical system may shut down. In the worst case, the system may cause an electric shock or start a fire.

FIGURE 7.6 Support Requirements for Service Entrances**STRATEGY**

Follow the mast, conduit or cable from top to bottom looking for places where water may enter.

7.2.9 Conduit or Cable Not Well Sealed at House Wall Penetration

IMPLICATIONS

The implications are—

1. water getting into the electrical system causing a shock hazard
2. water getting into the basement causing a nuisance
3. rusting of the service equipment

STRATEGY

Check Inside

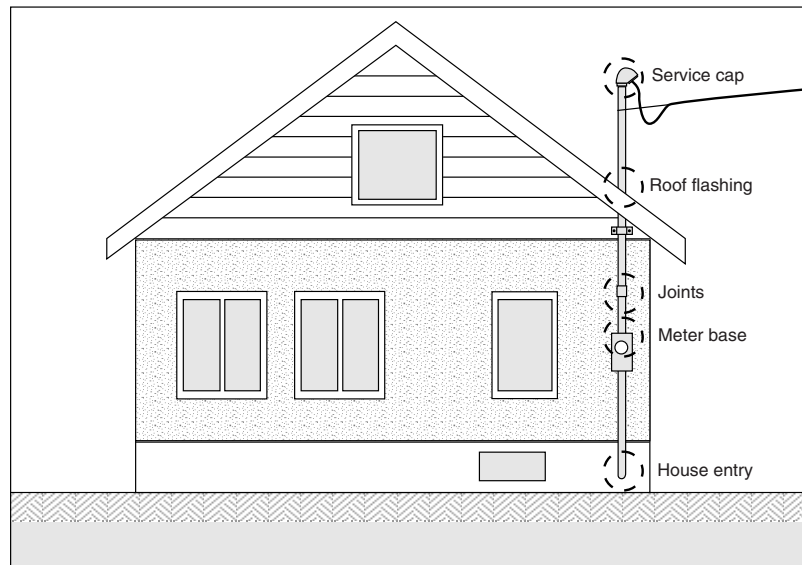
Make sure the wall/conduit junction is well sealed, if it's visible.

The conduit or cable may enter the house below grade. In this case, you won't be able to see whether the cable is well sealed where it goes through the wall. You'll have to check this inside when looking at the service equipment.

7.2.10 Service Entrance Cable Frayed, Damaged, or Covered by Siding

IMPLICATIONS

This can create a dangerous electrical condition. Anyone touching the cable could get a shock. If live portions of the cable touch metal siding, the entire skin of the building could become electrically charged. If the cable is covered by wood siding, for example, driving nails through the siding creates a shock hazard.

FIGURE 7.7 Service Entrance—Areas of Potential Water Entry**STRATEGY**

Where cable is used, carefully check its condition. Make sure it is accessible and visible over its entire length. Some authorities allow conduit to be run behind brick siding, for example, since the wires are protected from damage. Cable, however, should never be buried.

7.3 SERVICE SIZE

Voltage Is Always 240

Anyone buying a house will want to know the service size.

The size of the electrical service is a function of both the voltage and amperage. For 999 out of 1,000 houses, the voltage is 240 volts. 120-volt service is becoming very rare.

Ampacity Is the Key

The challenge is to know how much current the system is capable of carrying. People talk about this as the **ampacity** of the service or the **service size**. The common service sizes are 60-amps, 100-amps, and 200-amps, with some others thrown in between. While it's a little too simple to say that small houses are okay with 60-amp services, average houses are okay with 100-amps and big houses need 200-amps, it's not a bad yardstick.

We Don't Do Load Calculations

At the other end of the scale are load calculations, performed to determine exactly what a house needs. Home inspectors don't do these detailed calculations. They use general rules, and good inspectors explain to their clients how lifestyle is a big factor in determining whether the electrical service is adequately sized.

Let's clarify a couple of things:

Not a Safety Hazard

1. An undersized service is not a safety hazard. If you have a 60-amp service in a huge house with many electrical demands, what is likely to happen? If you use enough appliances at the same time, the total current draw through the red and/or black wires will exceed 60 amps and the main fuses or breakers will shut off the power. This is an inconvenience, but it isn't a safety hazard.

Simultaneous Use Is the Problem

- Undersized services are only a problem when many appliances are used simultaneously. Having an electric stove, an electric water heater, an electric clothes dryer and an electric central air conditioner doesn't mean much, unless you're trying to use them at the same time. Of course, the more large appliances there are, the more likely it is several will be used at one time.

New Construction

60-amp services are considered obsolete and not allowed in new construction. Adequate electrical capacity is tough to pin down, no matter how much calculation you do, because it's determined by how people use the electricity in the house. Let's look at some general rules.

7.3.1 100-Amp Service

Common

In most modern average-sized houses, 100-amp service is common. It works for most modern lifestyles in a typical three-bedroom, two-story house. It probably won't be enough in a big house. It probably won't be enough if you have electric heat, and it may not be enough if you have a big family and many electrical appliances, including central air conditioning, electric stove, electric clothes dryer, and electric water heater. Large saunas and electric swimming pool heaters may also result in a need for a larger service than 100 amps.

If there is a large workshop, pottery kilns, or some other craft that requires considerable electricity, 100 amps may also be insufficient.

7.3.2 125-Amp and Larger Services

Small Homes with Electric Heat or Bigger Homes

125- and 150-amp services are common in some areas on small houses with electric heat or larger houses without electric heat. They are also found on average-sized houses with more than a typical number of appliances. A 200-amp service would be found on an electrically heated home or a large house.

Very Large Home

In rare cases, you may find a 400-amp electrical service. This is the largest single-phase service available. It's the largest service most utilities will put into a residence. The residence is probably more than 6,000 square feet, or quite large and electrically heated.

7.3.3 Wire Sizes

Wire Gauge—Bigger Is Smaller (AWG)

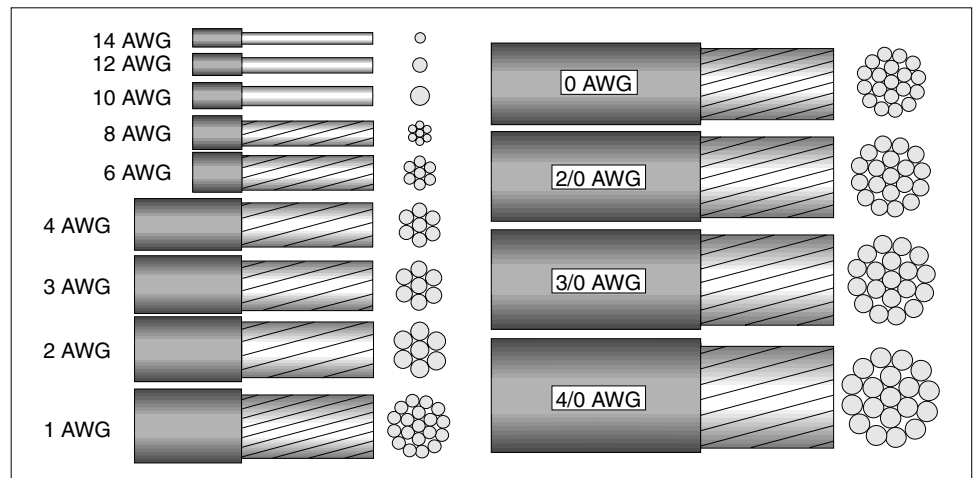
Logic would suggest that small wires have small numbers and as wires get bigger, the numbers get bigger. Actually, the American Wire Gauge (AWG) system works backwards. A **14-gauge** wire is smaller than a **12-gauge** wire and a **12-gauge** wire is smaller than a **10-gauge** wire.

Don't Use All Numbers

But it's more confusing than that. They don't use every number. For example, they only use 14, 12, 10, 8, 6, 4, 3, 2, 1, and 0.

One of the problems with numbering backwards is once you get down to **0** you don't have any numbers left for bigger wires. So, there is a wire size called **0** and the next bigger wire size is **00**, sometimes shown as **2/0**. The next size after that is **3/0** or **000**. The largest size you'll see in most houses is 4/0 (used in some areas for 200-amp services where the service entrance conductors are aluminum). Figure 7.8 summarizes the wire sizes you'll see most commonly.

Table 7.1 summarizes typical wire sizes for residential wiring in the United States and the amount of current (in amps) that can safely travel through each size. These

FIGURE 7.8 Common Wire Sizes

are general rules, and certain variables—such as the temperature rating of the wire and whether it is in free air or enclosed in a raceway or cable—can affect the wire's capacity. Where something doesn't fall into the general rules listed in Table 7.1, there may be a circumstance that makes it okay. Be careful with your criticism.

7.3.4 Determining Service Size

Now that we know how much current each wire can carry, let's see how we identify it in the field.

Check Service Entrance Wires

The best indication of service size is the service entrance wires themselves. Sometimes you can see enough of this wire to read the size on the sheathing at the top of the service entrance conduit. Where the service entrance wires are spliced to the service drop, there should be a drip loop. Not all wire has the size stamped on the outside of it, but if it does, you'll be able to see it here, if you can safely get close enough to look.

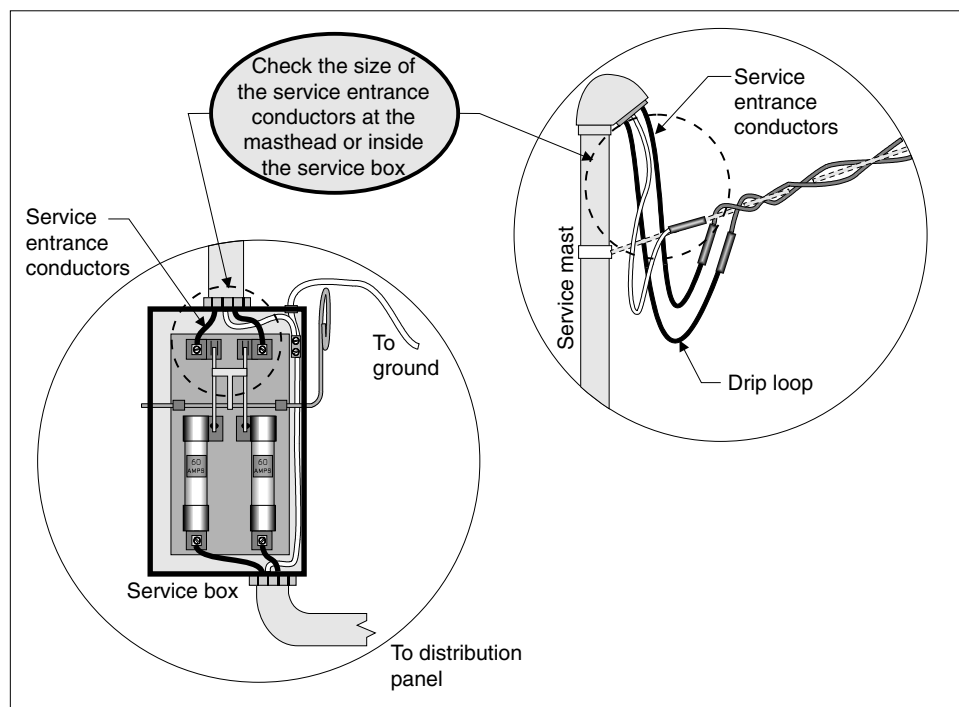
TABLE 7.1 Typical U.S. Service Sizes

Minimum Wire Size		Service Size (amps)
Copper	Aluminum	
10*	8*	30
6*	6*	60
6*	4*	70
4	2	100
2	1/0	125
1	2/0	150
2/0	4/0	200

Note: 1. *—not found on new work.

2. All wire gauges are AWG.

3. Several variations are possible depending on the type of wire, the temperature rating, etc.

FIGURE 7.9 Determining Service Size by the Service Entrance Wires

You may also be able to read the wire size at the other end of the service entrance conductors, where they come into the service box or panel (Figure 7.9).

Service Entrance Cable

It may be possible to read the size directly off the sheathing on a service entrance (SE) cable, since there is no conduit.

Copper or Aluminum

You may also be able to tell whether the wire is copper or aluminum. Aluminum wire always has the word ALUMINUM, the letters ALUM, or the letters AL written on the sheathing.

Not Reliable to Measure Bare Wire

It's difficult to identify the wire gauge by looking at the diameter of the bare wire, although some inspectors carry samples of the common wires and compare their samples to what they see in the field. Be careful not to touch any live electrical components.

Don't Measure Cable

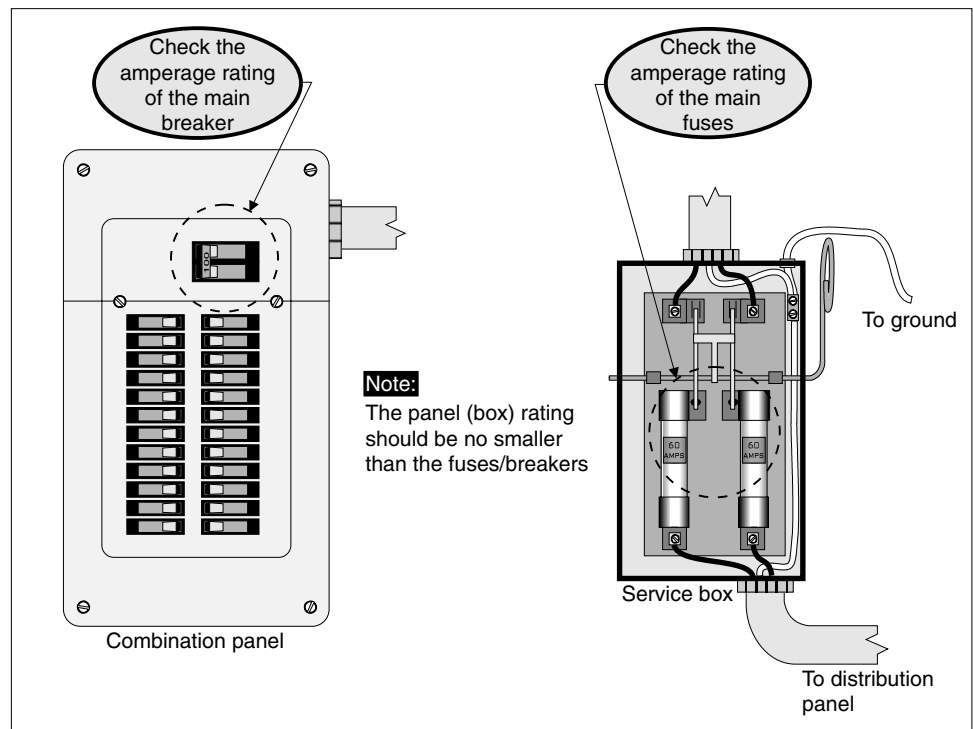
Measuring the outside diameter of the cable is risky, since different cable types have different sheathing thicknesses. Measuring the actual conductor size is usually not practical for the home inspector, although there are plastic wire gauges made for this.

Don't Measure Neutral

Sometimes the neutral wire is smaller than the red and the black wires. This is permitted, but can cause you to mistake the service size. You should be sizing the red and the black wires.

Check Main Fuses or Breakers

If you can't size the service entrance conductors, the next best indicator is the size of the main disconnect (assuming there is a single disconnect for the entire home) (Figure 7.10). If the main disconnect is fused, there will be two fuses in the service box. One fuse is for the red service-entrance conductor and the other is for the black. If each of these fuses is rated at 100-amps, you have a 100-amp service. **Be careful: you cannot add the two fuses together.** A 200-amp service would have two fuses, each sized at 200-amps. Where breakers are used, you should see a double-pole 200-amp breaker.

FIGURE 7.10 Determining Service Size by the Main Disconnect

The fuse size may be written on the fuse label, stamped into the body of the fuse, or stamped into one of the metal ferrules at either end of the fuse.

Don't Touch!

In some cases, the fuses will be mounted in the service box with the labels facing the back of the box. You may not be able to read them. Again, do not touch anything in the service box. **This can kill you.** Even if the power to the house is turned off, the service entrance wires coming into the service box will be live, and if any mistake was made hooking it up, the fuses can be live, too.

The fuse or breaker size **usually** indicates the service size, but not always. Some inspectors say, "Apparent 100-amp service," when all they can read is the fuse or breaker size. Some call for an electrician to verify the fuse size.

Limitations

If you can't size the service entrance wires and you can't see the fuse or breaker ratings, report that you could not identify the service size. Home service boxes have a cover that cannot be opened without shutting off the house power. **Do not shut off the house power.** Simply describe the limitation to your client.

Breaker Size Easy to Read

If the main disconnect is a circuit breaker, the breaker size is usually stamped or printed on the handles. You don't have to remove the service box cover to see this.

Sealed Service Box

Some boxes are sealed by the utility for the homeowner's safety and to prevent bypassing the meter or tapping off electricity upstream of the meter, in cases where the meter comes after the service box.

What If Wire Size Doesn't Match Fuse Size?

Determining the size of the breakers or fuses is not as reliable as sizing the service entrance conductors. The fuses or breakers and service entrance conductors **should** be a match in terms of their current rating, but they aren't always. Undersized fuses will cause you to report the service as smaller than it actually is, if you rely solely on the fuse size. This is not a safety concern, since the fuses will cut off the electricity long before the wire overheats.

Overfusing Unsafe

The fuses may be too large for the service entrance conductors. This would lead you to describe the service size as larger than it actually is, if you just read the fuses. This may be an unsafe situation because the fuses may not shut off the electricity quickly enough to keep the service entrance conductors from overheating.

The goal is to identify the service entrance conductor size and make sure it matches the fuses or breakers. We often have to base our determination on the rating of the main fuses or breakers because we can't size the service entrance wires. They are rarely wrong, but there is a slight risk in describing the service based only on fuse or breaker size.

Service Box and Distribution Panel Ratings

The service box and distribution panel have to be big enough to accommodate the service size. The box and panel can be rated for the same or more amps, but not less than the service size.

No Main Fuses or Breakers? Use Wire Size

In some houses, there is no single main disconnect and no service box. There are multiple disconnects to shut off all the house power. Up to six throws (switches or breakers) are permitted to shut off all the electricity in some houses. Your determination of the service size must be based on the service entrance wire size. Adding all of the disconnect ratings will not usually give you an accurate reading of the service size. This total is often more than the service size. It's allowed because not every circuit will be operating at its maximum rating at the same time. In reality, most circuits never operate near their maximum rating.

7.3.5 Advising Your Client

Service Is Adequate

If the service is adequately sized, using normal guidelines, you'll simply report the service size to your client and say that it is adequate for a typical home of this size. You're taking a small risk if you promise that it will be adequate for your client's needs, since you don't know their lifestyle.

Service Too Small

Where the service is undersized you'll report the service size and recommend increasing it. This includes replacing the service drop (or laterals if it's underground), the service entrance conductors, the service box, and usually, the grounding conductor and distribution panel.

Service Marginal

Where the service size is marginal, you'll report the service size and advise your client to live with the electrical service on a trial basis. Since you can't know their lifestyle, you won't be sure.

Tripping Main Breakers

Explain to the client that a marginally-sized service is not a safety issue. If the service box has circuit breakers, the breakers will trip when too much electricity flows. They'll have to turn off some of the electrical appliances and turn the main breakers back on.

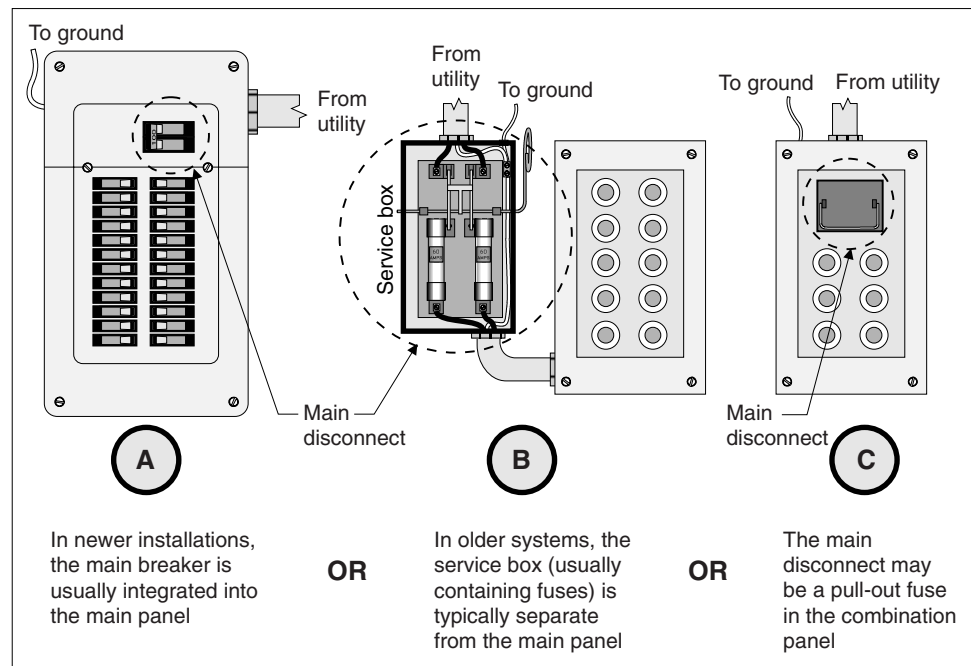
Blowing Main Fuses

If the service box has fuses, they may blow. Very few people can change their own main fuses (be careful, because there is still live electricity in the service box even when the house power is shut off). Never recommend that your clients do this. Utilities will come out to replace main fuses in many areas. If the utility won't change the main fuses, ask an electrician.

One Main Fuse Blows

With breakers in the service box, both will always trip. With fuses, one or both may blow. If one blows, roughly half the house will be without power, and none of the 240-volt appliances will work properly.

In the marginal situations, discuss the possibility of having to increase the service size, but explain that this is a convenience improvement rather than a necessity.

FIGURE 7.11 Breaker and Fuse Type Service Boxes

7.4 SERVICE BOX

Control Center

The service box (also referred to as the **service equipment** or **service panel**) is the main control center of the house electrical system. It's where all electricity can be shut off with a single switch. It's also where all the electricity can be shut off by the main fuses or breakers if the service is overloaded, or if there is a short circuit or ground fault.

Switch on Outside of Metal Box

Most houses will have a metal service box with a main disconnect switch and main fuses or breakers (Figure 7.11). On breaker systems, the main breakers are also the disconnect switch.

No Single Switch

We've mentioned situations where there may be no service box and there may be up to six throws to disconnect all the electricity in the house. Some inspectors recommend adding a service box with one main disconnect for the whole house. This is a safety improvement. It's easier to throw one switch in an emergency. A single disconnect is currently required for new work in many areas.

Combination Panel

Where there is a service box, it may be a stand-alone box or it may be a combination panel (also called service panel) that incorporates the main disconnect with the distribution panel and all its branch circuit fuses or circuit breakers.

Let's look at some of the problems we'll be watching for on service boxes:

1. Poor access or location
2. Loose
3. Rust or water in box
4. Unprotected openings (cover loose or missing, knockouts missing)
5. Damaged parts
6. Overheating

7. Incorrect fuse or breaker size; box rating too small
8. Service entrance wires exposed in the house
9. Poor connections
10. Improper taps
11. Neutral wire bypasses service box
12. Fused neutral wire
13. Fuses upstream of disconnect switch
14. Obsolete box
15. Exterior box not weather-tight
16. Box not rated for aluminum

IMPLICATIONS

Unless otherwise noted, the implications of these problems are electrical shock or fire.

7.4.1 Poor Access or Location

Service Box Location Locked

Service boxes can be inside or outside and are sometimes in garages. If they're outside, they have to be weather-tight. They can be surface-mounted or recessed into the wall. Service boxes on exterior walls are sometimes locked for security. This may make it impossible to shut off the power quickly in an emergency. If this practice is common in your area, find out what the position of the utility company is. The utility may suggest removal of the lock, or a specific location for the key. Distribution panels should not be locked.

Indoor Locations

Electrical equipment should be at least three feet from indoor gas meters and outdoor gas relief vents. We don't want sparks causing a gas explosion! Boxes should be mounted at eye level and should be easily accessible. As a general rule, the area 30 inches wide and 3 feet in front of the box should not be obstructed. When located inside the house, service boxes should not be in bathrooms, clothes closets, kitchen cupboards or stairwells, for example. Boxes in bathrooms are a shock hazard. Boxes in clothes closets or cupboards are a fire hazard because they may ignite storage or clothing.

IMPLICATION

The box may not be accessible in an emergency to shut off power. Poor locations are fire or safety hazards.

STRATEGY

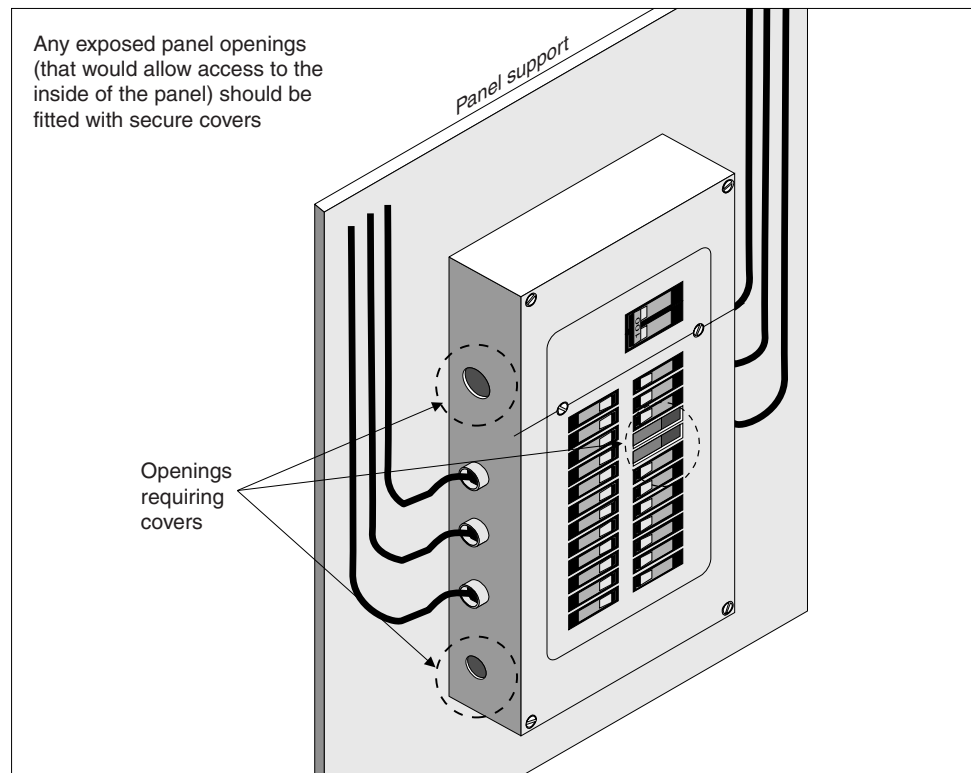
Make sure the box is accessible. If it's locked, everyone in the house should know where the key is kept.

Watch for boxes too close to gas meters or outdoor gas vents, or in bathrooms, stairwells, clothes closets or cupboards. Use common sense to decide whether the box location presents a hazard. If not, we do not recommend change. Most codes are not retroactive, and authorities also use their discretion in asking people to move service boxes.

7.4.2 Loose or Rusted

STRATEGY

Make sure the box is well secured to its support. Check both the exterior and interior of the box for evidence of rust.

FIGURE 7.12 Panel Openings

7.4.3 Unprotected Openings

The panel should not have any openings that people can reach into (Figure 7.12). Covers should be in place and be well secured. Box knockouts that are not used for cables should be intact or replaced with covers.

A person could be electrocuted putting their finger in the hole while fumbling in the dark to re-set a tripped breaker. A fire in the box is much more likely to be contained if there are no large openings.

Make sure all openings in the box are protected.

IMPLICATIONS

STRATEGY

7.4.4 Damaged Parts

Fuse holders, switches and the box itself can be damaged or broken.

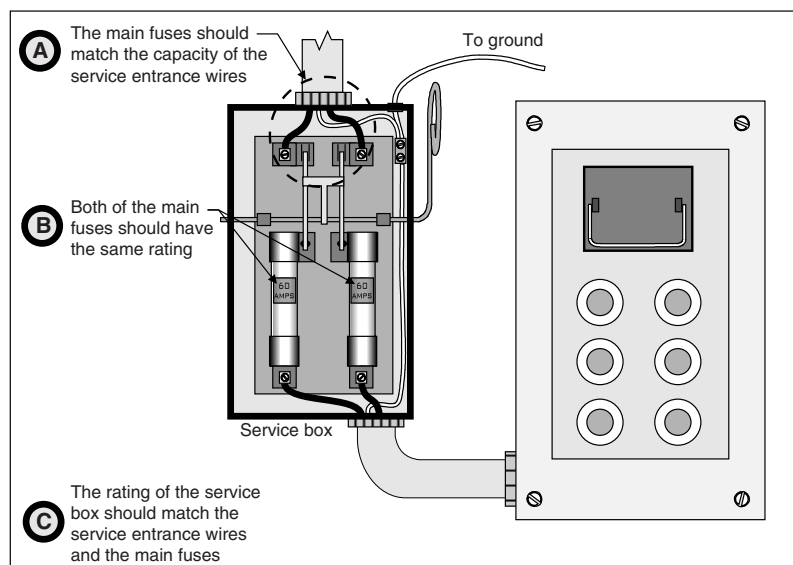
If you can open the service box, look for damaged or broken components. Pay particular attention to fuse holders and switch parts. Check the box and cover as well.

STRATEGY

7.4.5 Overheating

Look for blackened, melted, or charred surfaces. Wire sheathing is often discolored. The smell of burning insulation or sheathing is sometimes a clue. If the box is warm or there are wisps of smoke, there is an immediate and serious problem.

STRATEGY

FIGURE 7.13 Main Fuses Must Be Properly Sized

7.4.6 Incorrect Main Fuse or Breaker Size; Box Rating Too Small

The fuse or breaker should be (Figure 7.13)—

1. sized to adequately protect the service entrance wires
2. no larger than the rating of the service box

IMPLICATIONS

Fuses or breakers that are too small will shut off the house electricity although the system is still safe. They will unnecessarily restrict the amount of electricity available in the house. Oversized fuses or breakers will not shut off the electricity when the current levels become dangerous. Fire and electrical shock are possible.

STRATEGY

Try to make sure the fuses or breakers are the right size for the wire and for the box. The box rating must be at least as big as the fuses and service entrance wires. Where you can't be sure, let your client know.

Sometimes the fuses or breakers are slightly oversized because there is no breaker available that matches the wire ampacity. It's acceptable, for example, to have 200-amp breakers on wire rated for 180 amps, because you can't buy a 180-amp breaker.

Fuses Same Size

Both fuses should be the same size. If they're different, the client should be advised to have an electrician investigate and correct as necessary. At least one of the fuses is the wrong size.

Breaker Brand

The breaker should be the same brand as the box, in most cases. While there are exceptions, recommend further investigation where the breaker is a different brand than the service box.

7.4.7 Service Entrance Wires Exposed in the House

In most jurisdictions, the service entrance wires, even if in conduit, are not allowed to run through the interior of the house before going into the service box. These wires should come straight into the back of the service box.

STRATEGY

Where service entrance wires run inside the house before going to the service box, recommend further investigation. Again, the utility may have made an exception and permitted this arrangement. You should describe it as unusual and not commit yourself to a position that may make you look foolish.

7.4.8 Poor Connections

Loose connections cause more fires than any other electrical problem. Look at each connection. Sometimes it is clear from a visual inspection that they are physically loose. Sometimes you'll see evidence of overheating or melting as discussed earlier.

Loose connections also lead to excessive voltage drop. It's natural to have some loss, or drop, of voltage by the time you get to the end of a long circuit. Most codes allow this to be as much as 3% through a feeder or branch circuit, or 5% total through both. Loose connections lead to too much voltage drop, which you might notice as dim lights, slow motors, etc. It's useful to know this, but the Standards don't require you to test for it.

We do not recommend pulling on connections to make sure they are tight. You may create a loose connection, or worse, electrocute yourself.

Voltage Drop

7.4.9 Improper Taps

Wires for house services should come off the distribution panel, not off the service box. Where the service box is upstream of the meter (before the meter) any taps off the service box will not be metered and people are stealing the electricity (Figure 7.14).

FIGURE 7.14 Illegal Taps

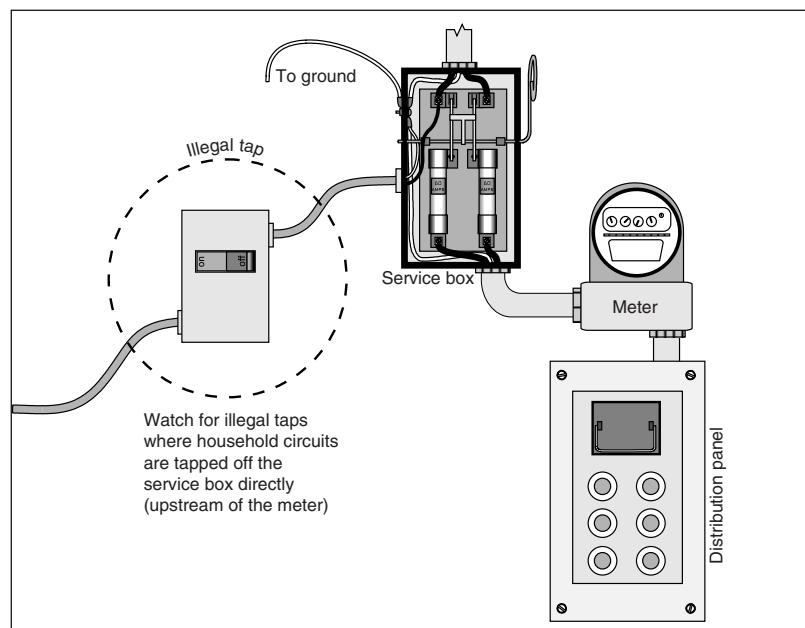
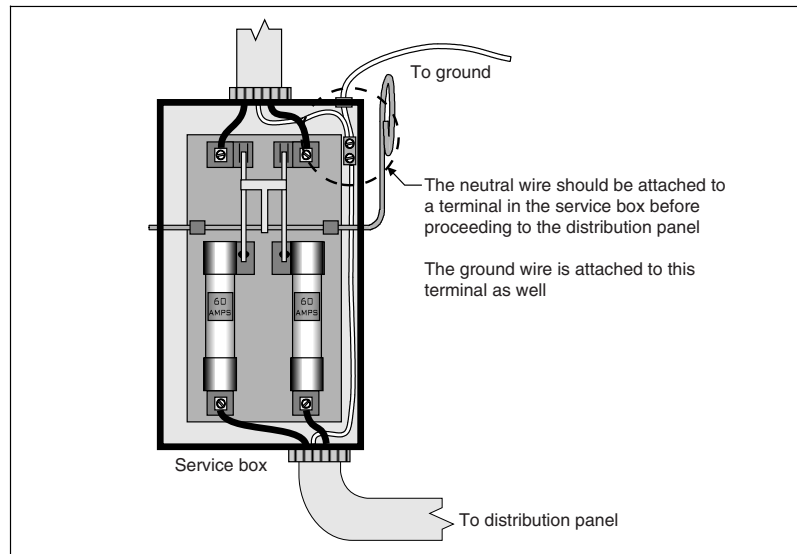


FIGURE 7.15 Neutral Wire Shouldn't Bypass the Service Box**IMPLICATIONS**

Service boxes are not designed for multiple taps (more than one wire for every fuse, breaker, or wire securement point). There is tremendous potential for loose connections. In addition, the possibility of legal action by the utility exists if people are taking electricity upstream of the meter.

STRATEGY

Look at the service box for wires that are tapped off the box directly.

7.4.10 Neutral Wire Bypasses Service Box

The neutral (white) service entrance conductor should be attached to a terminal in the service box, not in the distribution panel (Figure 7.15).

IMPLICATIONS

The service box is the point in the electrical system where the neutral and grounding systems are supposed to come together. We'll talk more about why this is done when we talk about grounding. If the neutral wire bypasses the service box, the electrical system is not as safe as it should be, because the neutral and ground systems are not connected (balanced), at least not at the proper spot.

STRATEGY

Check that the neutral service conductor is attached to a terminal bar in the service box. There will usually be a white wire, also attached to the terminal bar, which carries on to the distribution panel. There should also be a ground wire attached to the terminal bar directly, connected with a jumper, or bonded by the design of the box to the neutral wire. The ground wire connects to grounding electrodes such as a supply plumbing pipe and a ground rod.

7.4.11 Fused Neutral

The neutral (white) wire in the service box should not run through a fuse. The black wire should have a fuse, and the red wire should have a fuse, but the neutral wire should not.

Dangerous

A fused neutral creates a hazardous situation because the fuse for the neutral may blow before the fuse for the red or the black. People will think that since the fuse is blown and that the appliances don't work, the power to the house is not energized. However, if the white fuse has blown, voltage will be present through the whole house and back through the white wire to where the circuit is interrupted at the fuse.

Modern residential panels are arranged so it's hard to fuse the neutral; however, three-phase panels can be used inappropriately to fuse the neutral on a single-phase service. Some obsolete service boxes were designed with fused neutrals.

STRATEGY

Make sure that only the black and the red wires are fused in the service box. Recommend replacing old boxes with fused neutrals, and rearranging fused neutrals in modern boxes.

7.4.12 Fuses Upstream of Disconnect Switch

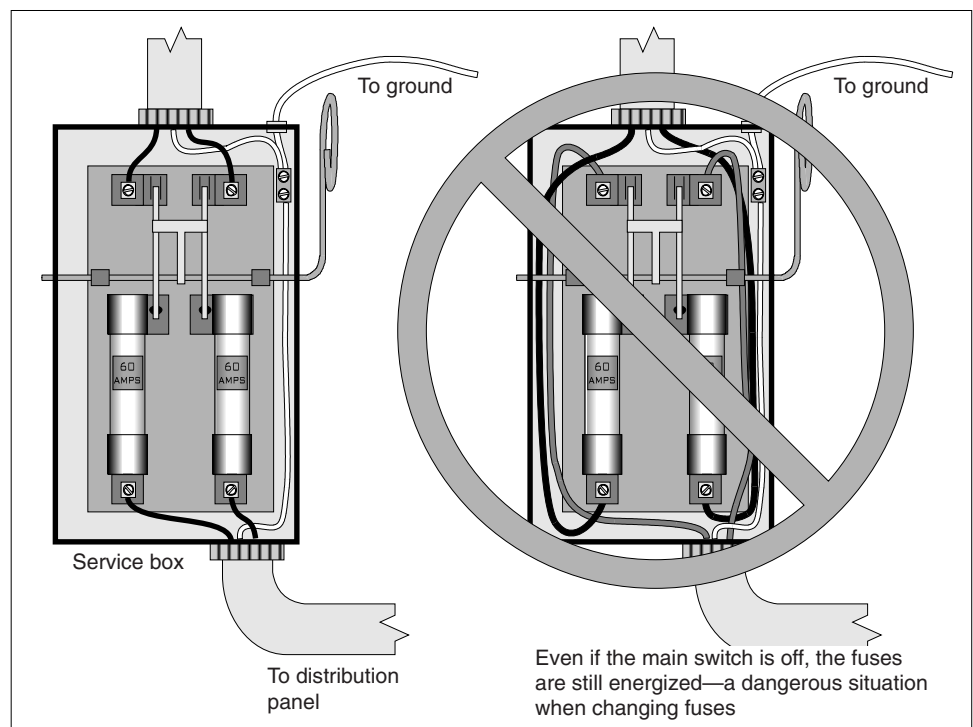
The proper arrangement for service box wiring is to have the black and the red service entrance wires attach to the main disconnect switch (Figure 7.16). The main fuses should be after (downstream) the switch. When the switch is thrown, the fuses should be de-energized.

It is possible to wire the service box so the electricity goes through the fuses first and then to the disconnect switch. This makes it more dangerous to change the fuses. This is incorrect and should be reported.

IMPLICATIONS

People may be electrocuted changing fuses, after shutting off power to the house, because they think the fuse holders are de-energized. If the switch is downstream of (after) the fuses, the fuses will always be energized, no matter what position the switch is in.

FIGURE 7.16 The Main Fuses Must Be Downstream of the Disconnect Switch



STRATEGY

Check that the service entrance wires are connected to the “line” side of the switch, rather than the “load” side.

7.4.13 Obsolete Service Box

Any service box with exposed electrical connectors is obsolete. This is typical of some very old ceramic systems. These were often located at ceiling level.

IMPLICATION

The implication is electric shock.

STRATEGY

Any switches not enclosed in a service box should be considered obsolete and replaced. You may be looking at a 30-amp service here.

7.4.14 Exterior Box Not Weather-Tight

Any service boxes mounted on the exterior of buildings should be weather-tight, unless they are mounted inside a weather-tight enclosure.

IMPLICATIONS

Moisture in the box can lead to electrical shock, overheating and fire.

STRATEGY

Where there is the possibility for, or evidence of moisture penetration, improvements should be recommended. Look for water, watermarks and rust in the box. Look for openings, especially on the top, where water could get in.

7.4.15 Box Not Rated for Aluminum

STRATEGY

Where aluminum wire is used in a service box, the box and its components, including breakers, should be rated for use with aluminum (CUAL). The rating may be spelled out on or inside the box.

Where you cannot verify that the box is rated for aluminum, recommend further investigation.

7.5 SYSTEM GROUNDING

Ground Wire Usually Idle

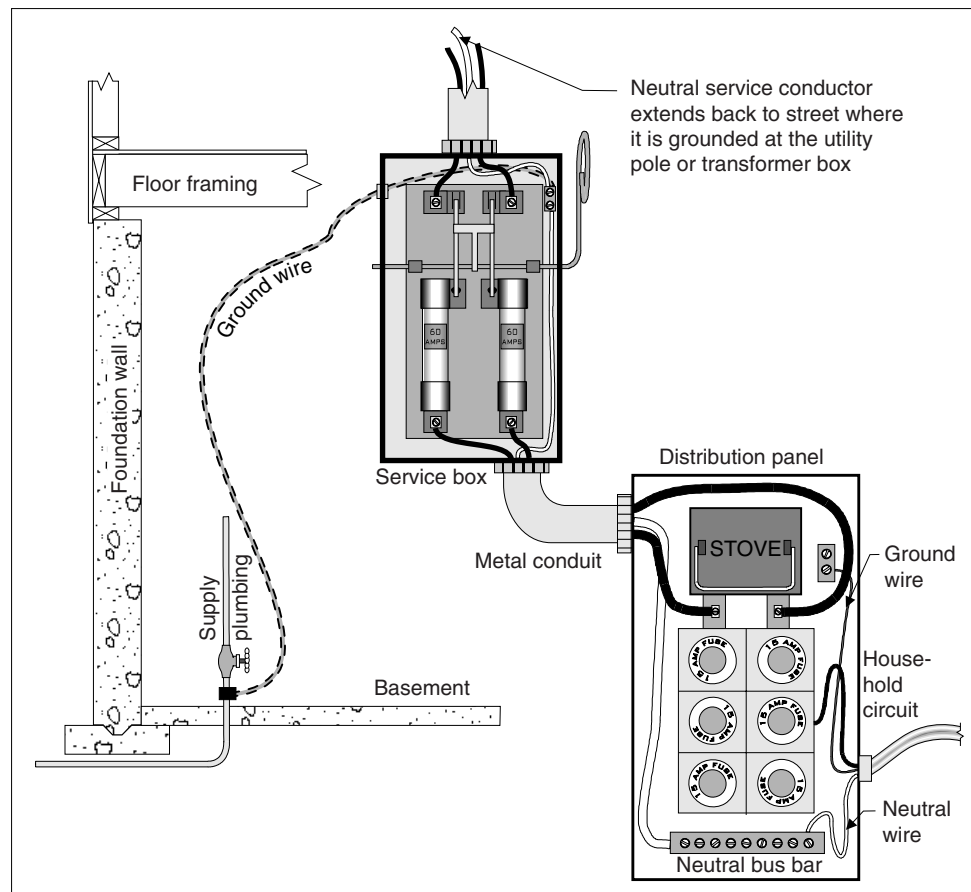
Electricity is confusing enough when we deal with wires that are supposed to carry the current through the house. It gets even more confusing when we talk about wires that normally don't do anything at all. Most people have some difficulty with the concept of **grounding** and its cousin, **bonding**. We'll try to keep it simple and take the mystery out of it.

Equipment Grounding System

There are two types of grounding in homes, with different functions. The **equipment grounding system** is the network of bare wires that runs through the home as part of the branch circuit wiring. This grounding system is connected to the white or neutral service wire coming in from the street. This equipment grounding system protects people from electric shock when electricity gets somewhere it shouldn't and allows fuses to blow and breakers to trip. Think of this as an emergency path for electricity.

Current Flows out to Street

The equipment grounding system connects to the neutral wire coming into the house at the service box. Current in the ground wires in the distribution system flows out the street transformer through the neutral service entrance wire.

FIGURE 7.17 Electrical Path for Ground and Netural Wires

Earth Grounding System

The **earth grounding system** uses a wire to connect the service box to the earth with water pipes, grounding rods, etc. This earth grounding system is a path for lightning or static electricity. It is not intended to carry the emergency current from the equipment grounding system to ground. The only time this earth grounding system would carry electricity from the home would be if there were a fault in the home causing current to flow through the ground wires in the distribution system **and** the neutral service wire out to the street were broken.

7.5.1 Grounding vs. Grounded Neutral Wire

Grounding (Ground) versus Grounded

Let's make sure we understand the difference between the grounding (ground) wire and the grounded neutral wire. The white neutral (grounded) wire carries current on an everyday basis as part of a normal circuit. The electricity flows through the neutral wires and is collected at the neutral bus bar in the distribution panel (Figure 7.17). It then flows to the neutral connection in the service box and is joined to the neutral service entrance wire. This neutral service wire then goes out to the transformer at the street. This is the way electricity is supposed to flow.

Electricity is not supposed to flow through the bare ground wire. The ground wire is normally at rest, available for emergency use only. If something goes wrong, the ground wire will carry electricity safely to the neutral service wire at the service box.

*Grounding Electrode Conductors**Equipment Grounding Conductors*

Let's clarify the terminology we'll use going forward.

We'll refer to the **earth ground wire**, which is the bare wire that connects the service box to the grounding rods or water pipes. It is technically called the **grounding electrode conductor**.

The ground wires on each branch circuit throughout the house are called **equipment grounding conductors**. We'll just call them **ground wires**.

7.5.2 Bonding

Grounding Defined

Grounding and bonding are frequently confused. Let's start with a simple way to separate them. Grounding is the act of connecting something to the ground (earth), so it has zero electrical potential. Everything that is grounded is connected to ground and can have no electrical energy stored in it.

Bonding Defined

Bonding is simply the act of joining two electrical conductors together. They may be two wires, a wire and a pipe, or they may be two tin cans. Bonding ensures that these two things will be at the same electrical potential. That means you won't get electricity building up in one and not in the other. No current flows between two bonded bodies because they have the same potential. A wire between two electrical outlet boxes bonds them. If a hot (black) wire touches one box, both boxes will have 120 volts of electrical potential and either could give you a shock.

Bonding to Ground

Bonding, itself, doesn't protect anything. However, if one of those boxes is grounded, there can be no electrical energy buildup. If the grounded box is bonded to the other box, the other box is also at zero electrical potential.

The Goal Is Ground

Whether it's a grounding or bonding connection, the goal in house wiring is to have no metal boxes or any other conductive materials near the wires become electrically hot. We want all these to be grounded so they have zero electrical potential and can't give us a shock.

Can't Ground to Gas Piping

The gas piping issue is confusing. You're not allowed to use gas piping to ground the electrical system. It's tempting because the gas piping in houses is usually metal. The gas piping usually goes outside underground and seems like a good way to connect to ground. However, much of the gas underground piping is plastic, which is not a good electrical conductor. This defeats the grounding function.

Bond the Gas Piping

We are not allowed to use the gas piping to ground an electrical system. However, gas piping must be bonded to the electrical grounding systems in most jurisdictions.

That's because we don't want the gas piping to develop an electrical charge. If it does, it could create an arc or spark to something nearby. If gas vapors are present, this could create an explosion.

Most municipalities require that the gas piping be bonded to the house electrical grounding system. Think of the gas piping getting rid of any stray electricity to the grounding system. Remember that we don't want electricity to build up anywhere except within the electrical system. Incidentally, bonding the gas piping usually bonds the furnace and ductwork too, because the piping touches the furnace. Bonding the gas piping is often accomplished with a wire attached to the cold (and sometimes hot) water supply pipe, near the water heater.

Arcing and Sparks

If we build up enough electrical potential in one ungrounded (hot) conductor and bring a second conductor close to it, we can make electricity jump through the air. The potential electrical energy is so great on the hot side that it sees a path to ground that isn't perfect but will bridge the air gap to get to it. This creates a spark. The electricity is said to be arcing between the conductors.

TABLE 7.2 Minimum Sizes for Ground Wires

MINIMUM SIZES FOR GROUND WIRES			
Service Wire (Copper)	Service Wire (Aluminum)	Grounding Wire (Copper)	Grounding Wire (Aluminum)
2 or smaller	1/0 or smaller	#8	#6
1 or 1/0	2/0 or 3/0	#6	#4
2/0 or 3/0	4/0 or 250 MCM	#4	#2

Note: All wire sizes are AWG except 250 MCM.

7.5.3 Where Does the Grounding System Go?

We've talked about protecting people in houses by attaching grounding wires to metal boxes, for example, that are close to electrical wires. The fuse panel, metal outlets, flip switches and screw-in light bulbs should all be grounded.

In the first half of the twentieth century, we didn't extend the grounding protection throughout the house distribution system. Since the early to mid-1960s, all new wiring extends the ground protection throughout the distribution network. All branch circuits now include a ground wire.

The ground wires are collected at the distribution panel and forwarded to the service box. At the service box, the house ground wires are connected to the service box equipment and enclosure. The whole assembly is connected to the service entrance neutral and to a grounding wire that goes to grounding electrodes (e.g., rod or plumbing pipe). In some cases, the system ground may be connected to the grounding electrode conductors at the meter, rather than at the service box.

This gives the electricity, in theory, two choices to flow to ground. Electricity that comes back to this point can either flow to ground through the neutral service wire, or through the ground wire to the grounding electrode.

The earth ground wires running from the service box to the grounding electrode are usually copper and are often bare (no insulation). Aluminum grounding wires are permitted if they're kept 18 inches away from the earth and away from moisture or other corrosive environments. Aluminum wires are not as good conductors as copper, so they have to be one size larger than copper wires.

7.5.4 Grounding Wire Size

Table 7.2 summarizes common copper wire sizes for grounding systems. **Note:** 6-gauge copper wire is the largest required if the connection is to a ground rod.

A home inspection does not include a check of the quality of the ground system. We're simply looking at the grounding wire, their connections, and the electrodes, where visible, for corrosion, damaged wire and loose connections.

7.5.5 Where Does the Grounding System End?

The goal is to get the electricity to flow to ground. There are several ways to do this, including—

1. through metal water supply pipes
2. through metal rods driven into the ground

Ground Wires through the House

Grounds Collected at Service Box

Two Paths to Ground

Ground Wire from Box to Pipe or Rod

We Don't Check Quality of Ground

3. through wires (often 1/2-inch reinforcing bar) buried in the footings of buildings (UFER ground)
4. buried grounding plates or rings
5. the frames of metal buildings (more common in commercial than residential construction)
6. the metal casings of private water supply wells

A New Issue

In some areas, metal water pipes running to the street water mains are being replaced with plastic. This defeats the grounding system, and is impossible to detect during a home inspection.

Rules Change

Various jurisdictions have different requirements, and like all code issues, the requirements have changed over time. For example, after 1987, grounding to a metal supply water pipe alone is not satisfactory. There must also be a ground rod, because plastic water piping is becoming so popular.

7.5.6 Conditions

IMPLICATIONS

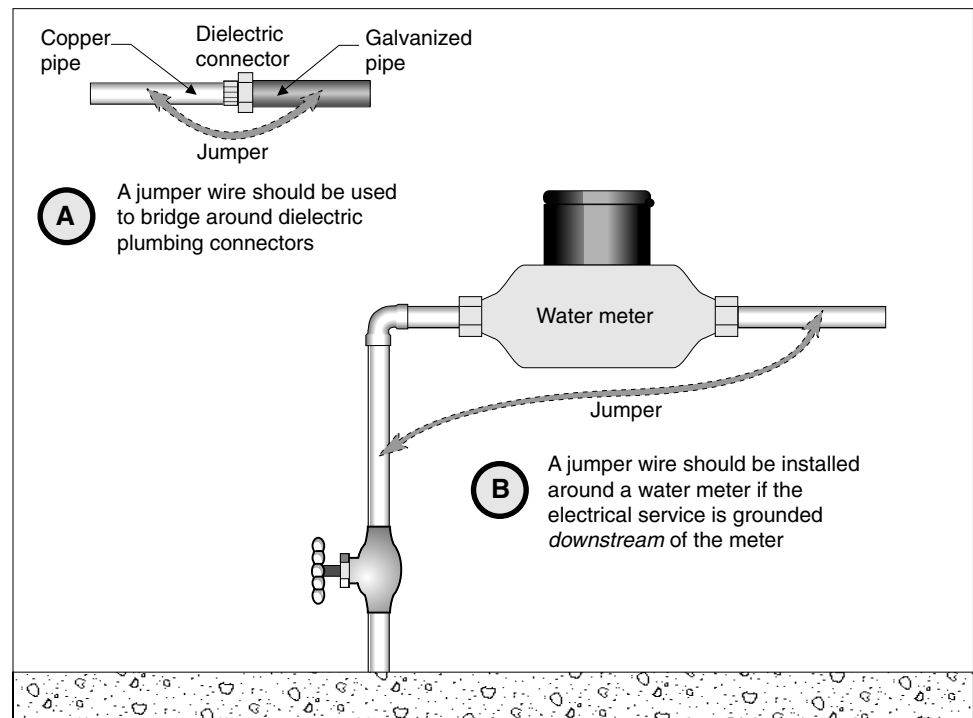
Let's look at what goes wrong with grounding systems. The implications are the same throughout. An ineffective or missing grounding system cannot perform its safety functions. There is greater risk of electrical shock, equipment damage, and fire in houses where the grounding system is not effective.

1. No grounding
2. Ground wire attached to plastic pipe
3. Ground wire after (downstream of) meters and valves, with no jumper
4. Spliced ground wire
5. Poor connections
6. Ground connections not accessible
7. Ground rod cut off
8. Corroded ground wire
9. Undersized ground wire
10. Neutral bonded to ground wire after (downstream of) service box
11. Neutral wires not bonded to ground wire at service box
12. Service box not bonded to ground wire
13. No ground in subpanel feeder wires

STRATEGY

No Grounding

In some cases, you won't find a ground wire from the service box to piping, to a ground rod or to anything else. If it is missing, this is a serious electrical defect and immediate improvement is called for. Clients, real estate agents and existing homeowners may have a tough time understanding how this is important since the house has been operating this way for some time. Explain that this is an emergency safety device and houses can operate for years without it, but the one time they need it (and it will happen) they won't have the protection they should. Is it worth risking someone's life?

FIGURE 7.18 Jumper Wires Needed**STRATEGY****Ground Wire Attached to Plastic Pipe**

Plastic piping does not provide a continuous path to ground. Plastic piping is an insulator rather than a conductor. This is the same as having no grounding and immediate improvement should be recommended.

Ground Wire after (Downstream of) Meters and Valves (No Jumper)

Meters, valves and dielectric connectors can interrupt electrical continuity.

Make sure the ground wire is connected as close as possible to the plumbing service entry and upstream of (before) any devices that might interrupt it. Where there are devices such as meters upstream, a jumper should be added around the devices, or the ground wire should be moved upstream of these (Figure 7.18). The jumper should be the same size as the ground wire.

In most areas the metallic natural gas piping must be bonded to the grounding system. This applies to new installations, or renovations involving gas piping or electrical service work. It is common to find this bonding wire connected to the plumbing system at the water heater, where the gas and water pipes are close together. In this case, a jumper should be provided around the water meter to maintain continuity of ground for the gas piping (although the electrical system may be properly grounded without this jumper).

Spliced Ground Wire

Ground wires should not be spliced since every splice is a potential poor connection.

Bonding Gas Piping

Follow the grounding wire where it's visible (where it's not entirely visible, inform your client that you couldn't see it all). Know what is accepted practice in your area. Write up splices if they are not allowed, or if they are poor quality.

STRATEGY

Loose or Poor Connections

The ground wire is typically connected to the service box at one end and to a pipe or ground rod at the other. Make sure the wire is securely fastened at both ends. Where ground wires are attached to pipes, it's not adequate to just wrap the wire around the pipe. The wire must be secured with a clamp approved for that purpose. The wire may be loose in the clamp or the clamp may be loose on the pipe.

Corrosion of the wire, clamp or pipe could result in poor conditions. Where you see this, recommend further investigation.

Ground Connections Not Accessible

Most codes require that the connection of the grounding wire to the piping or ground rods be accessible.

STRATEGY

Where you can't see how the grounding wire is connected, describe that limitation to your client. Advise your client to locate and inspect the quality of these connections, or have an electrician do it.

Ground Rod Cut Off

The ground rod should be a full 8 feet long.

STRATEGY

Look at the top of the ground rod. You should see the data stamped on it. If not, you should describe the ground rod as being suspect. Electricians can check the quality of the ground and it may be okay even if the rod has been cut. Utilities will sometimes allow people to do unconventional things with ground rods. Qualify your criticism, since the installation may be approved by the utility.

Corroded Ground Wire

Follow the wire through its entire length, if possible. Write up corroded wire or connections.

STRATEGY

Undersized Ground Wire

We included a chart earlier describing the minimum sizes for ground wires.

STRATEGY

Check the size of the ground wire against the size of the service wires. Since the wires are often bare, you may not be able to verify the size by reading the sheathing. Compare it to known samples or use a wire measurement device.

Neutral Bonded to Ground Wire after (Downstream of) Service Box

We'll talk about this more in the distribution system, but it's noted here as well, since it's a common defect.

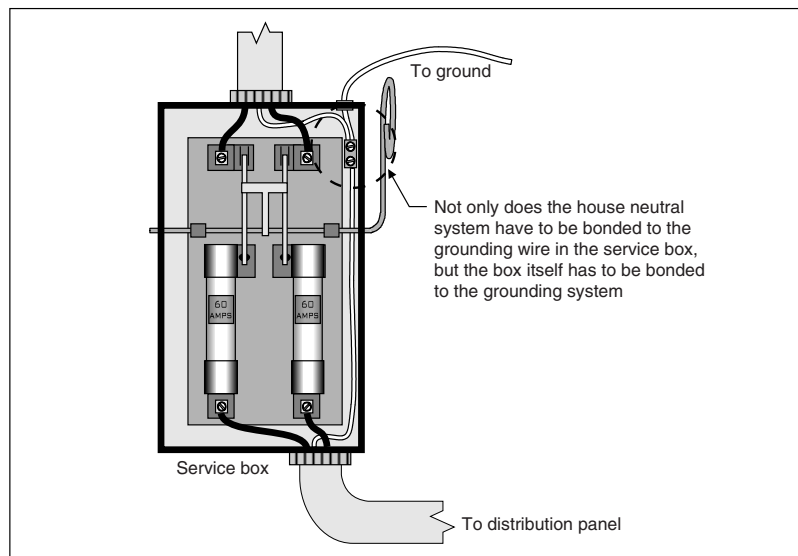
The only place that the neutral wire and the ground wire should be bonded together is at the service box.

The neutral wire carries electricity every day. The equipment ground wires should not. If the neutral and ground are bonded together out in the house distribution system (for example, at an auxiliary panel), the equipment ground wires may carry some everyday electricity. This is not safe.

Note: In combination panels (service box and distribution panel in one cabinet) this bonding is allowed in the distribution panel.

STRATEGY

This will be discovered most often at auxiliary panels. Check that the neutral and ground wires are separated.

FIGURE 7.19 Bond Service Box to Ground

Neutral Wires Not Bonded to Ground Wire at Service Box

The only place that the house neutral should be bonded to the ground is at the service box, but it must be done here. Failure to bond the neutral to the ground at the service box removes some of the emergency safety routes and protection that are available.

Make sure the equipment grounding wires from the house are grouped (usually in the distribution panel) and bonded to the house neutral at the service box. The neutral and ground terminals are often bonded by the design of the box. Some have a removable bonding jumper or screw. Make sure it's in place. If you aren't sure whether one is needed, note the make and model, and call the manufacturer.

STRATEGY

Service Box Not Bonded to Grounding Wire

Not only does the house neutral system have to be bonded to the grounding wire in the service box, but also the box itself has to be bonded to the grounding system (Figure 7.19). This is accomplished either with a bonding screw in the neutral terminal block, or with a jumper wire or screw to the box itself.

Check to make sure the service box itself is bonded to the point where the grounding system and neutral wires come together in the service box.

We have been talking about the service box as though it is separate from the distribution panel. This is true on older systems, but in many modern systems, a combination panel (commonly called a service panel) is used. The service box and distribution panel are in one cabinet, often with two separate covers.

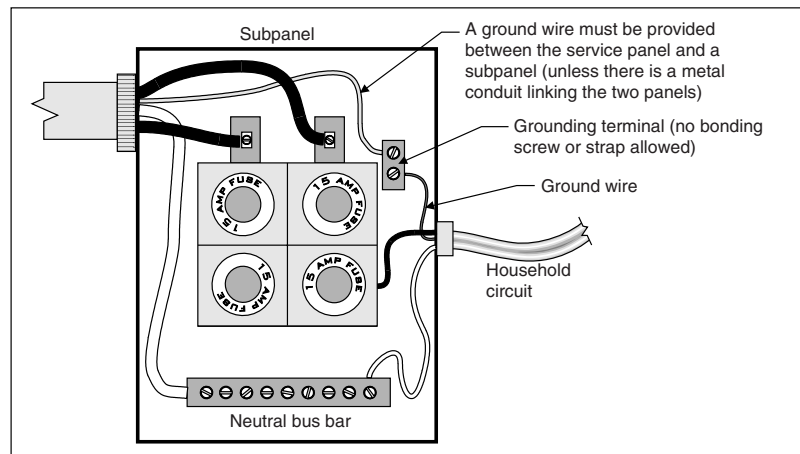
Where there are multiple main disconnects, they should be bonded together, and one should be connected to both the service entrance neutral and grounding electrode wire.

No Ground in Subpanel Feeder Wires

A ground wire in the cable from the service panel to a subpanel is usually needed so the subpanel and all of the branch circuits fed by the subpanel can be adequately grounded (Figure 7.20).

Combination Panel Includes Service Box and Distribution Panel

STRATEGY

FIGURE 7.20 Need Ground in Subpanel Feeder Wires**STRATEGY**

Check at the subpanel and at the main distribution panel for the presence of a connected ground wire in the subpanel feeder wire.

There may be an exception when the subpanel is adjacent to and bonded to the main distribution panel, most commonly by the metal conduit protecting the feed wires. Where the subpanel is bonded to the main panel by the conduit, a ground wire may not be needed.

7.6 DISTRIBUTION PANELS

It's the Fuse Box

These may also be called **service panels, panelboards, auxiliary panels, sub-panels, fuse boxes, fuse panels, or breaker panels**. Most people call it the **fuse box**, or the breaker box.

Tree Analogy

So far, we've been talking about bringing electricity into the house. Now we're going to talk about how to spread it around the house. You can think of the service drop and the service entrance conductors as the trunk of a tree. The distribution wires throughout the house are the branches of the tree. The distribution panel is where the branches attach to the trunk.

7.6.1 What Comes in and What Goes out

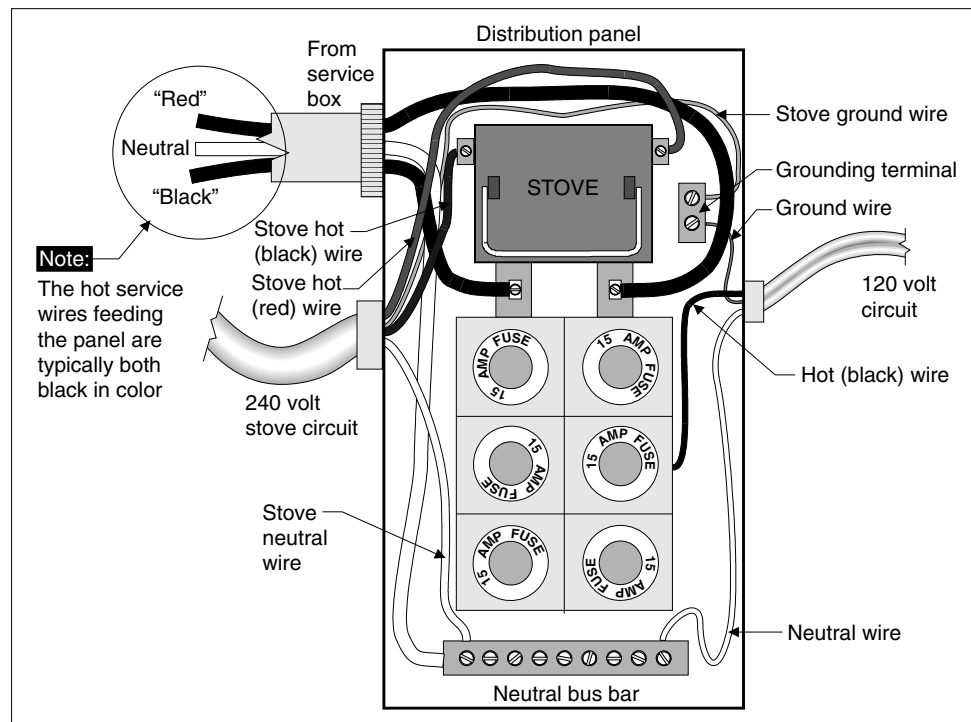
Wires from Service Box to Panel

Coming into the distribution panel from the service box is a black wire with 120-volt potential, a red wire with 120-volt potential, a neutral wire with no potential, and usually a ground wire (Figure 7.21).

Wires from Panel to Branch Circuits

What goes out of the distribution panel into the house is many black, red, and white wires. Bare ground wires also leave the panel with each circuit (installed after 1960, roughly). The black and the red branch circuit wires carry the electrical energy. The white wires complete the path to ground for each circuit, when the circuit is working. If the distribution system is grounded, the ground wire runs along each circuit ready to pick up any electricity that gets out of hand.

The black and the red wires are connected to fuse or breaker terminals. The white wires are connected to the neutral terminal or bus bar. The ground wires are connected to the panel enclosure (bonded to the panel).

FIGURE 7.21 Typical Arrangement of Panel Wires

7.6.2 240-Volt Circuits

240-Volt Circuits

Branch circuits can be one of two types. We have heavy-duty 240-volt circuits and we have smaller 120-volt circuits. The 240-volt circuits are used for large appliances, such as electric stoves, electric water heaters, electric clothes dryers, air conditioners and saunas. These large appliances use a red and a black wire from the panel. The potential from the black wire to the red wire is 240 volts.

Big Watts with Small Amps

240-volt circuits allow us to deliver adequate power to the larger appliances and keep the current flow (amps) and wire size relatively small.

Need Two Fuses or Breakers

One of the disadvantages of 240-volt circuits is that you need to have a fuse for the black wire and a fuse for the red wire. These 240-volt circuits have two fuses or breakers. It's important to tie these fuses or breakers together. Otherwise, people would pull out one fuse or shut off one breaker and think the power to the appliance was off. However, when they went to work on it, power would still be delivered through one wire. This can be fatal.

Tie Fuses or Breakers Together

To shut off power to a 240-volt circuit, you have to shut off two circuit breakers or pull out two fuses (Figure 7.22). The pullout blocks on stoves and clothes dryers, for example, force you to do just this. Before you can remove one fuse, you have to pull out the fuse holder, which has both fuses in it.

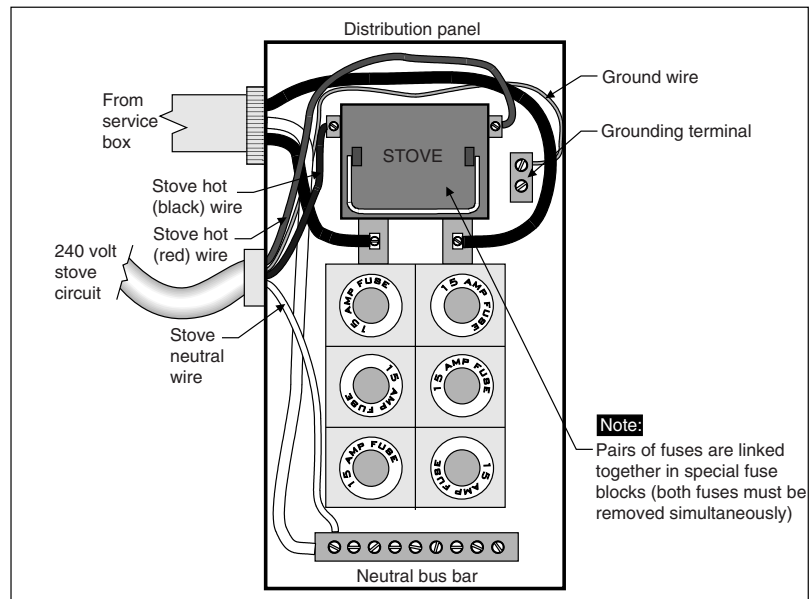
Linked Breakers

Circuit breakers use either a mechanical tie bar (link) that joins the two breaker handles together, or a fancy breaker that has a single handle but two separate poles.

Balanced 240-Volt Circuits

Most 240-volt circuits have a red wire, black wire and white wire. However, some have only a red wire and a black wire. How can a circuit work with no white wire?

We talked about 240-volt circuits much earlier, and said that when the red wire pulsed, the black wire was at rest, and vice versa. The red and black wires can act

FIGURE 7.22 Pull-Out Fuse Blocks for 240-Volt Circuits

as white wires, but only if the load is exactly the same on the red and black wires, as for a baseboard heater or water heater. Any appliance that has a 120-volt component will need a white wire to carry the unbalanced part of the load. Stoves have 120-volt lights, clocks and timers. Clothes dryers have 120-volt lights, timer and motors. Air conditioners may have 120-volt motors and crank case heaters. All appliances with both 240-volt and 120-volt components need a black, red, **and** white wire.

7.6.3 120-Volt Circuits

As Many As You Want

Circuit May Have 12 Outlets

You Won't Know How Many Things Are on Each Circuit

Typically 10 to 20 Circuits

Some 120-volt circuits use a black wire and a white wire (and a ground wire for safety purposes in newer work). Other 120-volt circuits use a red wire and a white wire. There is no limit to how many circuits you can have in a house. If you wanted to, every outlet could be on its own circuit with its own fuse or circuit breaker, and every light could have its own circuit. This would be expensive and wouldn't give you many advantages. Modern 120-volt circuits accommodate up to about twelve outlets and/or light fixtures on one circuit.

Home inspectors rarely know how many outlets or switches (called electrical **points**) are served by each circuit. It might take two people a couple of hours to figure it out. This obviously isn't practical during a home inspection, although this can be recorded as homeowners label their panels, writing out what each fuse or breaker controls.

We find anywhere from four to 40 general-purpose circuits in the house. These 120-volt circuits serve such things as electrical outlets (receptacles), lights, dishwashers, furnaces, garbage disposals, well pumps, garage door operators, and so on.

7.6.4 Wire Sizes and Fuse Sizes

Match Wire Size to Fuse Size

The wire size is determined by how many amps we need to run through the wire. Recall from Table 7.1 that every wire has a maximum number of amps it can carry safely. Therefore, every wire has a correct fuse (or breaker) size to go along with it. Most people remember some of these, and other ones they have to look up. The common numbers for copper wire are—

14 gauge wire needs **15-amp** fuses or breakers

12 gauge wire needs **20-amp** fuses or breakers

10 gauge wire needs **30-amp** fuses or breakers

8 gauge wire needs **40-amp** fuses or breakers

6 gauge wire needs **60-amp** fuses or breakers

There are exceptions, but most of the time the numbers above apply.

Most circuits for receptacles and lights use 14 gauge wire. Circuits for kitchens, pantries, bathrooms, and utility areas, including receptacles for washing machines and dishwashers, use 12 gauge wire. As a general rule, the 120-volt circuits use 14 or 12 gauge wire.

Wire Size for 240-Volt Circuits

The 240-volt circuits might use 14 or 12 gauge wire but they often use large wire. For example, water heaters, central air conditioners and electric clothes dryers might use 10 gauge wire. Electric stoves and ovens usually use 8 gauge wire. Electric heaters rated at 240 volts may use 12 or even 14 gauge wire. Our job at the distribution panel is to make sure that each fuse or breaker is the right size to protect each wire.

What If the Fuse Is Too Small?

If the fuse or breaker is too small, it will shut down the electrical system before it has to. A fuse that is too small will be a nuisance because it's likely to blow frequently, but it isn't a safety hazard.

What If the Fuse Is Too Big?

If the fuse or breaker is too big, it won't shut off the electricity soon enough. This may allow the wire to overheat and create a dangerous situation. Finding too large a fuse or breaker (often called **overfusing**) is probably the most common electrical problem you will come across.

7.6.5 Panel Descriptions

Combination Panel

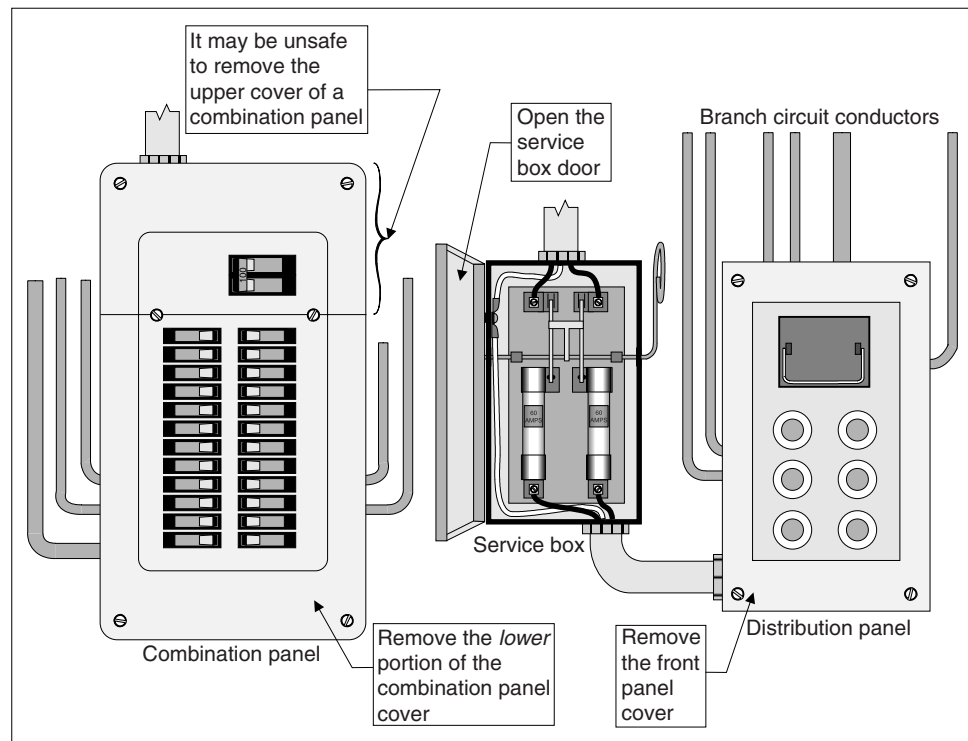
Distribution panels are typically steel boxes mounted on a wall, and with a door on the front of the box (Figure 7.23). They are designed to contain an electrical fire inside the panel. The panel can be by itself, or it can be attached to the service box. This **combination panel** or **service panel**, as it's known, has the service entrance wires coming into it. In the service box area, there will be a main disconnect and the main fuses or breakers for the whole house. The wires come out of the service box section into the distribution panel section. Electricity flows out through the various fuses or breakers to all the branch circuits.

Some Panels Have Doors, All Have Covers

Panels may have doors; some do not. All panels have covers that can be removed to inspect or work on the system. As home inspectors, we remove these covers. Homeowners do not need to remove covers to replace fuses or reset breakers.

Remove Panel Covers

The Standards ask us to inspect the interior of panels, so we have to remove the covers so we can match up the size of the wire and the size of the fuses or breakers. Opening the door in the front of the panel is not enough; you have to take the whole cover off. We'll talk about that a little more in Section 7.6.9.

FIGURE 7.23 Inspecting the Service Box and Panels

7.6.6 Location

Panels Can Be Inside or Outside

At Eye Level, Clear in Front for Access

Bad Locations

Distribution panels can be located outside, in garages, or inside the house, often in the basement. Indoor panels are usually located on an exterior wall, adjacent to where the service entrance conductors run down the outside of the house.

Distribution panels are supposed to be installed at about eye level so they are convenient. The area 3 feet in front and 30 inches wide should be kept clear.

There are certain places where you shouldn't find distribution panels, such as close to water, or where access is difficult when the power is out. Some codes prohibit distribution panels in bathrooms, clothes closets, stairwells or kitchen cabinets. These rules vary, depending on what jurisdiction you are in, and when the system was installed. Some areas prohibit panels within 3 feet of indoor gas meters and outdoor gas relief vents.

7.6.7 Subpanels or Auxiliary Panels

Can Be Anywhere

How Do I Know If There Is a Subpanel?

These panels can be located anywhere through the house and can be difficult to find. Sometimes they're beside the main panel, but they can be on the top story of the house, for example, and while they're not supposed to be tucked away in closets, bathrooms, kitchen cabinets, or the like, they often are.

Sometimes you won't know that there's a subpanel, but there are some clues. If the service panel has a 240-volt circuit with no apparent function, it could be going to a subpanel. Sometimes it's even labeled **Subpanel** or **Auxiliary Panel**. If the only major appliances you can find are a stove and electric dryer, and you have three large 240-volt circuits, you should suspect another panel somewhere.

7.6.8 Distribution Panel Size

The panel has to be big enough to handle all the electricity that it might see. For example, if the house has a 100-amp service coming into it, the service panel has to be rated for at least 100 amps. A house with a 200-amp service would need a panel rated for at least 200 amps. That's because the service entrance wires and main fuses or breaker will allow 200 amps, for example, to go through to the distribution panel.

The distribution panel rating can be larger, but not smaller than the service size because it is very likely to see that much current. The panel rating is usually on the inside of the panel cover, although it can be in a number of places.

7.6.9 General Strategy for Inspecting Panels

Be Careful

Although home inspectors are required by the Standards to remove electrical cover plates, we encourage you not to do it unless you have good understanding of electrical systems and are familiar with electrical panels.

Dangerous

Checking the distribution panels is one of the most dangerous parts of home inspection. We'd like to turn off the power in the house before removing the panel cover, but most times that's not practical because the home is occupied. Test the panel with a voltage detector before opening it.

Fuse Pullouts Have to Be Removed to Get Cover Off

A tricky configuration is one where fuse pullouts have to be removed before the panel cover can be pulled off. If you don't notice that until you've got the panel cover loosened, you'll find yourself in an awkward spot, holding onto a cover with no screws in it, but unable to lift it out over the fuse pullouts.

If you're familiar with this kind of panel, you will be tempted to remove the fuse pullouts so you can take the cover off. This has its own set of problems. We normally do not shut off the power in a house for fear of upsetting computers, timers, or life support systems. However, if you don't remove the fuse pullouts, you can't complete the inspection to the Standards. What do you do?

Shutting Off Power

If the panel circuits are labeled, you may know what circuits the pullouts control. For example, if it's a clothes dryer or a water heater, you will probably be fine removing the pullouts. You can check to see whether the clothes dryer is in use. If not, shutting off power to it for a minute or two is seldom a problem. Similarly, disconnecting a water heater for a couple of minutes is not a problem. The catch is, you can't always believe the labels on the panel. You should distrust the panel labeling. Many panels were labeled correctly originally, but subsequent changes are often made without relabeling. Be very careful with this. If the pullout is for a stove, and the timer is set to turn the oven on at 3:00 PM for three hours, you will ruin dinner if you pull the fuses!

Ask the Occupant

One solution is to ask the homeowner (if he or she is present) whether these fuses can be removed so you can perform your inspection.

Report Your Limitations

If you don't get permission, we recommend that you do not pull them out on your own. Write up the limitations to your inspection. Some home inspectors offer to come back and complete the inspection when people know it will be safe to turn off the power. This is a business decision you'll have to make.

Accidentally Tripping Breakers

There are some panels where it's difficult to remove the cover without accidentally tripping breakers. You might trip a breaker pulling the cover off or putting it back on. If you accidentally trip a breaker, you should let the homeowner know, either in person or by leaving them a note.

Swing-Out Panel Covers

Some panel covers are designed with one side that fits behind a couple of lugs, and the other side that rotates into place and is clipped there. Be careful with these

covers. Any cover that swings into place can easily pinch a wire. This is more likely to happen, of course, if the box is full or overcrowded with wires. Use considerable care here to avoid problems.

If we've made you hesitant about removing distribution panel covers, that's probably a good thing. It's great to do a good job, but electrocuting yourself is a waste of all those business cards you just had printed.

Inspector in the House: Dinner Will Be Late!

Removing the cover on the electrical panel is usually straightforward, but sometimes it can be a little tricky. With some panels it is easy to trip a breaker inadvertently when removing or replacing the cover. Our most memorable instance was the time we tripped the breaker for the range, and did not realize that dinner was in the oven, with the timer set to start the oven at 4:00 PM. Needless to say, the owners were not happy that we messed up their dinner plans.

7.7 CONDITIONS FOUND IN PANELS

IMPLICATIONS

Let's look at some of the problems we typically find in panels. The implications of most of these are electrical shock and fire, so we won't discuss the implications for each one unless there's something unique.

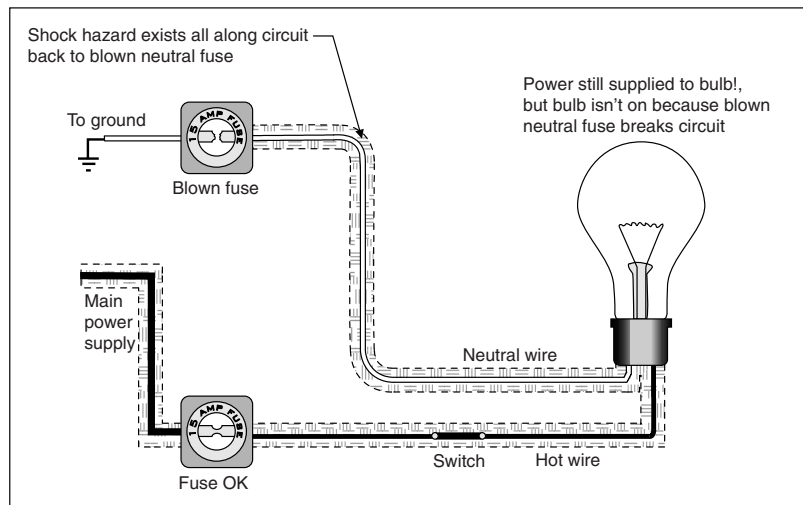
Because this is confusing enough already, we'll break the problems into six categories so you'll have a better chance of keeping them straight:

- Problems found in all panels
- Problems unique to subpanels
- Problems with fuses or breakers
- Problems with fuses
- Problems with breakers
- Problems with panel wires

7.7.1 Conditions Found in All Panels

Conditions found in all panels include—

1. obsolete and/or fused neutrals
2. damaged panel or components
3. loose or missing door
4. openings in panel
5. panel too small
6. overheating
7. rust or water in panel
8. circuits not labeled
9. panel crowded

FIGURE 7.24 Fused Neutrals in Old Wiring Systems

10. poor access
11. installed upside down
12. not suitable for aluminum wiring
13. poor location
14. outdoor panel not weather-tight

Obsolete and/or Fused Neutrals

Exposed Connections

Very old panels do not have covers or doors. Exposed wires can be touched readily. These panels, which are often white ceramic mounted on wooden boards on walls or ceilings, should be replaced.

Fused Neutrals

Another problem with old panels is **fused neutrals** (Figure 7.24). In early electrical work, people thought that if a fuse on the black wire was good, putting a second fuse in the circuit on the white wire would be even better.

Bad Idea

This was later recognized to be a bad idea and is no longer permitted. Why? If there's a 15-amp fuse on the black wire and a 15-amp fuse on the white wire, there are two fuses in the circuit. If the circuit is overloaded, one fuse will usually blow before the other. If the fuse on the white wire blows, the circuit will stop working, and the fuse on the black wire won't blow.

Dangerous

The receptacles won't be operable and the lights will go out. A homeowner would logically think that the power to that circuit was off. However, since the fuse on the black wire hasn't blown, there is actually power all the way through the circuit back to the blown fuse on the white wire in the panel. Someone working on that circuit will get a shock.

STRATEGY

Where you see old panels with exposed connections and/or fused neutrals, recommend replacement of the panels.

Damaged Panel or Components

If there is damage to the walls, door or cover of the panel, the damaged parts should be replaced. Similarly, fuse holders, breaker holders, bus bars, lugs, or terminal screws that are damaged or missing should be repaired or replaced.

STRATEGY

Look for damaged cabinet parts or components. Pay particular attention to older Bakelite (brand name for hard, dark brown, plastic-like material) fuse or breaker holders. These are often chipped or broken.

Loose or Missing Door

Some panels are designed with doors. Some are not. This section applies to panels intended to have doors.

STRATEGY

Remove the entire front cover (sometimes called the dead-front cover) from the panel. You should also operate the cover door, which homeowners will normally open and close. Make sure the door is present, secure, and operates properly.

Wood Screws Are Wrong

It's common to find that the appropriate machine screws have been lost and wood screws are inappropriately used to secure covers and cover doors. Wood screws have different threads, have pointed ends, are often too long, and can pierce wires. We recommend replacement of wood screws holding panels or doors in place.

Openings in Panel

When the dead-front cover is on, you shouldn't be able to reach into the panel. Knockouts (the metal plates designed to be pried out of the panel sides to allow wires to pass through) should be in place unless there's a wire passing through them. If there is a wire going through a panel, it should be clamped and protected from mechanical damage.

Fuse or breaker slots in the dead-front cover should not be open. Where a knockout or fuse and breaker slot has been removed and the hole is not used, the hole should be covered.

STRATEGY

Recommend that openings be covered as a priority improvement.

Panel Too Small

The panel rating must be at least the size of the service entrance so it can handle this current without overheating.

STRATEGY

Check the rating on the distribution panel and recommend replacement if it is smaller than the service size. If the panel rating is suitable, but there aren't enough circuits available, a subpanel can be added.

Overheating**STRATEGY**

Look for evidence of overheating, which includes—

1. warm panel
2. melted insulation, wires or terminals
3. charring or smoke discoloration
4. the smell of burned plastic

Wherever there is evidence of overheating, further investigation should be recommended, even if the panel is not currently warm.

Rust or Water in the Panel**STRATEGY**

Water in a panel is obviously dangerous since water is a good conductor of electricity. Rust on electrical components not only deteriorates the components but also can lead to poor connections. Where there is rust or water in the panel, repair, replacement or further investigation should be recommended.

STRATEGY**Circuits Not Labeled**

Many codes call for circuits to be labeled. In many cases, they are not.

Home inspectors should point out that circuit labeling is an important safety issue and recommend circuits be labeled. However, even if circuits are labeled, you won't know whether they are done correctly. Please do not rely on the labeling to be correct.

STRATEGY**Crowded**

Panels are designed with adequate room for the wires.

If it is hard to get the cover on the panel without pinching wires, it is probably crowded. A panel cleanup should be recommended. With little experience, you should be able to recognize crowded panels.

STRATEGY**Poor Access**

Panels should be readily accessible. Most codes call for a 30-inch wide, 3-foot deep area at the front to be kept open. The panel should be roughly at eye level and there should be no obstructions to opening the door fully.

Where panel access is restricted, you should recommend that the obstructions be removed or the panel be relocated.

STRATEGY**Installed Upside Down**

Check to see that the panel is installed in its correct orientation.

STRATEGY**Panel Not Suitable for Use with Aluminum Wire**

Where aluminum wiring is used in a panel, the panel should be rated for use with aluminum. This is normally on the panel data, often on the inside of the panel door. The panel often says something like, "suitable for use with CU or AL conductors," or "suitable for use with CU/AL rated circuit breakers."

Where you cannot verify that the panel is suitable for use with aluminum, recommend that this be checked. Don't say that the panel is not suitable for use with aluminum, because the labeling may have been removed or concealed. It's safer to write it up as a question.

STRATEGY

Use Common Sense

Poor Location

Watch for panels in bathrooms, clothes closets, stairwells, and kitchen cabinets. Watch also for panels too close to gas meters or gas relief vents.

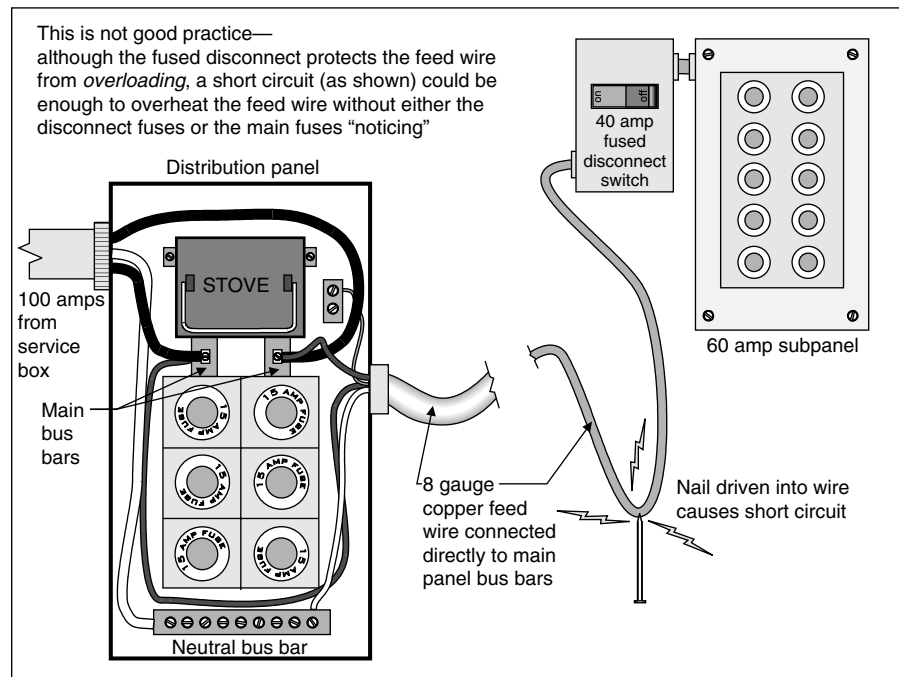
Use common sense to decide whether the panel location presents a hazard. If not, we don't recommend moving it. Codes change, and what used to be common practice is often not acceptable now. Codes will change in the future as well, but codes are not usually retroactive. Authorities use their discretion too, in deciding whether to ask homeowners to move equipment.

STRATEGY**Outdoor Panel Not Weather-Tight**

Look for outdoor panels to be protected against water entry. Most outdoor panels are flush-mounted, with covers that extend onto the wall surface. Recommend repair or replacement where you see rust or water in the panel.

7.7.2 Conditions Unique to Subpanels

All of the issues that apply to panels also apply to subpanels. The following problems are specific to subpanels:

FIGURE 7.25 Subpanel Wiring—Disconnect by Subpanel

1. subpanel and/or feeder wire not adequately protected
2. neutral and ground wires bonded at subpanel
3. subpanel not grounded

Subpanel and/or Feeder Wire Not Adequately Protected

Subpanels are usually smaller ampacity than the service size. For instance, a 100-amp service will have a 100-amp main distribution panel and may have a 40-amp subpanel. The 100-amp main panel is protected by the 100-amp fuses or breakers in the main disconnect. What protects the 40-amp subpanel?

Ideally, there are 40-amp fuses or breakers in the main distribution panel protecting the subpanel and the wire feeding it (which should be 8 gauge if copper, in this case).

When you see a subpanel—

1. check its rating
2. make sure the feeder wire to the subpanel is sized properly
3. look for fuses or breakers sized properly to protect both the feeder wire and the subpanel

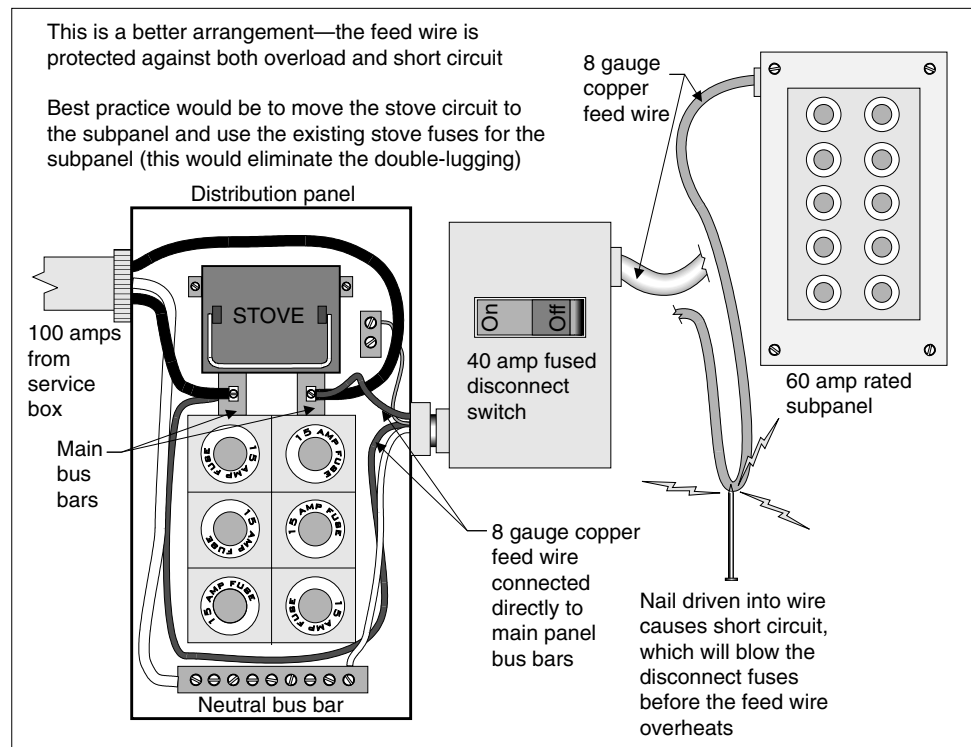
If there are no fuses or breakers to protect the panel and feeder wire in the main distribution panel, the only protection may be the 100-amp main fuses. This is not safe.

Sometimes there is a fused disconnect at the subpanel. In our example, someone might add a 40-amp fused disconnect adjacent to the subpanel (Figure 7.25). This will protect the 8 gauge wire in the event of an **overload**. If the subpanel draws more than 40 amps, these fuses will blow, protecting the 8 gauge wire. However, there is another issue.

Subpanels Need Protection

STRATEGY

Fused Disconnect at Subpanel

FIGURE 7.26 Subpanel Wiring—Disconnect by Main Panel

Fuses Are Best at Beginning of Circuit

Short Circuit Doesn't Go Through Fuses

Complex Rules

Call for Help If Unsure

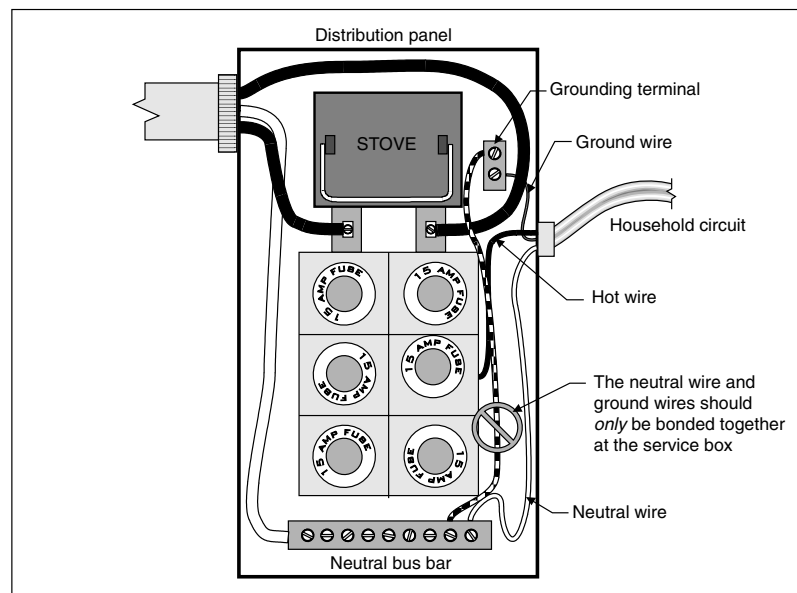
Putting the fuses or breakers anywhere in a circuit protects against an **overload** situation, but it does not protect against a **short circuit**. If the main panel is in the basement, and our subpanel with the 40-amp fused disconnect is on the third floor, what happens if someone drives a nail into the 8 gauge wire on the first floor? If the nail bridges the black and the white wires, we get a short circuit.

Let's assume 75 amps are flowing where we didn't expect it to be flowing. And it's not flowing through our 40-amp fuses on the third floor! Our 8 gauge wire will overheat and start a fire. The 40-amp fuses on the third floor can't help because they don't see the 75 amps and would never know anything is wrong. The 100-amp fuses in the main disconnect would not blow, because there is only 75 amps flowing.

It's best to put the fused disconnect at the beginning rather than at the end of a circuit (Figure 7.26). That way, the fuses or breakers will see the large current and will shut down the power. The 40-amp fuses should be in or right beside the main panel, not the subpanel.

Electrical authorities recognize this and prefer to see subpanel feeders fused at the main panel. Different codes make different exceptions, depending on how large the feeder is, how long the feeder is, and whether or not the feeder is mechanically protected (e.g., inside a rigid conduit). These rules become complicated, and are beyond the scope of this program. You may wish to discuss this issue with a utility inspector or electrician.

As a home inspector, you are probably justified in recommending further investigation on any system with a subpanel where the fuses or breakers are not at the main panel protecting it. Common sense works well.

FIGURE 7.27 Don't Bond Neutral and Ground Wires Downstream of Service Box

*Grounded Wires Should
Not Normally Be Energized*

STRATEGY

Neutral and Ground Wires Bonded at the Subpanel

This is not allowed (Figure 7.27). The neutral wire should carry the electricity back on an everyday basis. The equipment ground wire should be idle and should only be used in emergencies. If you join the neutral and the ground wires at the subpanel, the ground wire is likely to carry electricity. Its ability to work in an emergency is diminished. More importantly, since it is a bare wire and people don't expect it to be carrying electricity, there's an electrical shock hazard. Ground wires are connected to boxes and equipment cases. If the ground and neutral are joined, the boxes and equipment cases may be energized.

Check subpanels to make sure the ground and neutral are not bonded together. Some subpanels have bonding screws or jumpers that electrically connect the ground and the neutral bar. Make sure these are not connected. Where they are bonded, or you suspect that they are, recommend that an electrician correct this situation.

No Ground Wire

Sometimes the subpanel does not have an equipment ground wire in the feeder.

As discussed in Section 7.5 look for a ground wire, unless the main panel and subpanel are connected by metal conduit.

STRATEGY

7.7.3 Conditions Found with Fuses or Breakers

Common problems include—

1. Fuses or breakers too big
2. Double taps
3. Multi-wire circuits on one bus

TABLE 7.3 Common Wire Sizes for Larger Appliances

Appliance	Common Copper Wire Size (AWG)
Water Heater	#12, #10
Dryer	#10
Stove	#8
Central A/C	#12, #10, #8
Sauna	#10, #8

Fuses or Breakers Too Big

Overfusing

This is one of the most common electrical problems. You can only verify it by taking the cover off the electrical panel, and comparing the wire size to the size of the fuses or breakers. This is often referred to as **overfusing**, and is an electrical hazard.

STRATEGY

Check the sizes of fuses or breakers against the size of the wires. Where you can't verify they are properly matched, recommend that they be checked. Where fuse pullouts cannot be examined for fear of disconnecting power to an appliance, you should recommend a follow-up inspection of these fuses.

Checking Wire Size

You can usually determine the wire size by reading it off the sheathing. Where you can't, use a comparative technique, checking against known wire samples, either in the panel, or in samples you carry with you.

Color-Coded Fuses

Fuse sizes and breaker sizes can usually be read easily. Color coding on fuses and some breakers also helps. Blue is 15 amps. Orange is 20 amps. Red is 25 amps. Green is 30 amps.

Fuses Only Seem Oversized

Sometimes fuses for electric heaters seem oversized. They are often sized appropriately. For example, 14 gauge wire, which usually requires 15-amp fuses, is all right with 20-amp fuses in many cases with an electric heater. Similarly, 30-amp fuses on 12 gauge wire, which usually needs 20-amp fuses, may also be all right. This assumes that it is copper wire, and that the circuit is dedicated to the permanent electric heater.

Dedicated Circuits

Dedicated circuits for air conditioners often appear overfused, but are acceptable. The data plate on the air conditioner is what determines the fuse or breaker size, and the minimum circuit ampacity.

It's a good idea to know the common wire sizes used for the major appliances. This may allow you to recognize when the wire is inappropriate for the appliance. The table above lists common wire sizes for larger appliances.

Further Evaluation

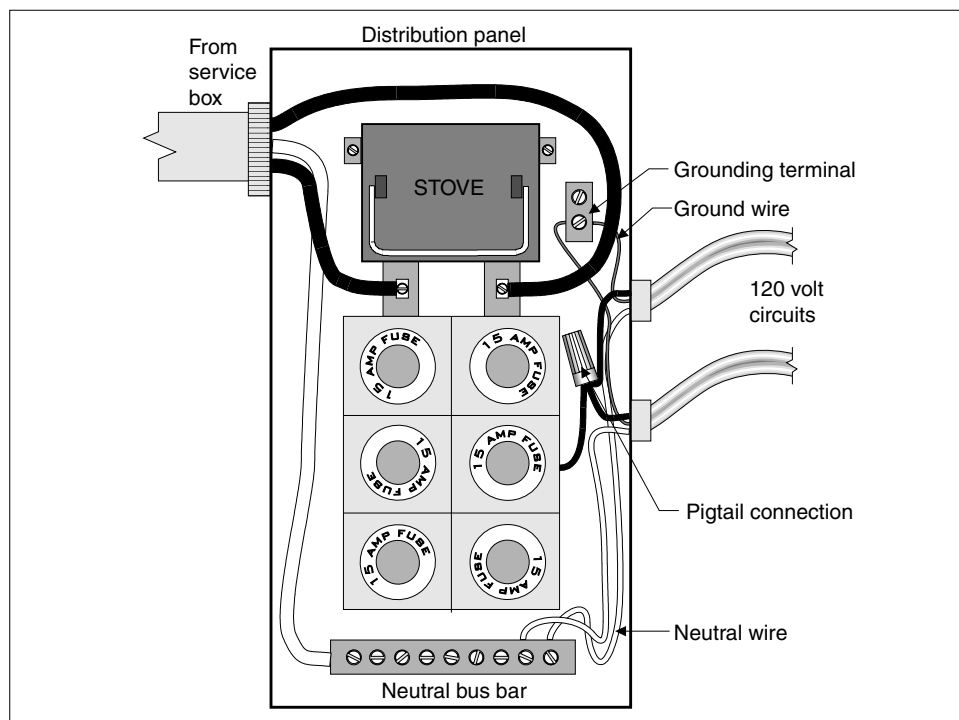
If you suspect the wire is too small for the load, recommend having it checked by an electrician. If the breaker is correct for the wire, it may trip frequently. In addition, an undersized wire can lead to excessive voltage drop, which will cause the appliance to run incorrectly.

When the Fuse Size Looks Right but Isn't

Sometimes you can be fooled when checking the size of the fuse against the size of the wire at the panel. If you have a 30-amp fuse and a 10 gauge wire, everything will be fine. What you may not know and won't likely discover during a home inspection is that the 10 gauge wire may be spliced with 14 gauge wire at a box somewhere along the circuit. The 14 gauge wire circuit is not adequately protected by the 30-amp fuse.

Watch for Mixed Wire Sizes

Electrical codes don't allow this practice for obvious reasons, but that doesn't mean it's never done. You'd have to be very sharp, and perhaps lucky, to pick this up during

FIGURE 7.28 Pigtailing to Avoid Double Taps

a home inspection. Some inspectors try to follow the entire length of the larger (240V) appliance circuits to check for splices or connection points.

Double Taps

Double tapping (double lugging) is the term for attaching two wires to a single fuse or breaker. Most terminals are not designed to accommodate two wires. The risk of a poor connection is greater with double taps.

Some electrical inspectors will allow double taps if wires are the same size and if the connection looks good. As inspectors, you should always note double taps. Explain that some authorities will accept this situation, but suggest that they have it checked out. Speak to your local utility, inspectors and electricians to see what their position is.

Sometimes double taps are solved by taking the two wires that are attached to a terminal screw and removing them both. The wires are then **pigtailed** together with one end of a third short wire (Figure 7.28). The other end of the third wire is attached to the fuse or breaker terminal. This eliminates the double lugging, although it creates one more connection in the box.

Most utilities will accept this solution, although every additional splice is another possible poor connection. Quality work uses the minimum number of connections.

Sometimes double taps are accepted where the second wire is a low voltage transformer for a door bell or telephone system, for example. However, double taps are considered more problematic where the two wires are different sizes.

Some special breakers are designed to hold two wires. In this case, double taps are often permitted. To be on the safe side, you should note all double taps, and let an electrician make the final call.

STRATEGY

Pigtail

Fewer Splices Are Better

Transformers

Special Breakers

Multi-Wire Circuits Fed by One Bus

Multi-wire branch circuits including circuits for split receptacles (two 120-volt circuits), and 240-volt circuits, have two hot wires. The two hot wires should be a red and a black. In some cases, circuits will be fed with two blacks or two reds accidentally.

240-Volt Circuits

On a 240-volt circuit, the appliance won't work properly since the appliance is seeing only 120 volts. This situation is usually corrected promptly because the appliance doesn't work properly or doesn't work at all. You won't often see this in the field.

Split Receptacles

The more subtle situation is the multi-wire 120-volt branch circuit, including split-receptacle circuit. Split duplex receptacles have the top half on one circuit and the bottom half on another. A dishwasher and garbage disposal, for example, can be plugged into the same receptacle without blowing a fuse or tripping the breaker. The top half of the receptacle is on one fuse or breaker and the bottom half is on another.

3 Conductor Cable

Normally, electricians run three-conductor cable to these receptacles. This is a cable with a black wire, a red wire and a white wire. The black wire may feed the dishwasher, and the red wire may feed the garbage disposal.

It is important that the black wire and red wire are fed from different bus bars at the panel. That is, the black wire must be fed by the black service entrance conductor and the red wire must be fed by the red service entrance conductor. Why does this matter?

White Wire Serves Red and Black

The black wire and the red wire are out of synchronization. This becomes important when we look closely at how the circuit is completed in a split-receptacle multi-wire circuit. We said that the cable had a black, a red and a white wire. The white wire is the return path for both the black and the red wire. Yet it isn't any bigger than the black or the red wire. Won't the white wire be overloaded?

The answer lies in the black and the red wires being out of synchronization. The current flowing through the neutral wire is much smaller than you would expect.

STRATEGY

Breaker Panels Have Staggered Buses

You can usually identify which bus the breakers are attached to. Many modern breakers are finger-jointed so that adjacent breaker positions are on opposite bus bars. You have to be careful, though, because while this applies to most panels, it does not apply to all. Also, there are some panels where **most** of the adjacent breaker positions are on opposite poles, but not all of them. This is something we'll have to leave to the electricians to verify, since we don't touch or dismantle the electrical equipment within the panel.

Twin Breakers

The same situation applies to fuses. In most situations you'll be able to tell whether the two fuses in a pullout are on different poles. Occasionally you won't be sure.

Twin breakers have two breakers that occupy a single slot in a panel. These twins are effectively adjacent breakers on the same pole. Multi-wire branch circuits should not be connected to twin breakers.

One of the clues you can use is to look closely at the neutral wire. The chances are high that if the black and the red both come off the same pole, the neutral wire will have overheated. Look for discoloration or melted insulation, for example, on the white wire. Look closely where it joins the neutral bar in the panel.

7.7.4 Conditions Found with Fuses Only

The items in Section 7.7.3 apply to both fuses and breakers. These problems are specific to fuses:

1. fuses loose
2. fuse holders loose or broken
3. fuses for multi-wire circuits not linked

STRATEGY

Fuses Loose

Where you notice that fuses are loose, they should be written up. We are not supposed to correct deficiencies we find in houses, although you may be unable to resist the temptation to simply tighten the fuse rather than to write up a loose fuse as a defect.

Fuse Holders Loose or Broken

Sometimes the supports for the fuses are worn or broken and will not stay securely in their position in the panel. In other cases, they won't hold the fuses securely. This is particularly true of pullouts, often associated with 240-volt circuits.

STRATEGY

Where fuse holders are loose, or there are pieces obviously broken, replacement of the components or the entire panel should be recommended.

Fuses for Multi-Wire Circuits not Linked

A multi-wire circuit has a red and a black wire. This is typical of all 240-volt circuits. It's also true of 120-volt circuits going to split receptacles, for example. We talked earlier about the danger of these multi-wire circuits. Since there are two conductors that are both hot (black and red) going to the same place, you need two fuses. Removing one fuse and having the circuit not work can be very misleading. People will think there is no electricity in the circuit and start to work on it. Obviously, the other wire will still be hot and people can be electrocuted.

The solution is to make sure that when you disconnect the circuit, you have to pull out two fuses at the same time. Multi-wire circuits should be on fuse pullouts if the two circuits end at the same location (e.g., a duplex receptacle). Where this is not done, recommend rearrangement.

STRATEGY

Look for red wires at the panel. Every one should be on a pullout fuse block. If in doubt, recommend that it be checked. There are some exceptions, so don't take too strong a position.

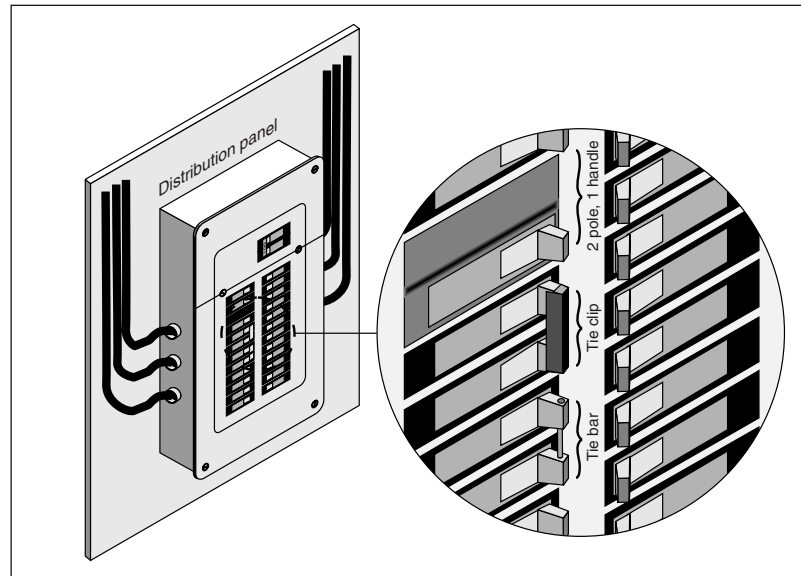
7.7.5 Conditions Found with Breakers Only

Common problems include—

1. multi-wire circuits not linked
2. too many breakers
3. wrong breaker in panel
4. loose breakers

Multi-Wire Circuits Not Linked

We've talked about requiring fuse pullouts for multi-wire circuits that end at the same location. On circuit breakers, we have to mechanically tie the two breakers together to ensure that both the red and the black are disconnected before we work on the circuit.

FIGURE 7.29 Special Circuit Breakers for 240 Volts

These ties or links may be a metal cap that slips over the breaker handles (Figure 7.29). In some cases, it is a metal bar that slips through the holes in the breaker handles. In other cases, the breaker may have a single handle (throw) but double poles. Any of these is satisfactory.

STRATEGY

Again, watch for red wires in the panel. These will usually be part of a three-conductor cable and part of a multi-wire or 240-volt circuit. When in doubt, get it checked since there are exceptions.

Too Many Breakers

Every panel is designed to accommodate a given number of circuits. Some manufacturers provide half-sized **twin breakers** so you can get two circuits into a spot designed for one breaker in the panel. This is convenient because it allows you to squeeze an extra circuit in, but it can lead to too many circuits in the panel.

STRATEGY

There should not be more circuits used in the panel than the panel is rated for. This information is usually on the panel label. A total of 42 single-pole circuits is the maximum allowed in the largest panel available.

Wrong Breaker in Panel

Every panel manufacturer makes breakers for its panels. Only those breakers designed for that panel should be used in it. Different breakers may not fit properly and may create dangerous, loose connections.

STRATEGY

Question the installation of breakers made by one company in a panel made by a different company. Again, note this as a question rather than a conclusion. The situation should be described as unusual and the possible problem discussed with the client (loose connections, overheating, electrical shock, etc.).

STRATEGY

*Don't Energize Circuits
That Are Off*

Loose or Damaged Breakers

One breaker that is obviously misaligned from the rest is probably loose. While we do not recommend touching components inside the panel, you should be able to touch the breaker handle safely. Homeowners have to do this to turn the circuit breaker on and off. Any breakers that are loose should be noted for replacement.

Breakers with damaged bodies or handles should be replaced. What do you do when you find a fuse blown or missing? What if you find a breaker tripped or in the “off” position? **Leave it!** There is usually a good reason it is off. You can electrocute someone or start a fire by turning the circuit back on.

Loose or damaged breakers should be reported as a priority item for safety reasons.

7.7.6 Conditions Found with Panel Wires

We haven't talked about branch circuit wiring yet, so we'll only deal with issues you'll find in the panel.

1. Sheathing not removed
2. Overheating
3. Loose connections
4. Damaged
5. Not well secured
6. Wires crossing bus bars
7. Abandoned wires in the panel

Sheathing Not Removed

The sheathing should be stripped off wires inside the panel. The sheathing should remain where the wire is secured by cable clamps as it enters the panel. However, the sheathing should not project more than about an inch into the panel. This makes the wire hard to work with, allows heat to build up and adds fuel to a fire in the panel.

Note any cable sheathing extending more than one inch into the panel. Also make sure that the sheathing does extend into the panel. The sheathing protects the wire where it is clamped in place coming into the panel.

STRATEGY**Overheating**

We've talked about overheating and how you should look carefully for it on the neutrals of multi-wire circuits. You should also be looking at the termination point of each wire for evidence of overheating: discoloration, melting insulation, smell, and actual heat.

STRATEGY**Loose Connections**

A loose connection is dangerous. These cause more electrical fires than any other electrical problem. Look for these at the fuse or breaker terminals, the neutral bars, the ground connections, any splices in the panel and at the lugs on the bus bars.

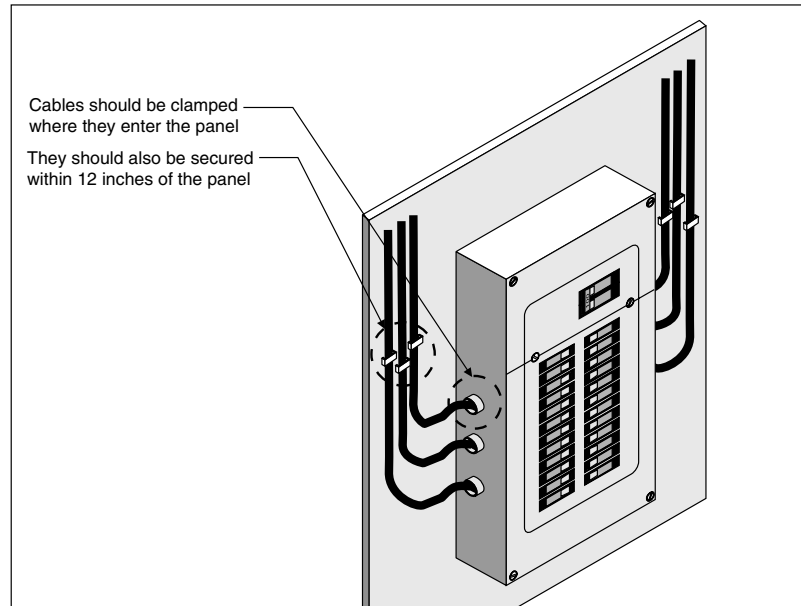
A loose connection may cause excessive voltage drop in the circuit. This may show up as a dim light, for example.

Some inspectors pull gently on the wires to check that they are secured. We consider this dangerous and do not recommend it.

STRATEGY

Voltage Drop

Don't Pull on Wires

FIGURE 7.30 Securing Wires**STRATEGY****Damaged**

Where insulation is missing or split, the wire should be repaired or replaced. Many electricians consider replacement the only respectable repair.

Insulation can also be brittle and broken due to age or overheating. Again, this should probably be replaced.

Not Well Secured

Cables should be clamped where they come into the panel (Figure 7.30). They should be held securely in place, but not pinched between sharp metal edges.

STRATEGY

Look at each cable as it enters the panel and make sure it is well secured. Some inspectors pull lightly on each cable outside the panel to make sure it is well connected to the panel itself.

Wires should also be secured to wall studs or some other firm substrate within 12 inches of leaving the panel.

Wires Crossing Bus Bars

Wires should not be run across the face of bus bars. Wires should be arranged to run around bus bars down the sides and across the bottoms of panels.

STRATEGY

Rerouting wires within the panel is usually not a big problem unless they have been cut too short. In this case, the section of wire will have to be spliced or replaced.

Abandoned Wires in the Panel**STRATEGY**

One is never sure whether the wire ends are live. Abandoned wires should be removed from the panel.

7.8 BRANCH CIRCUIT WIRING (DISTRIBUTION WIRING)

*Power from Panel to
Point of Use*

These wires carry the electricity from the main distribution panel or subpanels to the points of use. These include receptacles, lights, switches, and appliances such as stoves, dishwashers, garbage disposals, furnaces, and air conditioners. The job of branch circuit wiring is to get the electricity from the source to the destination safely.

7.8.1 Types of Wires

Copper and Aluminum

The most common household wiring is **copper**. **Aluminum**, which is silver colored, was a popular alternative from the mid 1960s until the late 1970s. While aluminum wires aren't used much for branch circuits in new work, you'll see them in many existing homes. Aluminum wires are still commonly used for service entrance conductors, and some of the large 240-volt circuits still use aluminum wire (stoves and clothes dryers, for example). We'll talk more about aluminum wire shortly.

7.8.2 Cable Types

NM Cable

There are different types of cable for different uses. The most common distribution wiring in houses is NM (which stands for **Non-Metallic** sheathed cable). This is also called Romex or Loomex, brand names for this type of wire. This cable has a paper, cloth or plastic (most modern) sheathing. This type of wire is widely used, but is not permitted in some areas. Check your rules!

UF BX

UF (underground feeder) cable is rated for underground use. BX cable (or **AC-90** cable) is a flexible metal-jacketed cable. Some people refer to this as **armored cable**. It is typically used where the wire might be subjected to mechanical damage. It was often used to feed furnaces, boilers and water heaters, although most modern codes don't require that.

7.8.3 Wire Insulation and Sheathing

Insulation

The **insulation** refers to the material wrapped around the individual wires. On old wire, it was rubber (knob-and-tube, for example). On more modern wire, it is plastic.

Sheathing

The **sheathing** forms the cable, wrapping around the insulated black, white and/or red wires and the uninsulated ground wire, if there is one. The sheathing is what you can see and touch, typically in the unfinished areas of homes and garages.

Functions

Older wires have cloth or paper sheathing. Newer wires have plastic sheathing.

The function of the insulation is to keep the wires from touching each other within the cable. The function of the sheathing is to provide mechanical protection for the conductors and their insulation.

Certain types of wires are typical of certain time periods. Allowing for regional differences, we can generally say the following wires were used during the periods described in Table 7.4.

TABLE 7.4 Wires and Time Periods of Use

Wire Type	Dates of Use
Knob-and-tube	1920 to 1950
Cloth-sheathed, two- or three-wire cables with no ground wire	1945 to the early 1960s
Cloth-sheathed cable with an integral ground wire	Early 1960s to early 1970s
Cloth-sheathed aluminum cable with integral ground	1964 to 1978
Plastic-sheathed aluminum cable with integral ground	1974 to 1978
Plastic-sheathed copper cable with integral ground	1974 to date

7.8.4 Number of Conductors

Two-Conductor Cables

The knob-and-tube are single-wire cables. Modern wires are typically two or three wires (black, white and sometimes red) and after the mid-1960s, they also included an uninsulated ground wire.

Two-Conductor Plus-Ground Cables

The **two-conductor-plus-ground** cables are used for most normal household circuits (Figure 7.31). (Remember, **conductor** is just a fancy word for **wire**.) The three-conductor-plus-ground cables are typically used for 240-volt appliances and multi-wire branch circuits, including split receptacles.

7.8.5 Wire Sizes

14 and 12 Gauge Wire

Fourteen gauge wire is the most common in residential branch circuits (Figure 7.32). Twelve gauge wire with 20-amp fuses is currently required for at least two circuits.

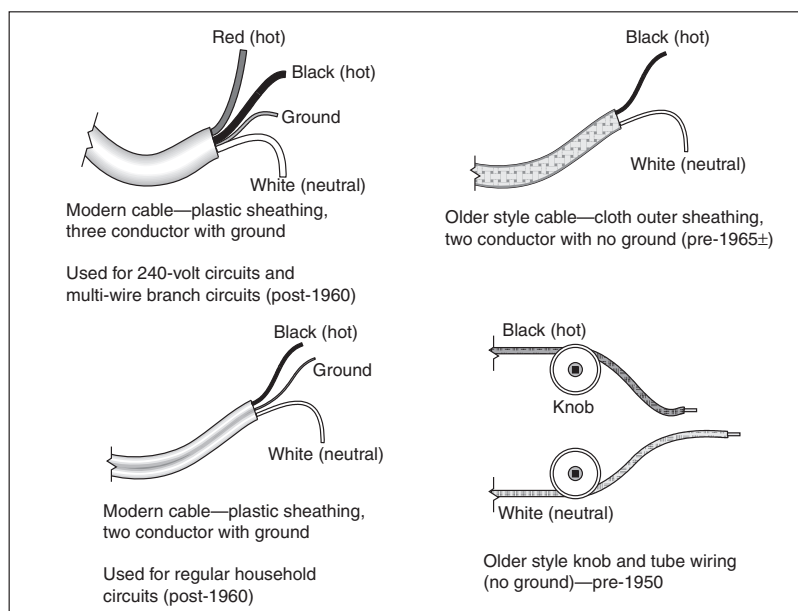
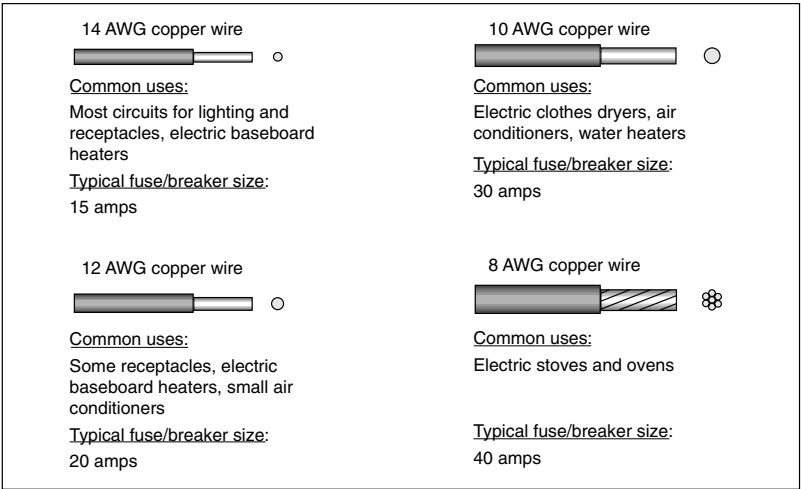
FIGURE 7.31 Number of Conductors

FIGURE 7.32 Common Household Wire and Fuse Sizes



12 Gauge

Twelve gauge wires are used for some electric baseboard heaters, water heaters, dish washers, garbage disposals, and private well pumps.

10 Gauge

Ten gauge wires are typically used for air conditioners, saunas, electric clothes dryers, some electric water heaters, some cooktops and ovens.

8 Gauge

Eight gauge wire is used for stoves and large air conditioners.

Note: This is all copper wire we’re talking about. Aluminum wires are usually one size larger, since aluminum does not conduct electricity as well as copper.

7.8.6 Number of Circuits

12 Points per Circuit

There is no magic number of circuits required in a house. No more than 12 outlets and/or lights should be on a single branch circuit. The acceptable number of circuits is largely a function of lifestyle.

10 to 20 Circuits Typical

The distribution panel and subpanel (if there is one) will tell you the number of circuits in a house. You may have as few as four 120-volt branch circuits or as many as 42. We often see 10 to 20 such circuits. You have to make a judgment call about whether that’s adequate.

One Outlet per Room
Not Enough

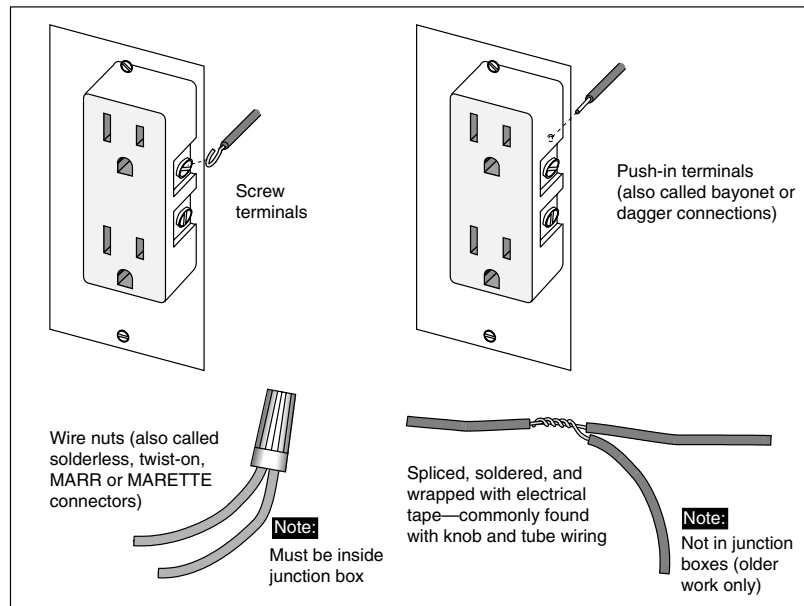
You will get some clues as you move through the house. For example, if there is only one electrical receptacle in each room, that isn’t enough. An insufficient number of plugs may encourage people to overuse extension cords, which isn’t safe. That usually corresponds with a small number of branch circuits at the panel. People should add more receptacles, and will often have to add more branch circuits.

One Kitchen Counter Outlet
Not Enough

If there is only one kitchen counter receptacle, the distribution network may have to be expanded. Remember that this has nothing to do with service size. We’re simply talking about adding more branches to the tree. This doesn’t change the size of the tree trunk.

More Outlets and Lights
May Mean More Circuits

There is a relationship, of course, between the number of branch circuits and the number of outlets and lights. The more outlets and lights you want, the more circuits you should have.

FIGURE 7.33 Types of Connections

7.8.7 Connectors

No Open Splices

Wires should be connected to each other or to fixtures only inside electrical boxes. Connections can be in the panel box, junction boxes, switch boxes, receptacle boxes, and light boxes, for example. You should not see any open splices in the wire. Knob-and-tube is the only exception to this, and then only on original (pre-1950) work. Knob-and-tube wire should be joined to modern 2-conductor cable in a junction box.

Wires Secured to Terminal Screws

The wires themselves are usually attached to **screw terminals** (Figure 7.33). The wire insulation is stripped off for about three-quarters of an inch. The wire is turned clockwise around the screw head about three-quarters of a turn, and the screw head is tightened to hold the wire in place.

Push-In Connections

There are also **push-in terminals** where about one-half inch of the insulation is stripped off the wire and the wire is pushed straight into a hole, usually in the back of a switch or receptacle. These are also called **bayonet** or **dagger** connections. Many electricians are critical of the quality of push-in terminals, although they do make the work go faster. Some say the wire is easily nicked with this type of connector.

Soldered and Taped Connections Wire Nuts

Wires may also be joined by twisting together, soldering, and wrapping the connection in electrical tape, all inside a junction box. This is rarely done anymore. Most taped connections do not have any soldering and suggest low-quality work. Wires can also be joined with **wire nuts** (also called **solderless connectors**, **insulated twist-on connectors** or **Marr connectors**). **Marr** and **Marrette** are brand names for a common type of wire nut. Again, these should be inside boxes.

Loose Connections

While there are many ways to secure wires, loose connections are a possibility with any of them. Watch for loose connections and the overheating that results. Be suspicious of taped connections, even inside boxes.

7.8.8 General Strategy for Inspecting Branch Circuit Wiring

Follow the Wiring

Follow the wires where you can, from the panel through to their termination points. You'll typically be testing receptacles, switches and lights as you go through the house. Where wiring runs through unfinished areas, look at the visible wiring.

Look for wiring in places where it shouldn't be (exposed to damage or heat), such as along baseboards or inside closets.

Don't Do Flex Test

Some inspectors recommend flexing wires to make sure they are still pliable. Some say wires should be able to be bent 180° between your fingers. We feel that this is a dangerous test, since if the insulation is brittle and breaks off, you may find yourself holding a hot wire. This can also be a destructive test if the insulation cracks while you're flexing it.

We believe this kind of test is better left to an electrician, who can disconnect the power before testing the flexibility of the wire, and can repair any damage that results.

7.8.9 General Conditions

Let's look at some of the branch circuit wire problems you'll commonly see.

1. Damaged
2. Not well secured
3. Loose connections
4. Open splices
5. Wires too close to ducts, pipes, vents, chimneys and flues
6. Wires too close to the edge of studs or joists
7. Wires run through steel studs without protection
8. Exposed wires on walls or ceilings
9. Exposed wires in attics
10. Indoor cable used outdoors
11. Buried cable not rated for buried use
12. Household wiring used as extension cord or extension cord used as permanent wiring
13. Undersized wire
14. Improper color coding
15. Abandoned wire

Note: We're going to talk about knob-and-tube and aluminum wiring separately. The conditions we'll discuss here apply to knob-and-tube and aluminum wiring as well. Special conditions applying only to those will be discussed separately.

The implication for all of these is possible electric shock or fire.

IMPLICATION

Damaged Wire

Damaged wire includes wire that has been cut or pinched, wire that has been chewed on by pests, burned wire, and wire with cracked (brittle) insulation.

STRATEGY

Follow as much of the wiring as you can to make sure it is intact. Pay particular attention where wire passes through materials, especially if they are sharp. Look

closely where wire is held by staples to make sure the staples haven't damaged the wire. Exposed wires are particularly susceptible to damage.

Wires are most likely to be chewed in attics where pests are commonly found.

Wire not Well Secured

Wire should be held in place with **strain relief clamps** wherever it goes into a box. Wire should also be secured within 12 inches of leaving a box, usually with a staple. Wire running through the house should be secured (this includes supporting it by a hole through a joist or stud) every 4½ feet.

STRATEGY

As you follow the wire, make sure it's well supported. It's usually easy to add support. Poorly supported wires are a sign of amateurish work, and should make you look more closely.

Loose Connections

Wherever wires are secured, whether with terminal screws, push-in connectors or solderless connectors, they should be tight. You'll normally just get a look at connections at the panel. The Standards do not require removing cover plates from receptacles or light switches. Many inspectors do this in selected areas to get a look at wall insulation, to determine the type of wire feeding the receptacle, to determine whether the receptacle is suitable for aluminum wire, to look for evidence of overheating, to determine whether a receptacle is a split receptacle, or to determine whether 20-amp rated receptacles are fed by 12 gauge wire.

STRATEGY

Wherever you remove cover plates, including at the panel, look at the connections. Be very careful what you touch. We do not recommend that you test connection tightness by touching or pulling on the wires. You should be looking for evidence of loose connections, which is most often overheating.

Open Splices

All connections or splices should be in junction boxes.

STRATEGY

Note any connections that are not in boxes. Adding a junction box is usually not a difficult task.

Wires Too Close to Ducts, Pipes, Vents, Chimneys, and Flues

Wires should be at least one inch away from ducts and pipes. Often a piece of insulation is used to separate the wire from the ducts or pipes. This is fine. The distance from vents, chimneys and flues, depends on the fuels. Typically, for a single-wall metal vent connector, the wire should be 6 inches away if the fuel is gas, 9 inches (18 inches in some areas) away if the fuel is oil, and 18 inches away if the fuel is wood. There may be local variations, but this is a good starting point.

STRATEGY

As you follow the wire through, make sure the wire won't overheat because it's too close to these heat sources. Pay particular attention to wires run through furnace rooms.

These clearances apply to conduits as well as cables. Plastic conduit is obviously susceptible to high temperatures. Metal conduit may also conduct heat and transfer it into the wires, diminishing their ability to carry electricity.

Wires Too Close to the Edge of Studs or Joists

Wires should be 1¼ inches away from the face of studs or joists.

STRATEGY

Where you see wires closer than this to the edge, recommend that protective metal plates be provided over the edge of the joists or studs. There are many places this may happen where you'll never know.

Wires Run through Steel Studs without Protection

The raw edges of steel studs can cut non-metallic sheathed wire. In commercial installations, BX (armored) cable or rigid conduit is used. In single family homes, most jurisdictions allow the use of non-metallic sheathed cable with special grommets that protect the wire where it passes through studs.

Wires cannot be fastened directly to metal studs. They must stand off the studs with an insulator at a distance of at least one-quarter of an inch in most areas.

STRATEGY

Where you can look at metal studs, check for damage to the wire. Check for plastic sleeves or grommets where the wires penetrate the studs and make sure wires are not attached directly to the studs.

This is an area that may change since steel frame construction is becoming popular. Other solutions may be developed, and rule changes are expected. Again, you may want to be cautious with your criticism. It may be best to recommend further investigation to confirm your suspicions.

Exposed Wires on Walls or Ceilings

Wires that are not in conduit or are not armored should not be run where they can be damaged by normal day-to-day activities. Wires shouldn't be run along baseboards, around doorframes, in closets, etc. Where these are found, they should be relocated or protected. There are covers available to protect surface-mounted wire.

STRATEGY

Wires should not run on walls where they can be mechanically damaged; for example, anywhere within 5 to 7 feet of the floor. We encourage you to use common sense. No matter where the wire is, if it's likely to be damaged, recommend protection.

Wires run to appliances such as furnaces and water heaters have to be in armored cable in many jurisdictions because they are exposed to damage where they run along beside the appliance. Know what is required in your area and watch for improper work.

Exposed Wires in Attics

Wires should not be run on the tops of ceiling joists in attics. It's easy to step on the wires here, especially if they're concealed below insulation. It's better to run the wires through the ceiling joists. Cables within six feet of the access hatch should be protected from mechanical damage. If there is a stairway or built-in ladder to access the attic, wires should be protected in attic areas with seven feet or more of headroom.

STRATEGY

Check what is accepted in your area. This is a rule you'll find broken all the time, and frankly, most clients won't follow your advice on this one. However, if you don't mention it, an electrical inspector who comes along later will mention it. More importantly, if you don't mention it and someone in the attic electrocutes themselves, chances are good you'll hear about it.

Indoor Cable Used Outdoors

NM cable is suitable for dry environments. It can't be used outdoors.

Look for indoor-rated cable on exterior wiring.

STRATEGY

Buried Cable Not Rated for Buried Use

Where you can see buried cable coming out of the ground, check to see what type of cable it is. It should be rated for underground use. If you're not sure, jot down the type of wire and phone an electrical supply house or check another reference.

STRATEGY

Household Wiring Used as Extension Cord or Extension Cord Used as Permanent Wiring

NM type cable should not be used as portable extension cords. It is not designed to be coiled and flexed the way extension cords are.

Extension cords should not be fastened in place or concealed in walls, floors or ceilings.

STRATEGY

Extension cord arrangements may disappear before your client takes possession of the house. However, if you see this installation, make a note of it and explain the problem to your client.

Undersized Wire

The wire should be a match for the fuse or breaker size and should be continuous throughout the circuit. It is not appropriate to step down the size of the wire after leaving the box.

STRATEGY

Where wiring is visible, check that its size is not reduced in downstream sections. In addition, wiring in the panel should be proper for the fuse size. Where you have a 30-amp fuse, 12 or 14 gauge size is not appropriate. You may have to change the fuse or breaker, or, depending on the appliance served, you may have to change the wire. Another way of saying that the circuit is **overfused** is saying the wire is **undersized**.

Improper Color Coding

Normally, black and red wires signify hot wires and white wires are neutral. But life isn't always simple.

Sometimes white wires can act as black wires. (Don't you just hate it?) Let's look at an example.

*When White Wires Are
Black or Red*

Sometimes electricians want to run 240-volt balanced-load circuits (for example, an electric baseboard heater or water heater). They don't need to use three-conductor cable because the black acts as white when it's at rest, and the red acts as white when it's at rest. We don't need a white wire! However, electricians often don't have two-conductor cable with a black wire and red wire with them. They do, however, almost always have black and white two-conductor cable, so they run a 240-volt circuit with the black wire attached to one breaker and the white wire (acting as a red) attached to another breaker.

*Wrap Ends of White Wire
with Black Tape*

This is acceptable as long as the white wire is color coded at each end to show that it is acting as black or red. Wrapping the last inch or two of wire with black electrical tape is a common way to accomplish this.

STRATEGY

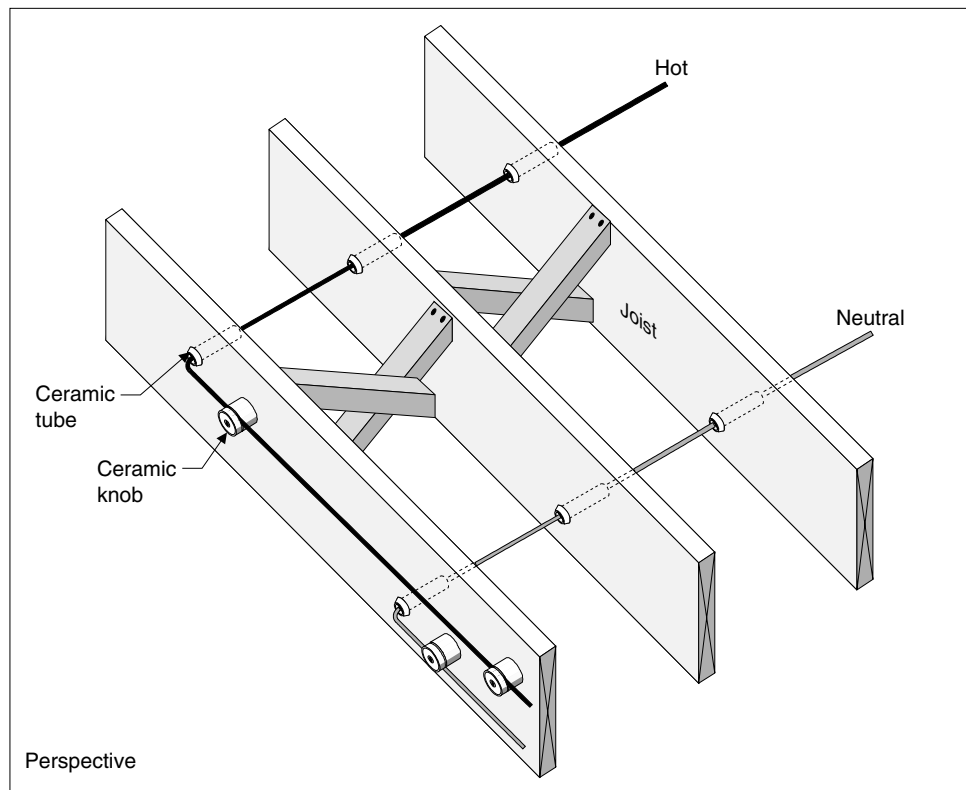
If you look in a panel and see a white wire attached to a fuse or breaker terminal, your first instinct is that it is wrong. This is a good instinct. It might be all right, but if the white wire is acting as a hot (black or red) wire, it should be color coded black at the ends to indicate this. Where this is not done, you should write up the situation as suspect and call for checking by an electrician.

Abandoned Wire

Abandoned wire should always be removed.

STRATEGY

When looking at the ends of an abandoned wire, you can't tell whether it's live. Many people assume it's not. If you're wrong, there's a big price to be paid. The best practice is to remove any abandoned wire so there is no possibility of leaving a live wire exposed.

FIGURE 7.34 Knob and Tube Wiring

7.8.10 Knob-And-Tube Wire Conditions

Most of the problems we have been talking about with respect to panels and wiring also apply to knob-and-tube systems. Some issues are more common with knob-and-tube wiring.

Identifying Knob-and-Tube

Knob-and-tube wiring is easy to identify (Figure 7.34). Two separate cables run to each electrical point, one black, one white. Sometimes the sheathing is black on one and is white on the other, although this isn't always the case. The ceramic knobs and tubes used to support the wire also clearly identify this kind of wiring. Knob-and-tube wiring was common until 1950 and used into the 1960s in some areas. Knob-and-tube wire was made in several gauges. Branch circuit wiring was often 12 gauge.

Remember that there will be no original equipment ground wires anywhere in the distribution system with knob-and-tube wiring. Original connections are often not in junction boxes. Connections are made by splicing, soldering and taping the wires. Knob-and-tube wiring was no longer used by the time multi-wire branch circuits were introduced.

Problems specific to knob-and-tube wiring include the following:

1. Connections not in boxes
2. Brittle wire, insulation or sheathing
3. Wire buried in insulation
4. Fused neutrals

Connections Not in Boxes

The original spliced connections did not have to be in junction boxes. This is acceptable as long as the connections are original, since we have some confidence that the splices are appropriately made by twisting the wires together, then properly soldering, and finally taping with the old, dull black, cloth tape.

STRATEGY

Modern plastic electrical tape (the black shiny stuff) on a knob-and-tube connection, or taped connections with knob-and-tube wiring joining modern two-conductor cable, are not acceptable to most authorities. The only knob-and-tube connections that don't have to be in junction boxes are original ones. Knob-and-tube wire should join modern two-conductor cable in a junction box.

Brittle Wire, Insulation, or Sheathing

Older Wire

Another issue with knob-and-tube wiring is the age of the wire. While the copper does not fail under normal work loads, a house wiring system that's been around for a long time may well have been overfused at some time in the past, and the wire may have been overworked. The older it is, the more likely it is to have been abused at some point.

STRATEGY

Insulation and Sheathing

The rubber insulation or the cloth sheathing may deteriorate over time, become brittle and start to fall off. This is a function of age and high temperatures. The high temperatures may be high ambient temperatures, or due to overfusing. In either case, watch for insulation problems on knob-and-tube wiring. Resist the temptation to flex the wire to see if it's still supple. You may get a shock or damage the wire.

Problems with the sheathing and insulation are the most common failings of knob-and-tube wiring, other than amateurish modifications and extensions of the circuits.

Wire Buried in Insulation

Hard to Dissipate Heat

This leads us to the discussion of knob-and-tube wiring buried in attic insulation or in wall insulation. This is a controversial issue and different authorities have taken different positions. Some maintain that knob-and-tube wiring shouldn't be buried in insulation because that inhibits its ability to dissipate heat. It's true that all wire can work more comfortably if it's able to dissipate the heat generated as electricity flows through it. But there is a counterpoint.

Separate Wires May Be an Advantage

Since the black wires and white wires are usually separated by several inches in different cables, knob-and-tube might be better able to dissipate heat than modern cable. Modern two-conductor cable has the black wire and white wire less than one-quarter of an inch apart inside a plastic sheathing. It is harder for modern cable to dissipate heat than for knob-and-tube. This is probably only an issue where several cables are bundled together and there is heavy continuous loading on each cable.

STRATEGY

Nonetheless, you should find out what is accepted practice in your area. In some jurisdictions, knob-and-tube wiring is only permitted in insulation if the insulation is non-combustible. Find out what your authority thinks is non-combustible. Some consider fiberglass insulation combustible (it really is at certain temperatures). Insulations made of cellulose, wood shavings, shaved leather, most plastics, etc., are combustible. Insulations made from vermiculite, gypsum slag and asbestos are not combustible, although asbestos has other issues.

Fused Neutrals

Because knob-and-tube installations are old, they may connect to old panels. Some of these old panels have **fused neutrals**, which can be dangerous.

TABLE 7.5 Breaker Sizes for Aluminum Wire

Wire Size	Breaker or Fuse Size
12 gauge	15 amps
10 gauge	25 amps
8 gauge	30 amps
6 gauge	40 amps
4 gauge	55 amps

STRATEGY

When you see knob-and-tube wiring and an old ceramic panel, look for fused neutrals. We discussed these when we talked about problems in panels earlier. If the neutral wire and hot wire are both fused, only one is likely to blow, shutting off the circuit. If the neutral fuse blows, the circuit is live through its entire length. This is dangerous for anyone working on the circuit. Recommend replacement of panels with fused neutrals.

7.8.11 Aluminum Wire Conditions

The common problems specific to aluminum wire are—

- 1. Connectors not compatible with aluminum
- 2. No anti-oxidant grease on stranded wires
- 3. Overheating

Can Be in Old Houses

Aluminum wiring was used from the mid-1960s to the late 1970s. You probably won't find aluminum on branch circuit wiring in houses built after 1980. However, you will often find it in old houses. Electrical work performed in old houses from 1965 to 1978 commonly used aluminum wire.

The appropriate fuse or breaker sizes for common aluminum wire sizes are outlined in Table 7.5.

Connectors Not Compatible with Aluminum

Aluminum Problems

When aluminum wiring was introduced in the 1960s as a less expensive alternative to copper, people had already recognized that it didn't conduct electricity as well as copper. That problem could be handled by using slightly larger wire to carry the same amount of electricity. This was a good decision.

Some problems specific to aluminum weren't recognized for a few years: Aluminum wire—

- 1. tends to creep out from under terminal screws
- 2. forms a rust (oxide) that is an electrical insulator
- 3. is softer than copper and easier to damage when working with the wire
- 4. was, in some of the early wire, a low quality material, prone to weakness and breaking

Because of these issues, some house fires were attributed to aluminum wiring. By about 1970, the industry had redesigned many connectors specifically for aluminum.

CUAL

These were designated **CUAL**. They worked well, except on 120-volt branch-circuit receptacles (outlets). Some of the CUAL receptacles did not solve the over-

CO/ALR

heating problems with aluminum wire. A subsequent version known as **CO/ALR** was introduced in the early 1970s, and used tin-plated terminals. This version has worked well.

Switches

In some jurisdictions, CUAL is acceptable for switches. Other jurisdictions prefer a CO/ALR designation on switches. There are also jurisdictions which don't require any special connectors for switches because of the infrequency of problems. Again, learn what's accepted in your area.

*Panel and Breakers
Should Be CUAL*

Check for aluminum-compatible equipment elsewhere. At the panels, you should verify that the panels and their components are rated for use with aluminum. CUAL is an appropriate designation for service equipment, panels, circuit breakers, dryer receptacles, stove receptacles, etc.

*Aluminum Compatible
Wire Nuts*

Special wire nuts (also called solderless connectors or twist-on connectors) are designed for use with aluminum. It may be hard to identify them during a home inspection. If you see aluminum wiring in a home, you won't be able to check every connection anyway and you should be recommending an electrician check all connections, including the wire nuts. The best approach to aluminum wiring is to use equipment that is compatible with aluminum. A second choice, allowed in some areas, but discouraged in most, is **pigtailing**. The process involves using a piece of copper wire between the aluminum wire and the receptacle, for example. Pigtailing has the disadvantage of making boxes more crowded and adding more connections.

Copalum

The aluminum-to-copper wire connection may be made with a proprietary system known as **Copalum**. These are not used at all in some areas. In other areas, they are quite common. Many consider this the best way to work with aluminum wire.

Split-Bolt Connectors

Split bolt connectors can be used to join copper and aluminum wires, although these are more common in industrial applications.

Bad Reputation

In any case, by the time the industry developed these changes to solve the problems with aluminum wiring, aluminum had received such a bad name that most manufacturers stopped making it.

Overheating

Overheating usually is the result of loose connections, slightly corroded connections and/or damaged wires. These problems occur at the ends of the wires where they are connected to fuse or breaker terminals, appliances, wire nuts, or equipment such as receptacles or switches. Receptacles are the most common problem area.

STRATEGY

General Strategy for Inspecting Aluminum Wire

Make sure you identify aluminum wiring if it is visible. Check for appropriate special connectors, or at least pigtail. Know what approaches are accepted by your local electric company.

Remove Covers

Watch for overheating, particularly at receptacles, but at all connections. Consider removing covers from at least some of the kitchen, bathroom and workroom receptacles. Look for anti-oxidant grease on stranded large wire connections (8 gauge and larger).

Recommend Inspections

Recommend a detailed inspection by an electrician, unless you've evaluated every aluminum wire connection. This may include checking the torque on the breakers.

Recommend annual inspections of the wiring to ensure connections do not become loose.

Replace Small Amount

If there's only a little aluminum wiring in the house, rather than going through a lot of work to investigate, correct and maintain it, it is often simpler to replace it with copper.

Rewiring a whole house with copper is usually not practical or necessary.

Inspector in the House: The Tale of the Underfed Outlets

A few years ago, I was inspecting an older suburban bungalow. The listing bragged about the newly finished basement. The workmanship in the basement showed evidence of a do-it-yourself job—poorly fit trim, doors that did not close properly, and flooring that was not well sealed at the joints. Although there was not much to see, I had my suspicions and dug a little deeper. Removing cover plates from accessible electrical outlets showed an alarming consistency—rather than 12 or 14 gauge wire, the outlets were powered with telephone wire!

The lesson: trust your instincts. If it feels wrong, it probably is. Put another way, if the work that shows is bad, the work that doesn't is often worse.

7.9 LIGHTS, OUTLETS, SWITCHES, AND JUNCTION BOXES

In this section, we'll talk about checking things by taking covers off electrical outlets, light switches and junction boxes.

7.9.1 Conditions at Lights

Let's start by listing some of the common lighting problems we find.

1. Damaged or loose
2. Overheating
3. Inoperative
4. Obsolete
5. Not grounded
6. Missing
7. Poor stairway lighting
8. Conventional lights used in wet areas
9. Improper recessed light installations
10. Improper closet lighting
11. Heat lamps over doors
12. Isolating links needed on pull chains

IMPLICATIONS

The implications will be inadequate lighting and/or shock or fire hazard unless noted otherwise.

We're talking about lights that are permanent fixtures to the house. We do not test floor lamps, table lamps, etc. Testing lights is easy—you simply use the switch to turn them on and off.

STRATEGY**Damaged or Loose**

Light fixtures that are damaged or loose may or may not be staying with the house. However, they may be a shock hazard or may be in danger of falling. Recommend they be repaired or replaced, as necessary.

STRATEGY

Hot

Overheating

Light fixtures are often out of reach and you may not detect an overheating situation. Where you are close to a light, you may be able to feel the heat, smell the burning insulation, or see evidence of charring or scorching. Many light fixtures get very hot under normal operation, and you have to be careful not to overreact.

Light bulbs are hot even when working properly. A 100-watt incandescent bulb has a surface temperature of roughly 380°F. A tungsten halogen bulb can be 480° or higher.

STRATEGY**Inoperative**

There are many reasons why a light may not work. We should simply describe it as **inoperative** and leave the diagnosis to others.

STRATEGY**Obsolete**

Light fixtures with exposed wire connections are obsolete and should be replaced. Any fixture with a bare wire secured to a terminal should be replaced.

STRATEGY**Not Grounded**

All modern installations (post 1960) should have grounded light fixtures and light support boxes. However, it is beyond the scope of a home inspection to dismantle light fixtures to determine whether they are grounded.

STRATEGY**Missing**

A light fixture should be provided in most rooms. Use common sense here. If part of the house cannot be lit, or cannot be lit from where you enter the room, recommend rearranging the lighting. Wall switches that control receptacles are an acceptable alternative to light fixtures. Lights are not needed in clothes closets, unless it's a walk-in closet.

It is also important that every door into the house has an exterior light. This is a convenience and safety issue.

STRATEGY**Poor Stairway Lighting**

If the stairs have more than six treads, you should be able to turn the stairwell lighting on at either the top or the bottom. You may not need switching at both the top and the bottom for an unfinished basement. However, wherever somebody is likely to go up or down the stairs and stay until it gets dark, it makes good sense to provide three-way light switches.

STRATEGY**Conventional Lights Used in Wet Areas**

Light fixtures in showers, saunas, or outdoors exposed to the weather, should be suitable for damp environments. These fixtures can usually be identified by a heavy gasket around the diffuser. If it looks like water could get into the light, you should question the installation and have it checked further.

Two Types

Improper Recessed Light Installations

As discussed in Chapter 5, **recessed light fixtures (potlights, can lights, high-hat light fixtures)** can be a fire hazard with respect to insulated ceilings.

Wrong Type of Bulbs

Another problem with recessed lights is the use of conventional light bulbs. Any light bulb that burns with the socket at the top and the bulb facing down is going to have heat collect around the socket. This is not an ideal situation. Light bulbs generally burn cooler and last longer if they burn with the base down. The problem is compounded if we don't use the proper bulbs, which are designed to reflect the light and heat downward.

Oversized Bulbs

Bulbs of the incorrect wattage can also be a problem with recessed lights. Bulbs that are too large build up heat and create a fire hazard. Some fixtures are rated for one size of bulb if they are not in an insulated ceiling, and a smaller bulb if they are in an insulated ceiling.

Many modern recessed lights use small halogen bulbs. These also can get quite hot. The halogen fixtures use small transformers at each light. These transformers should be removable for servicing. The wires on the lights themselves and on the cable coming in should be at least 12 inches long to allow maintenance.

STRATEGY

Be careful with recessed lights and treat all of them as a potential fire hazard. Check the type of fixture, whether it's in an insulated ceiling and for proper recessed light bulbs.

STRATEGY**Improper Closet Lighting**

It doesn't make sense to put a light in a closet immediately above a shelf, where things will likely be stored around the light. This can be a fire hazard, and there is a danger of breaking the light and injuring someone with the broken glass.

Closet lights should be mounted on the ceiling away from the shelf, or on the wall above the door.

In some jurisdictions, bare bulbs are not permitted and the lights must be in fixtures. Cages around the bulbs are not acceptable in some areas. Check your local rules.

STRATEGY**Heat Lamps over Doors**

Heat lamps have some of the inherent problems we talked about with recessed lights. They can also set fire to things underneath them. Doors should not be able to swing under heat lamps. Heat lamps should not be mounted above shower rods or shower doors. Towels thrown over the top of doors or rods can be ignited by heat lamps. Watch for this, and where you see heat lamps installed inappropriately, recommend they be rearranged.

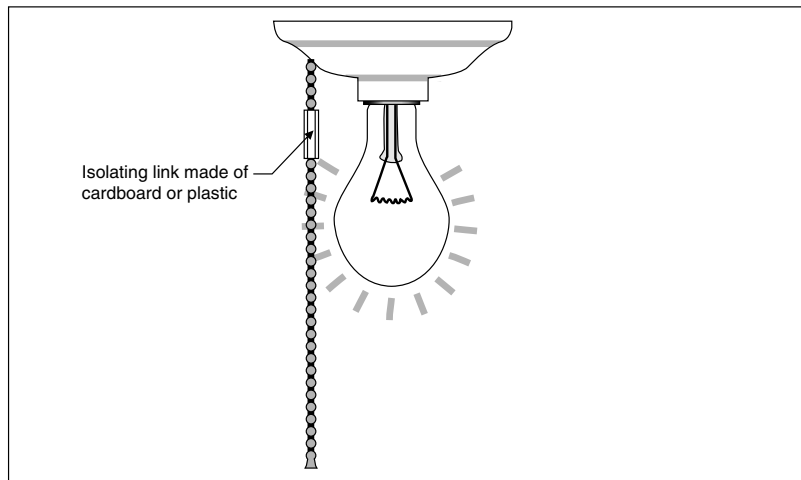
STRATEGY**Isolating Links Needed on Pull Chains**

Lights with metal pull chains to turn them on and off should have isolating links to prevent the pull chains from being alive electrically (Figure 7.35). You can usually see these insulating links. Heavy cardboard or plastic replaces some of the metal links, usually close to the fixture itself. If the pull chain does not have these links, you should question how the chain is electrically isolated from the light. Recommend further investigation.

7.9.2 Conditions at Receptacles (Outlets)

For this part of the inspection you'll need an inexpensive circuit tester, available at hardware or electrical supply stores.

They typically have three lights that indicate wires that are not connected, or are connected improperly. Some testers come with a push button to trip test GFIs. We don't use these models, for reasons discussed later in this section (See "Testing GFIs").

FIGURE 7.35 Isolating Links Needed on Pull Chains

Now, let's look at branch-circuit outlets and some of their common problems:

1. Damaged
2. Loose
3. Overheating
4. Ungrounded outlets
5. Open neutral or open hot connections
6. Reversed polarity outlets
7. Inoperative
8. Wrong type receptacle
9. No GFIs
10. Overheated neutral on split receptacles
11. Worn receptacles
12. Broken pin or blade in slots
13. Not enough receptacles
14. Too far from basins

The causes will be self-evident for the most part, including mechanical damage, wear and tear or installation error. The implications continue to be shock and fire hazards.

Damaged

STRATEGY

Outlets or their cover plates that are visibly damaged should be replaced. This is often mechanical damage caused by furniture.

Loose

STRATEGY

Receptacles that are not well secured to their boxes can be dangerous. The hot terminals are often on the sides of outlets. If the outlet is loose in a metal box, it can touch the box, creating a very dangerous situation. When you use a circuit tester (sometimes called a circuit analyzer or plug-in tester), you'll get a sense of whether receptacles are loose.

STRATEGY**Overheating**

Evidence of overheating on electrical outlets may include discoloration or partial melting of the face. Overheating may also be apparent at the wire connections (melted insulation, charring or scorching, a smell of burned plastic, or warm to the touch) if the cover is removed.

Overheating is more common on aluminum-wired outlets. Overheating may also be more likely on push-in (backwired) receptacles. With aluminum or push-in receptacles, keep overheating in mind as a possibility at all the outlets you look at. In addition, focus on outlets used for heat-generating appliances, such as hair dryers, curling irons, kettles, toasters and microwave ovens.

Ungrounded Outlets

Ungrounded receptacles with only two slots can only receive two-prong plugs. It is clear to everyone that the circuit and appliance are not grounded.

There is a safety issue if the box itself is not grounded, as was a common installation practice until the early 1960s. If a hot wire or terminal from the outlet contacts the metal box, the live box becomes a shock and fire hazard.

Ungrounded outlets create another safety issue if the old two-slot receptacle is replaced with a modern U-ground receptacle (with a third slot for the ground pin). These are often installed in old houses where there is no ground wire in the branch circuit wiring. A false sense of security is created since the outlet looks grounded, but is not.

It wasn't until the mid 1990s that the addition of a GFI (Ground Fault Interrupter) was considered an acceptable alternative to grounding in an existing installation.

GFI is not as good as grounding, in some respects. Let's look at an ungrounded receptacle with a GFI. If you plug in a power drill (with a grounded three-prong plug) and the hot wire inside the drill touches the metal case, the case will become live. When you pick up the drill, you will get a shock. If the ground pin in the drill plug was connected to a house ground system, you would not have gotten a shock.

However, while you'll get a shock, it will only be for a very brief interval, since the GFI will sense the imbalance in the circuit (as electricity flows through the black wire, through you into ground, but not through the neutral wire) and the breaker will trip.

In practice, very few people ground their receptacles in an old house unless there is an appliance that relies on grounding. Any appliance with a three-prong plug relies on grounding. Home computers, for example, dissipate **static charges** through the ground wire. The GFI solution will not be helpful here. We need to ground the outlet.

Most inspectors point out ungrounded outlets but don't recommend complete updating. Clients should be helped to understand the issue and then make their own decisions. Partial updating is common.

When You Think Ungrounded Outlets Are Grounded

Sometimes your tester will tell you that a circuit is grounded even if it's not. For example, if a clothes washing machine is plugged into the outlet and you plug in a tester, there may be an inadvertent path to ground through the washing machine. The flow to ground would be from the ground pin in the tester, through the outlet, through the washing machine plug, back to the washing machine and possibly to the ground below the washing machine or through the water connections to the water piping out to ground.

The path to ground is not through a designed ground connection, but an accidental path through an appliance.

Two Slot Receptacles

Box Not Grounded

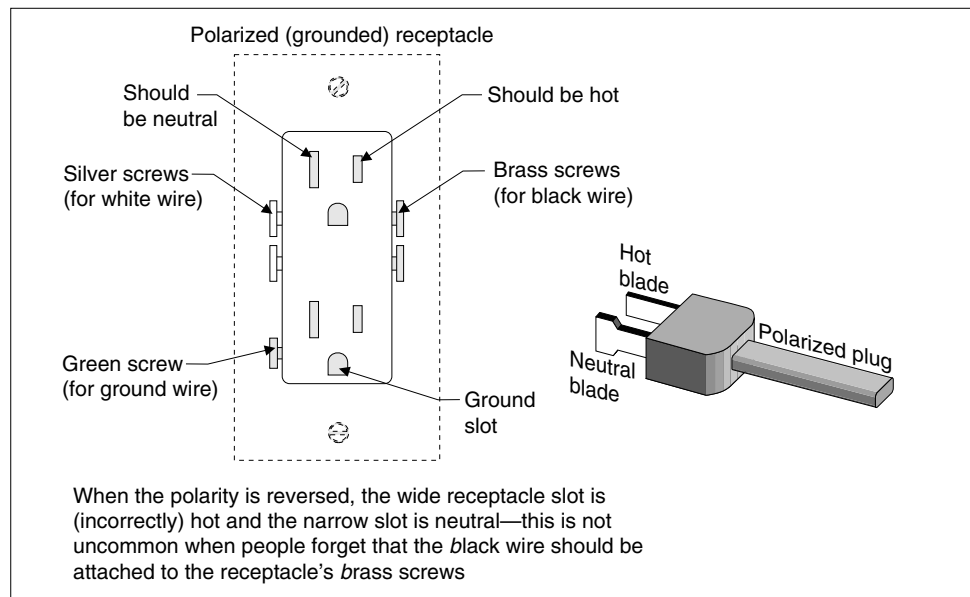
Three-Slot Receptacles on Ungrounded Circuit

GFI Instead of Grounding

Ground Only Old Outlets Where Needed

Partial Update

STRATEGY

FIGURE 7.36 Reversed Polarity

You can avoid being fooled by unplugging appliances from outlets before testing. However, be careful that you do not disrupt household appliances by unplugging them. Home computers and VCRs, for example, should not be unplugged. Most inspectors don't unplug appliances to test the ground.

Open Neutral or Open Hot Connections

When you use your electrical tester, sometimes it will show **open neutral** or **open hot**. Open means not connected. In either of these cases, the outlet won't work. This is obviously the result of a wiring mistake and the outlet should be rewired. It usually means the wire is not attached to the outlet.

Reversed Polarity Outlets

Your electrical tester will sometimes show **reversed polarity**. What does this mean?

Polarity is a confusing issue. Within 120-volt household circuits, if the polarity is correct, the black wire is the hot wire (ungrounded conductor) and the white wire is the neutral wire (grounded conductor). If the polarity is reversed, the white wire is hot and the black is neutral (Figure 7.36).

Polarity matters with appliances that have switches. When the appliance is plugged in, power should only go as far as the switch. If polarity is reversed, power will go through the entire appliance back to the switch. If a wire comes loose in the appliance, the entire case of the appliance may be "hot" electrically, even though the appliance is not on. This is a shock hazard.

Home inspectors should check the polarity of receptacles. If the receptacles are not polarized, this is not as issue. Polarized appliance plugs will not fit into non-polarized receptacles because the slots are both narrow. The wide blade for the neutral side will not fit in. We trust none of you will ever admit to filing off the wide flanges on the neutral blade, since this defeats the protection of polarization.

Where you find polarized receptacles, which include some of the two-slot receptacles and all of the modern U-ground receptacles, we should be checking for polar-

Polarity

Switched Appliances

Test Polarity

Black Wire to Brass Screw

ity with a portable circuit analyzer. You should describe **reversed polarity** receptacles as needing rewiring.

One of the ways to identify reversed polarity, other than with a tester, is by removing the outlet cover plate. The black wire is typically connected to the brass screw terminal on the outlet, and the white wire is connected to the silver. This color convention is useful but not 100 percent reliable for identifying reversed polarity.

Sometimes the outlet will appear to be wired properly when you take the cover off. The black wire correctly goes to the brass terminal and the white wire correctly goes to the silver terminal. The polarity may have been reversed somewhere upstream, perhaps right at the panel itself. That's why it's not completely reliable to look at the receptacle wiring connections to see if polarity is reversed.

Reversed polarity outlets often go unnoticed for a long time. Many appliances will work just fine. However, the polarity has to be right to provide the protection designed into the system.

STRATEGY*There May Be Power in Box***Inoperative**

If the outlet is apparently unpowered during your test, it should be written up for improvement. Do not assume there are no hot wires to the outlet. Simply describe it as not working and in need of attention.

Removing the cover will sometimes show why the outlet is not working. (Occasionally there will be no wires coming into the box!) However, in many cases, it won't be clear and you should not get into a trouble-shooting mode. You can be fooled if the outlet is controlled by a wall switch. Many rooms do not have overhead lights but have a switch on the wall. Make sure you flip the wall light switch before describing the outlet as inoperative. In some cases, both halves of a duplex receptacle are controlled by the wall switch. In other cases, only the top or bottom half of the receptacle is switched.

*Switched Outlets***STRATEGY****Wrong Type Receptacle**

Some 20-amp receptacles have a neutral slot at right angles to the hot slot. Conventional plugs will not fit into the slots. Others have one T-shaped slot that will accept 15-amp and 20-amp plugs.

In modern construction, 20-amp circuits are required for at least two circuits. These are often in the kitchen and dining room. Receptacles on these circuits should be rated at 20 amps.

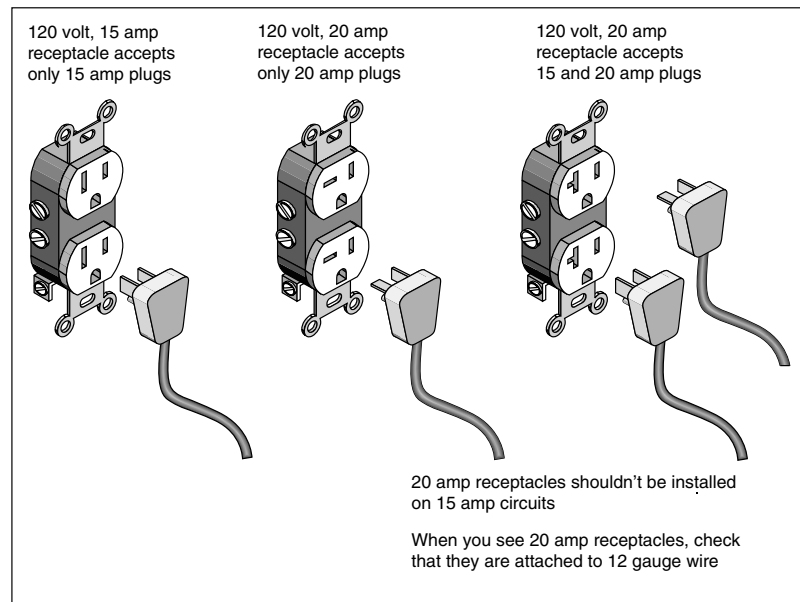
Outlets rated at 20 amps should not be used on 15-amp circuits. If they are, recommend that these outlets be replaced. 20-amp outlets can be identified by a slot that is T-shaped rather than straight (Figure 7.37).

*20-Amp Outlets Need 12 Gauge Wire and 20-Amp Fuses***No GFIs**

A ground fault interrupter (**GFI**) is a special receptacle with ground fault protection. There are also ground fault circuit interrupters (**GFCI**) which are circuit breakers installed at the panel to provide ground fault protection for the whole circuit. Let's make sure we don't get confused between grounding, ground fault circuit interrupters and fuses or breakers.

*GFI and GFCI**Grounding*

The equipment grounding wire in the branch circuits protects us by providing an emergency path for electricity when the electricity gets where it shouldn't. The ground wire presents a path that's more inviting to the electricity than flowing through us. If grounding systems work properly, we won't get electrical shocks even when things go wrong.

FIGURE 7.37 Wrong Type Receptacle*Fuses and Breakers*

Fuses or breakers are safety devices too, but they don't do the same thing as ground wires, nor do they do the same thing as GFCIs. Fuses or breakers shut off power to the circuit if more electricity flows through the circuit than it was designed for. If there is a 25-amp short circuit, for example, in a common household circuit, the 15-amp fuse or breaker will shut off the power. However, if there were an 8-amp electrical current flowing from the black wire through the ground wire (short circuit), the fuse or breaker would be perfectly happy.

GFCI Don't Replace Fuses

Circuits with GFCIs need to have fuses or breakers to shut off the electricity in the case of an overload. So what does a GFI do? A GFI finds electrical **leaks** and shuts off the power if leakage is occurring. It finds leaks by comparing the current flowing out through the black wire to the current flowing back through the neutral (white) wire (Figure 7.38). Where it sees less current coming back through the white than went out in the black, it assumes there is a fault to ground (a leak), and shuts down the circuit.

*Find Leaks**You Create a Short Circuit*

If you create a short circuit by touching a black and white wire, the GFI may not trip because the current in the black and white wires would be the same. This helps explain why GFIs aren't quite as good as grounding, but are much better than nothing.

Where Are They Used?

GFCI receptacles are currently required for bathrooms and washrooms, outdoors, in garages, above counters for a kitchen sink or wet bar, and in unfinished basements and crawlspaces.

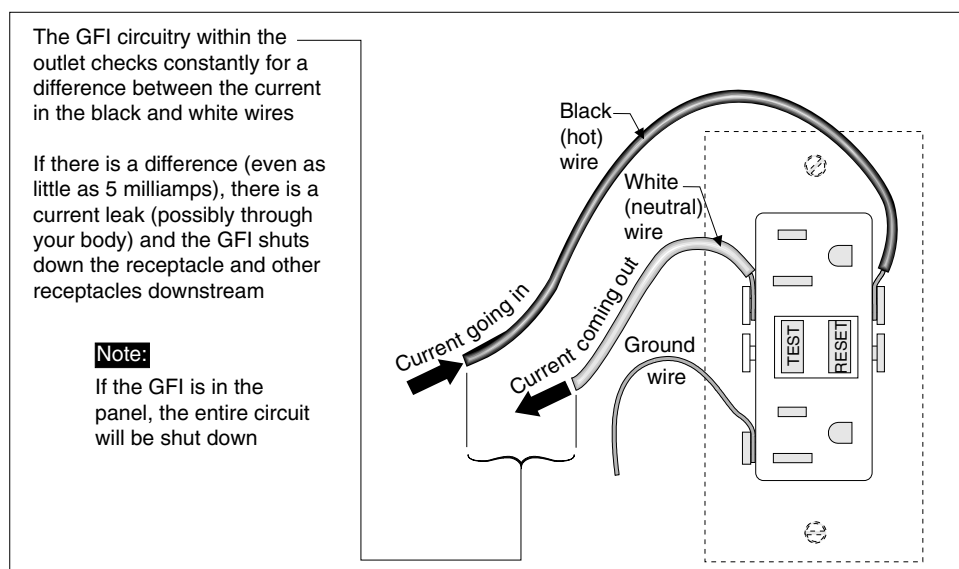
Only the outdoor outlets within 6½ feet of grade need GFIs.

*Only Needed Close to Grade Outside**Swimming Pools
Whirlpool Baths*

You also need GFCIs for swimming pool equipment and whirlpool baths, although there are usually no receptacles involved.

Exceptions

There are exceptions to these rules in some cases and so you should be careful how you apply these.

FIGURE 7.38 Ground Fault Interrupter*Codes Have Changed*

Since we're not doing code compliance inspections and since the requirements for GFIs in different locations came into force in different years, you'll have to make your own decisions as to how far you are going to go. While you could recommend GFIs in all currently required locations in all houses, most inspectors do not. You are wise to get some guidance from the local authorities. What do they recommend on existing homes?

*GFI Outlets Protect Ordinary Outlets Downstream***Is It on a GFI or Not?**

Sometimes it's difficult to know whether an outlet has **ground fault protection**. Ordinary outlets can be protected by GFIs either at the panel (**a GFCI breaker**), or by a **GFI outlet** upstream. A GFI outlet will protect all outlets wired downstream of it. This is because all electricity flows past the GFI outlet to the other receptacles. The GFI will check all of the electricity flowing through the black wire and measure all of the electricity coming back through the white.

Many electricians put a GFI outlet in a first-floor bathroom in new houses, and put ordinary outlets in other bathrooms in the house on the same circuit. These outlets don't look like they have GFI protection, but they do.

Testing Procedure

There are a couple of ways for you to test this, but all of them involve some walking. If you trip the GFI breaker in one washroom, you can leave it off and check the receptacles in the other washrooms and bathrooms to make sure they are off. Then you can come back, reset the tripped GFI in the first washroom, and check the receptacles in the other bathrooms for proper wiring.

Client Help?

Some people employ their clients as assistants on this kind of test to save some steps. Other inspectors take the extra steps to make sure it's done right.

Testing GFIS

There are two ways to test GFIs within the scope of a home inspection. The first is to use the test button on the GFI receptacle. The second is to use the test button on a plug-in circuit tester. (Not all circuit testers have the ability to test GFIs.)

If you press a test button and the power to the outlet is not cut off, the GFI is faulty or wired improperly. Your circuit tester should show if there is reversed polar-

STRATEGY*Test Button on Outlet*

One Test Works, Other Doesn't

ity, open ground, open hot, etc. The tester also shows whether the power shut off when the test button tripped.

Sometimes the test button on the GFI works, but the test button on the circuit analyzer does not. This is usually because the circuit is ungrounded. This should show up on your circuit analyzer. The test button on your circuit analyzer tests the GFI by creating a small fault to the ground wire. If the receptacle is not grounded, this circuit isn't closed and no electricity flows. The tester won't trip, and you will incorrectly report the GFI as defective, if you aren't careful.

Use Outlet Test Button

Most GFIs use an internal test that does not rely on a ground wire. The test button on the GFI outlet is a better test than the one on an external tester. Most electrical authorities will accept GFIs without grounding. In using your circuit analyzer to test GFI receptacles, make sure you test the entire outlet and use the test button on the GFI to help you come to the right conclusion.

*Line and Load Sides
Reversed (Not Reversed
Polarity)*

One common problem with GFIs is that they can be wired backwards. The line side of the GFI should be connected to the power supply. The load side of the GFI should be connected to outlets downstream. If it's wired wrong, with the power coming in to the load side, the GFI protection is defeated at this outlet. Outlets downstream will be protected, but not this one. The dangerous part is that the test button will trip but there will still be power to the outlet! That's why it's a good idea to have your circuit tester plugged into the outlet when you trip the GFI test button.

*Not on Multi-Wire Branch
Circuits*

GFIs can't work on multi-wire branch circuits because the current through the black or red wire is usually different from the current through the white. The GFI would trip constantly.

STRATEGY

Worn Receptacles

Receptacles that are older and have been heavily used may have loose slots that won't hold plugs tightly. Plugs that tend to move or fall out of receptacles are dangerous because of the poor connection and overheating that can result. They are also dangerous because live parts of the plug blades may be exposed if the plug is not tight in the outlet.

When you plug in your tester, you can often sense whether the slots are loose. Once the tester is put in, try to move it side to side and up and down. The lights should stay on, and the tester should not move easily.

Where you find worn receptacles, recommend their replacement.

STRATEGY

Broken Pin or Blade in Slots

Where you find an outlet with something broken off in a slot, recommend that it be replaced. This can be dangerous, since whatever is sticking out of the slot may be hot. Furthermore, you can't plug in an appliance.

Not Enough Receptacles

Outlets Every 12 Feet

In older houses, there will usually be fewer receptacles than in newer houses. In modern construction, there should be one outlet about every 12 feet along walls. Any wall 2 feet or longer should have an outlet.

*Shouldn't Have to Walk More
Than 6 Feet to an Outlet*

The actual rules say that any point you walk to on a wall must be within 6 feet of an outlet to one side or the other. Since most appliance cords are about 6 feet long, you shouldn't have to go farther than 6 feet to find a receptacle for your appliance.

Kitchen Counter Outlets

Above kitchen counters, outlets in new construction should be arranged so you don't have to go more than 24 inches in either direction to find an outlet. If the counter space is 12 inches long or more, an outlet is needed above it.

Island Outlets

In new work, outlets are required on work islands in kitchens. Outlets are most often installed on the face of cabinets below the island, or they can be mounted on

STRATEGY*Codes Not Retroactive*

the island in an upright position on a pedestal (tombstone). They should not be surface-mounted on countertops, because water may get into the outlets.

It is a somewhat subjective call to determine how much electricity is enough for your clients. Most older houses won't have the electrical distribution found in new houses. We recommend that you point this out. We encourage people to see where their electrical needs are focused after they move into a house, rather than make an arbitrary and expensive upgrade of the electrical distribution throughout the house.

Code requirements are usually not retroactive, and you don't have to upgrade electrical distribution on an existing house. Wherever renovations are undertaken, it obviously makes sense to improve electrical distribution at that time since walls are open and running wires is straightforward. It's often required to meet current codes when renovating, anyway.

We'll normally cite a deficiency if there's only one outlet in a bedroom, for example. If there's one outlet on each wall of the bedroom, we usually consider it adequate.

Adding Outlets

Each branch-wire circuit can have 10 to 12 points. (This is a combination of light fixtures and receptacles. Switches are not included.) In many houses, existing branch circuits are extended to add more outlets.

STRATEGY*Outlets Beside Basins
Away from Tubs***Too Far from Basins**

Electrical authorities like receptacles in bathrooms to be close to the basin, but they shouldn't be right over the basin since the cord may droop into the water.

Receptacles should be on either side of the basin. The authorities recognize that this is where people use hair dryers, curling irons, electric toothbrushes, electric shavers, and so on.

Inspector in the House: Locked Out of a Garage by a GFI

While inspecting a new town house, I cut the power to the exterior electrical outlet at the back of the building with the GFI (Ground Fault Interrupter) button on my circuit tester. This confirmed that this outlet was protected by a GFI upstream somewhere. So far, so good.

Knowing that these homes typically have one GFI type exterior outlet protecting the other exterior outlets, I went to reset the GFI at the front porch electrical outlet. But there was no outlet there! No problem, I thought; there must be a GFI breaker at the panel.

I was surprised and a little disturbed to find no GFI breaker panel! I did see a label on the panel saying "Exterior/Garage GFI," and I realized the GFI must be in the garage. That makes sense, I thought.

This garage did not have a door into the house—the only access was through the vehicle door. I found the push button for the vehicle door opener but it did not work, because the opener was plugged into the GFI outlet in the garage, and I had tripped the circuit! Naturally, there was no exterior disengagement mechanism for the vehicle door opener.

I could have cut a hole in the garage door and pulled the manual release handle, or cut a hole through the closet beside the garage, and squeezed into the garage. Neither option was appealing, so I finished the inspection and left an apologetic note for the homeowner to call me.

I arranged to go back the next day and, in the basement, found the wire going through the wall to the garage outlet. I punched a small hole through two layers of drywall, and was able to reach through and press the reset button on the GFI. Problem solved!

I patched the small holes in the drywall and have changed my routine for town houses with garages—I never trip an exterior outlet unless I know I can get to the GFI reset.

7.9.3 Conditions at Switches

Common switch problems are listed below. Causes include installation problems, mechanical damage, wear and tear, and mechanical failure. The implications are shock and fire hazards.

1. Damaged
2. Loose
3. Overheating
4. Inoperative

STRATEGY

Damaged

Watch for switches or cover plates that have been mechanically damaged. Recommend replacement. This is a safety issue, since people will be touching the switch handle regularly.

STRATEGY

Loose

Recommend replacement if the switch handle is loose, if the whole switch is loose in the box, or if the box is loose in the wall. These are shock and fire hazards.

STRATEGY

Overheating

Any switch that is warm to the touch, or shows evidence of charring, discoloration or melted plastic, should be considered a serious safety hazard and should be immediately improved. One exception is dimmer switches which are warm to the touch under normal use.

Wattage Rating of Switch

Where a switch controls several bulbs (e.g., a chandelier or several light fixtures), the wattage of the light bulbs may exceed the wattage rating of the switch.

STRATEGY

Inoperative

If the switch does not work, don't try to solve the problem. Just report that the switch does not operate the appliance it is supposed to. The switch could be fine and a light bulb could be burned out, for example.

7.9.4 Conditions at Junction Boxes

All Connections

Junction boxes are used to contain and support switches, outlets and lights. They can be metal or plastic, and come in a large variety of shapes and sizes. Junction boxes may also simply contain wire splices. All modern connections should be made inside a junction box. An electrical panel is considered a junction box.

Accessible

Junction boxes should not be buried in walls, floors or ceilings. They're supposed to be accessible. The presence of many junction boxes acting as splice containers suggests amateur work. Good electrical installations include very few junction boxes.

Covers

Junction boxes should have appropriate cover plates. Often, this is the switch or outlet cover plate. With light fixtures, the light fixture itself forms the cover for the box.

Let's look at the problems with junction boxes. Causes include installation problems and mechanical damage. The implications are, of course, shock and fire hazard.

1. Damaged
2. Missing
3. Loose
4. Not grounded
5. Cover loose or missing
6. Grounded
7. Concealed boxes

STRATEGY

Damaged

Damaged boxes should be repaired or replaced.

STRATEGY

Missing

Electrical splices made without a junction box are incorrect and a box should be added. Exceptions include original knob-and-tube connections. Modern wire attached to knob-and-tube wiring should have connections in boxes.

STRATEGY

Loose

Junction boxes that are not well-secured, or have components such as cover plates that are loose, should be written up for improvement.

STRATEGY

Not Grounded

You won't pick up this one often, but if the cover plate is missing, you may be able to see whether the junction box itself is grounded. This is only an issue on metal boxes.

STRATEGY

Cover Loose or Missing

Missing or loose cover plates on junction boxes should be replaced or re-secured.

STRATEGY

Crowded

This is something you'll only see if the cover plate is missing. A common sense approach is all you need here. While there are several specific rules for various boxes, with little practice you can tell whether a box is crowded. If it's difficult to put the cover on without pinching a wire, it's overcrowded.

STRATEGY

Concealed Boxes

Boxes have to be accessible. This is a problem you'll rarely discover, since if the box is concealed, you won't see it.

7.10 SMOKE DETECTORS

Function

Smoke detectors protect people by notifying them of fire in the home, especially when people are sleeping. They usually provide an audible alarm, and some have a visual alarm consisting of a flashing light.

Inspecting Smoke Detectors

The Standards ask us to report the absence of smoke detectors, but do not ask us to test them. We suggest you make it clear to your client whether or not you tested the smoke detectors.

How Many Detectors?

What if there is one detector in the home, but you don't believe that's enough? The Standards only ask you to report the absence of smoke detectors. Our approach is to advise that detectors should be on every floor level of a home, and should be near all sleeping areas. We also advise that interconnecting detectors is a good idea, but expensive to do in an existing home. We stress to clients that maintenance and testing of these safety devices is important.

REVIEW QUESTIONS

1. Service entrance conductors run from the service drop to the service box or service panel.
True False
2. The service size is determined by the amperage and voltage ratings of the service.
True False
3. Is the service drop overhead or underground?
Overhead Underground
4. Are service laterals overhead or underground?
Overhead Underground
5. The drip loop is typically part of the service drop.
True False
6. The wire size for the service drop must be the same as the wire size for the service entrance conductors.
True False
7. How many wires would you typically find in a 240-volt service drop?
8. What would four wires mean?
9. What would two wires usually mean?
10. List seven common problems with service drops.
11. List clearances above ground, swimming pools, and roofs.
12. The service entrance conductors run from the _____ to the _____.
13. All service entrance conductors must be in conduit.
True False
14. All service entrance conductors must be attached above roof level.
True False
15. The drip loop should be as close to the roof surface as possible.
True False
16. List ten common problems with service entrance conductors.
17. List four common areas of water penetration into service entrances.
18. List two functions of the drip loop.
19. Roughly 999 out of 1,000 houses have _____ volts available (insert a number).
20. Roughly 999 out of 1,000 houses have _____ service entrance wires (insert a number).
21. Load calculations are part of a home inspection.
True False
22. Implications of an undersized service include—

a. shock hazard	d. melting insulation
b. fire hazard	e. nuisance tripping of the main fuses or
c. shock and fire hazard	breakers

23. What is the largest single-phase service that you are likely to find in a home?
24. What size copper wire would you expect to find on the service entrance for:
 - a. A 60-amp service?
 - b. A 100-amp service?
 - c. A 200-amp service?
25. Which of the following is a reliable way to identify the service size?
 - a. the size of the service drop wires
 - b. the distribution panel rating
 - c. the service box rating
 - d. the service entrance conductor size
 - e. the service entrance conduit size
26. Aluminum is an acceptable service entrance conductor material.
True False
27. A fused 200-amp service has two 100-amp fuses in the service box.
True False
28. If you can't read the size of the fuses in the service box, you should remove them to try to get a better look.
True False
29. Which of the following is a safety issue?
 - a. The main fuses have a lower rating than the service entrance wire.
 - b. The main fuses have a higher rating than the service entrance wire.
 - c. The service box has a higher rating than the service entrance wires.
 - d. The service box has the same rating as the service entrance wires.
 - e. The distribution panel has a larger rating than the service entrance wires.
30. The service box can be inside, outside or in a garage.
True False
31. There can be up to 10 throws to disconnect all of the electricity in the house.
True False
32. The service box may stand alone or may be combined with the distribution panel.
True False
33. List 16 common conditions you might find in a service box.
34. The functions of earth grounding systems include _____.
35. The grounding electrode conductor may also be called _____.
36. You are allowed to ground to the gas piping.
True False
37. The grounding systems are connected to the house neutral at the _____.
38. From the point where the ground and the neutral wires connect, there are two paths to ground. What are they?
39. List five destinations where ground wires can terminate.
40. List 12 common problems found with grounding systems.

41. Distribution panels are also called (list six other names if you can) _____.
42. Wires run directly from the service box to the branch circuits.
True False
43. What is the advantage of a 240-volt circuit?
44. What size copper wires are adequately protected by:
 - a. 15-amp breakers? _____
 - b. 20-amp breakers? _____
 - c. 30-amp breakers? _____
 - d. 40-amp breakers? _____
45. What size wire is found on most general purpose outlets for lights and receptacles?
46. Which of the following is a safety concern?
 - a. a fuse that is too small
 - b. a fuse that is too big
47. Subpanels must be immediately adjacent to the main panel.
True False
48. Typical distribution panels have 60 circuits.
True False
49. Do we normally shut off the power when looking at an electrical system in the house? Why or why not?
50. List 14 conditions that may be found in all panels.
51. Explain why fused neutrals are a bad idea.
52. What is the implication of a panel that is too small for the service?
53. Should the neutral and the ground wires be bonded together at the sub-panel?
54. How should the feeder wires to a subpanel from the main panel ideally be protected?
 - a. No fusing is necessary.
 - b. The main house fuses or breakers will protect the wire.
 - c. Fuses should be provided at the subpanel.
 - d. Fuses should be provided at the main panel.
 - e. None of the above.
55. What are the implications of oversized breakers?
56. What does the term “double tap” mean?
57. The red and black wires of multi-wire circuits should be on separate bus bars.
True False
58. List four common problems found only with breakers.
59. List six problems found with wires in panels.
60. What is a split receptacle?
61. What is meant by **linking** in multi-wire branch circuits?
62. List two different branch wire materials you might see.

63. Describe three different cable types you might see.
64. What is the difference between insulation and sheathing?
65. The most common wire size for branch household circuits is _____ gauge.
66. What are the issues on running wire through steel studs?
67. List five common branch circuit wire problems.
68. Give four examples of damaged wire.
69. List three methods for securing wire ends.
70. What is meant by **open splice**?
71. Wire should be _____ inches away from the edge of studs.
72. A grommet is _____.
73. Wires should not be run on wall surfaces without _____.
74. What is the problem with running wire across the top of the attic ceiling joists?
75. Abandoned wire should be _____.
76. Describe how knob-and-tube wiring is visibly different from modern cable.
77. Does knob-and-tube wiring have a ground wire associated with it?
78. List four common problems with knob-and-tube wiring.
79. List three common problems found with aluminum wiring.
80. How would you identify aluminum wiring?
81. When was it used?
82. What caused the problems? (list at least three things)
83. Where is the anti-oxidant grease typically used?
84. List 12 common lighting problems found in houses.
85. How are lights tested?
86. Why should heat lamps not be located over doors?
87. List 14 common receptacle problems.
88. Describe how you would test a GFI receptacle during an inspection.
89. Explain the problem with reversed polarity.
90. Can an outlet be protected by a ground fault interrupter without it being apparent?
91. Outlets should be close to basins.
True False
92. List four common problems found with switches.
93. All junction boxes should be concealed.
True False
94. List seven common junction box problems.
95. Knob-and-tube wiring was commonly installed without junction boxes.
True False
96. One smoke detector is needed in each home, near the kitchen because that's where most house fires start.
True False

